Supporting Information: Substituting Reference σ-Profiles Degrades Solubility Prediction; on Activity Data, Bounding a Deployed COSMO-SAC Against Its Inputs Returns a Map, Not a Verdict

N. L. Polomoshnov[*,†,‡] and A. V. Rudik[‡]

†*Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University, Moscow, Russia*

‡*V. N. Orekhovich Institute of Biomedical Chemistry, Moscow, Russia*

E-mail: nikitapol@fbb.msu.ru

This document is the Supporting Information for the article named above. Its own sections, tables, figures, equations and lemmas carry an S—§S1, Table S3, Fig. S1, Eq. (S1), Lemma S1—and a reference without one, such as §3.5 or Table 4, is to the article, whose Table 1 defines the terminology used here, including IDAC, infinite-dilution activity coefficient. With one exception, every result here is one the article states: for each of them the claim, the set it is measured on, its uncertainty and its scope are in the article's own text, and what this document adds is the proof, the per-row table, the robustness grid, the control and the negative behind each. The exception is the loss-simplification ablation of §S1, a pre-registered check that the paradox is not an artifact of the objective, whose numbers appear in this document alone; it contradicts nothing the article states, and the article draws nothing from it.

S1 Supplementary methods

COSMO-SAC constants. The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area $a_{\text{eff}} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\text{kcal}\,\text{Å}^4\,\text{mol}^{-1}\,\text{e}^{-2}$, hydrogen-bond coefficient $c_{\text{hb}} = 85580$ with cutoff $\sigma_{\text{hb}} = 0.0084\,\text{e}\,\text{Å}^{-2}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$; the σ-profile grid is 51 bins on $[-0.025, 0.025]\,\text{e}\,\text{Å}^{-2}$. The segment-activity fixed point is damped ($\lambda = 0.7$) and run for 16 iterations at train time and 30 at evaluation.

A usage pitfall in the reference implementation. Supplying the 2002 kernel with a typed three-profile base produces a silently-wrong result that the library returns without warning. The reference implementation[S1] pairs its 2002 model with the single-profile Virginia Tech database; the demonstrated input is one untyped, Mullins-averaged profile. When that 2002 model is instead supplied with a typed three-profile Hsieh base — the configuration that arises when a 2002 and a 2010 arm share one profile source — it silently reads the non-polar channel only, discards the donor and acceptor channels, keeps the total cavity area as prefactor, and returns a number without error. The prescribed way to recover a single 2002 profile from the typed set is instead their *sum*, $p = p_{\mathrm{nhb}} + p_{\mathrm{OH}} + p_{\mathrm{OT}}$ (Eq. 9 of Ref.[S1]); the nonpolar-only read is neither that sum nor a documented mode. This single-channel arm scores MSE 1.263 on the VT-2005-matched set, whereas the *fair 2002* arm under the full convention (the same molecule supplied either to our differentiable layer or to `cCOSMO` with the full profile placed in the single nonpolar channel it actually reads) scores 1.757/1.758. This is an issue for the library maintainers (a silently-wrong return where an error would be safer), not a defect of any published benchmark. The kernel arms of §3.5 are deposited across two artifacts, and which file holds a value does not follow the set. The dispersion-*off* values (1.757 and 0.765 on the VT-2005-matched set, 1.911 and 1.978 on the broad IDAC set), all under the full convention in which the kernel comparison is run, are our differentiable layers on each kernel's native profiles. Both dispersion values are the reference implementation's own dispersion-on call: the VT-2005-matched set's 0.785 from the kernel-flip artifact, where the reference returns no dispersion for the three bromine/iodine pairs and that arm therefore scores on n=57; the broad IDAC set's 1.855 from the dispersion-on field of the fidelity-lever artifact, whose VT-2005-matched counterpart in the same file is instead our own layer's 7.43. That 7.43 is the measure of the gap: our transcribed dispersion term is not validated against the reference, which is why the dsp arm is deferred to that implementation throughout. The input-fixed control of §S3(a′) is a third deposited file: it runs the 2002 kernel on the summed typed profile and re-derives the 1.757 and 0.765 arms of the first file as its own gate.

The differentiable COSMO-SAC-2010/dsp layer, and the scope of its dispersion caveat. The 2010 layer implements the four mechanisms of:[S2,S3] the donor/acceptor-typed three-profile representation, the temperature-dependent electrostatic misfit $c_{\mathrm{ES}}(T) = A_{\mathrm{ES}} + B_{\mathrm{ES}}/T^2$, the typed hydrogen-bonding kernel, and an optional dispersion term in the tabulated parameter ϵ/k_{B}. With dispersion *off* it reproduces the NIST COSMO-SAC-2010 reference to RMSE $1.4 \times 10^{-4} \ln\gamma$. That figure is quoted from the validation run's console output: unlike its 2002 counterpart, whose comparison is deposited, the 2010 validator wrote no artifact, so this one figure cannot be checked against the deposited record. The dispersion term is not validated to that standard, for two reasons we state rather than absorb. It carries the tabulated coefficient at a single sign, the class-dependent sign reversal of Ref.[S3] being unimplemented and unexercised on the matched set, which holds no water or carboxylic-acid pair; and no ϵ is tabulated for the three matched pairs containing bromine or iodine, which our layer scores at ϵ=0. That alone accounts for the MSE 7.43 our layer records over all 60 VT-2005-matched pairs; on the remaining 57 it returns the reference 0.785, showing no divergence where both are defined. Every dispersion-off contrast in the paper is therefore computed on our layers and cross-checked against the NIST `cCOSMO` reference, and the *2010, dispersion on* runs are quoted from `cCOSMO` itself (§3.5).

Solver, curriculum and loss. The SLE fixed point is iterated as the damped map $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln\gamma_2(x_2^{(k)}, T)) + (1 - \lambda)x_2^{(k)}$ with closure-specific kernels. Gradients use the implicit-function theorem at the converged $x_2^\star$ (with $\eta = \partial \ln\gamma_2/\partial x_2$) rather than backpropagation through the iterations: exact at the fixed point, memory-constant in the iteration count, and well conditioned unless $1 + x_2^\star\eta \to 0$. The iteration counts above follow a convergence audit: a shorter n=8 schedule fails a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs. The three-phase curriculum is (P1) auxiliary-property pretraining on crystal, Hansen and $\ln\gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. Of the weighted loss, few terms are load-bearing for this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the σ-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\mathrm{fus}}$, on $\ln\gamma_2^\infty$ and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remainder — channel orthogonality, contrastive, mixture-of-experts balance, priors — is auxiliary. The grounded arms begin with a warm-up on the σ pool of up to 40 epochs, early-stopped on a held-out tenth, and the pinned recipe freezes the σ head for P2/P3, so that stream enters the warm-up and P1 only.

The closure itself, and the remaining terms of the 2002 layer. The COSMO-SAC layer of §2 maps a profile pair to $\ln\gamma_2$ through a damped fixed point for the segment activity coefficient Γ_S,

$$\Gamma_S(\sigma_m) = \Big[\textstyle\sum_n p_{\text{norm}}(\sigma_n)\,\Gamma_S(\sigma_n)\,\exp\big(-\Delta W(\sigma_m,\sigma_n)/RT\big)\Big]^{-1}, \tag{S1}$$

with a precomputed exchange-energy matrix ΔW (electrostatic misfit plus a hydrogen-bonding term), a restoring contribution assembled from those segment activities, and an optional Staverman–Guggenheim combinatorial term. The restoring contribution is $\ln\gamma_2^{\text{res}} = (A_2/a_{\text{eff}})\sum_m p_2^{\text{pure}}(\sigma_m)[\ln\Gamma_{S,\text{mix}}(\sigma_m) - \ln\Gamma_{S,2}^{\text{pure}}(\sigma_m)]$, so the cavity area A_2 sets the residual term's prefactor as well as normalising the profile. The optional Staverman–Guggenheim combinatorial term needs a second size factor $r_i = V_i/r_0$, and where that volume comes from differs between the two places this closure is used, which matters for reading the oracle arms below. In the *decomposition*, where every input is a tabulated one, V_i is the reference COSMO cavity volume of the same calculation that produced A_i, so the two share a cavity basis. In the *trained model* the reference volume is never read: the volume channel is off in every arm this paper reports the paradox on (the volume channel is disabled, so no V_i reaches the closure and the combinatorial term is not evaluated at all), and in the one arm where it is on—the + Staverman–Guggenheim row of Table S3—V_i is the auxiliary head's predicted molar volume, detached. Small hydride solvents such as water carry explicit hydrogens so that the polar and hydrogen-bond channels activate. The pair representation is passed through an MLP and may be augmented with RDKit descriptors, Morgan fingerprints and an ionic-context feature; the crystal head additionally offers Walden-entropy and group-contribution T_m modes. The two NRTL control closures of §2 are the Non-Random Two-Liquid model,[S4] the most expressive closure here, which fits interaction parameters per pair — predicting two interaction energies with a $1/T$ law and a non-randomness factor, optionally regularised by a UNIFAC group-contribution prior[S5] — and its symmetric one-parameter collapse $\ln\gamma_2(x_2,T) = \ln\gamma_2^\infty(T)(1-x_2)^2$, in which the identifiability analysis of §2.1 is cleanest. The crystal head's bounded parametrisations are below.

Crystal head. $T_m = 100 + 600\,\text{logistic}(\cdot) \in [100, 700]$ K, $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot)$; the group-contribution T_m prior is affine- calibrated on the train split before entering the head.

Architecture. The headline COSMO-SAC model uses a message-passing encoder (a shared backbone with a 2-layer residual role-specific branch) of hidden width 64 over 3 message-passing layers, on 35-dimensional node and 8-dimensional edge features; a set2set readout (3 processing steps) pools each graph, solute and solvent interact through a 3-layer cross-attention block, and the 512-dimensional pair representation $[\,g_s, g_v, g_s\odot g_v, |g_s - g_v|\,]$ (dropout 0.012) feeds the crystal head, the 51-bin σ-profile head, and the parameter-free COSMO-SAC closure. The full model has 5.40M trainable parameters. Whether that capacity is the binding constraint is not measured here: an in-sample linear probe does not bear on it (§S6.1, where isotropic noise features reach a *higher* train R^2 than the trained encoder does), and the expectation that capacity is not the binding constraint is carried by the arm comparisons of §3.1 instead. The DirectGNN control shares this encoder, interaction, and readout verbatim, and adds a thermometer encoding of T in place of the temperature dependence the physics arm receives through $\Phi(T)$ and the closure. Figure S1 draws the two arms and marks where each of the two error terms enters.

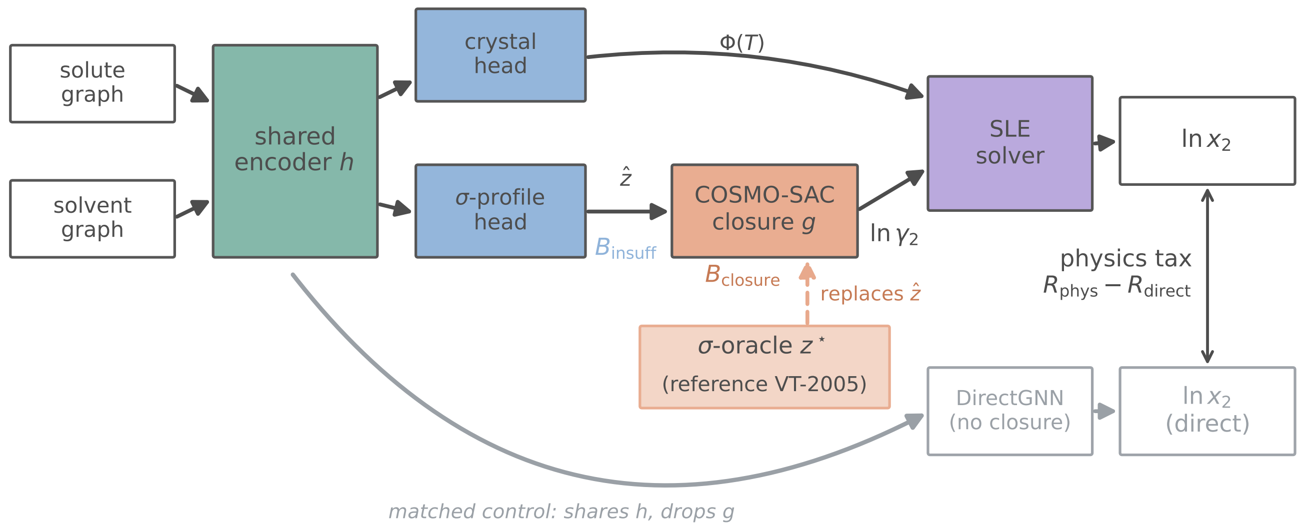

Figure S1: The model and the oracle substitution. Input insufficiency B_{insuff} enters at $\hat{z}$ and closure misspecification B_{closure} where the fixed map g reads $z^\star$, so B_{closure} is the error of map and profile calculation together; the σ-oracle (salmon) replaces $\hat{z}$ with the reference VT-2005 profile $z^\star$. DirectGNN (grey) is the capacity- and budget-matched control, separately tuned, and the double-headed arrow between the two $\ln x_2$ outputs is the physics tax $R_{\text{phys}} - R_{\text{direct}}$.

What the oracle substitution replaces, and what it does not. The deposited reference table carries 1432 rows, each a SMILES string, a cavity area in Å^2, the 51 area-weighted profile bins in Å^2, and a COSMO cavity volume in Å^3. The loader canonicalises the SMILES with RDKit, giving 1424 distinct keys, and returns for each key exactly two objects: the 51-vector $p^\star$ and the scalar $A^\star$, the tabulated cavity area. It does not return the cavity volume, and nothing in the model path reads it.

For each molecule the σ-head emits a shape $\hat{p}^{\text{shape}}$ (a softmax over the 51 bins, so it sums to 1), a strictly positive area $\hat{A}$ (a softplus scaled by 75 and offset by 20 Å^2), and their product $\hat{p} = \hat{A}\,\hat{p}^{\text{shape}}$. The learned area is therefore the integral of the learned profile by construction, and the reference area is the integral of the reference profile as well: across the whole table $|\sum_m p^\star(\sigma_m) - A^\star|/A^\star$ is at most 7×10^{-16}. The substituted object is thus one object and not two, an area-weighted profile, and the substitution is $(\hat{p}, \hat{A}) \mapsto (p^\star, A^\star)$ on every row whose canonical SMILES is a key of the table, applied to solute and solvent *independently*, each on its own match. That single replacement is what carries the reference area into both places the closure reads an area: the residual prefactor A_2/a_{eff} and the normalisation p_2/A_2.

Nothing else is replaced. Not the size-factor volume—the combinatorial term is not evaluated at all in these arms, as above, so no volume of either provenance enters. Not the crystal head: it reads the solute-only pre-interaction embedding, which the substitution does not touch, so its T_m and ΔH_{fus} are the same in both evaluations of a given checkpoint, and the separate crystal override of §2.4 is not applied in any oracle arm. Not the solver.

One coupling is worth stating rather than glossing, because "only the profile changes" is not quite exact. The bounded correction reads a compact summary of the solver's inputs which includes the closure's own τ, so a substituted profile perturbs the correction, and through it the corrected crystal parameters the solver receives. The perturbation is small. Over the three seeds, on the rows where the substitution applied, it moves T_m by at most 0.025 K against a median of 450 K, ΔH_{fus} by at most $4.6\,\text{J mol}^{-1}$ against a median of 3.7×10^4, and Φ—the quantity that actually enters the SLE—by at most 7×10^{-4} against a median magnitude of 4.7. On the rows where the substitution did not apply the same quantities still move, by 1.2×10^{-4} K, $0.045\,\text{J mol}^{-1}$ and 6×10^{-6}, which is the residue of forwarding one checkpoint twice on a GPU and sets the floor these figures should be read against. So Φ is not bit-identical between the two evaluations, and the fourth decimal of a per-row Φ comparison between them is not meaningful; nothing at the scale of the +0.41 turns on it.

All three oracle applications—evaluation-only, the training-time injection, and the channel swap—use the same table and the same both-sides rule; the two training-time arms differ from the evaluation-only one in *when* the replacement is made and in nothing about *what* is replaced.

What the match rule reaches, and which side carries the effect. On the 5608 labelled test rows the table matches the solvent in 5571 (99.3%; 64 of the 70 distinct solvents) and the solute in 297 (5.3%; 7 of the 147 distinct solutes). Every solute match is also a solvent match, so the rows split into 37 with neither molecule replaced, 5274 with the solvent only, and 297 with both; the substitution flag carried by the deposited per-row files is exactly the union, 5571 rows. That asymmetry is a property of the reference database, which covers common solvents and few drug-like solids, and it is worth stating because it decides which side the reported gap is a statement about. Comparing the σ-oracle arm against the grounded arm it is evaluated from, row class by row class and averaged over the three seeds, the MAE difference is -0.000 ± 0.000 on the 37 neither-matched rows—the two are the same checkpoint forwarded twice, and agree to 5×10^{-6} there—$+0.431 \pm 0.036$ on the 5274 solvent-only rows, and $+0.006 \pm 0.025$ on the 297 both-matched ones. The +0.41 of §3.1 is therefore carried by the solvent profile. On the rows where the reference solute area A_2 is the one entering the prefactor, the two arms are not separated; with 297 rows over 7 solutes that is a failure to resolve a difference and not a demonstration that the substitution is harmless there, and we draw nothing from it beyond the scope it sets on the headline number.

Encoder backbones. The shared encoder has three interchangeable backbones: the headline message-passing network (MPNN); a hybrid local-message/global-attention graph transformer (GPS[S6]); and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), which splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen parameters to prevent channel collapse. TIMP does not outperform the plain MPNN across our runs. That is an observation and we do not explain it: the linear probe of §S6.1 has no untrained-encoder arm to read against and supports no explanation.

Optimisation, and the two arms' settings. The physics arm uses Adam (weight decay 5×10^{-4}, gradient clipping at 2.0, batch 64) under the three-phase curriculum of §2: 30 epochs of auxiliary pretraining (P1), 70 of full SLE training (P2), and 10 of low-rate stabilisation (P3), at learning rates 2.6×10^{-4}, 8.5×10^{-5}, and 8.5×10^{-6} respectively. The DirectGNN control was tuned separately and its settings differ: dropout 0.174 against 0.012, P2 learning rate 8.6×10^{-4} against 8.5×10^{-5}, weight decay 4.2×10^{-5} against 5×10^{-4}, gradient clipping 2.54 against 2.0, over 110 P2 epochs equal to the physics arm's total budget at the same width (64) and depth (3). Width, depth, batch size and epoch budget are therefore matched and the optimisation is not, so Δ_∞ and the physics-versus-control ΔMAE are differences between two separately tuned pipelines and are not attributable to the thermodynamic bottleneck alone (§2.4). A hyperparameter-matched control was not run.

Loss. The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hard-wired to 0 and two more (bridge, solubility variance) default to 0 through the configuration—disabled experimental scaffolding, released with the code for completeness. Table S1 lists the 13 carrying a non-zero default, together with the solubility-variance term and the σ-profile earth-mover term, which is applied outside the registry when the grounding stream is on; only the top two groups are load-bearing for the claims, and the *Loss-simplification ablation* paragraph below tests which.

Table S1: The 13 registry terms carrying a non-zero default weight, with the solubility-variance term (zero through the configuration) and the σ-profile earth-mover term, which is not a registry entry.[a] A further 17 registered terms are hard-wired to 0, and one more (bridge) is zero through the configuration.

role	term	default wt.
task	solubility (bin-weighted Huber on $\ln x_2$)	1.0
crystal grounding	melting point T_m	0.3
	fusion enthalpy ΔH_{fus}	0.3
σ/act. ground.	infinite-dilution $\ln \gamma_2^\infty$	0.5
	finite-composition $\ln \gamma_2$	0.5
	Hansen parameters	0.2
	σ-profile EMD (grounding stream on)	†
regulariser	solubility monotonicity in T	0.1
	physics preference	0.1
	adaptive-correction residual	0.01
	NRTL τ	0.01
	solubility variance	cfg
control coupling	DirectGNN NLL	0.2
	DirectGNN regulariser	0.05
architecture	mixture-of-experts balance	0.02

[a] † marks the σ-profile earth-mover term, active only when the grounding stream is on. Per-phase weight overrides are pinned in each run's resolved configuration.

Loss-simplification ablation. For a diagnostic result the elaborate objective is a potential confound: the paradox could be an artifact of loss tuning. The test registered in advance of the run trains the end-to-end grounded arm with only the task ($\ln x_2$ Huber), the crystal grounding ($T_m, \Delta H_{\mathrm{fus}}$), and the σ-profile warmup, *dropping* the activity-grounding auxiliaries ($\ln \gamma_2^\infty$, $\ln \gamma_2$, Hansen) and every regulariser, and then reads the sign of the paradox (oracle vs. learned) and the surrogate drift. The registered prediction was that both persist, the mechanism being end-to-end training through a misspecified closure rather than the auxiliary terms, and both do at the level the ablation measures. The ablation re-measures the solubility *paradox* (reference minus learned $\ln x_2$ MAE), the observable effect, not the profile drift directly: on the full test set it is $+1.13$ (95% CI $[0.88, 1.38]$) under the minimal objective, *larger* than the $+0.30$ ($[0.20, 0.39]$) under the full one. This is the *overall*, solvent-dominated paradox, and it strengthens once the activity-grounding auxiliaries—the only pressure toward physics—are removed; on the both-reference subset (n=297 rows over the $\approx$7 VT-covered solutes) the sign reverses (-0.23 minimal vs $+0.22$ full), so the ablation is evidence that the overall observable effect is loss-robust, not subset-level confirmation of the mechanism. It is *consistent with* the drift being intrinsic to end-to-end training through a misspecified closure rather than a regularisation artifact. It is an implication of the amplified paradox, not a direct re-measurement of $\|\hat{\sigma} - \sigma\|$ under the minimal run (the minimal-loss checkpoint was not retained, and the larger reference-arm error is consistent with more profile drift but does not isolate it from closure-path recalibration). The minimal model trains normally (validation $\ln x_2$ MAE 1.87, within 0.1 of the full-loss base), ruling out a degenerate-training reading (seed 42).

S2 Proofs

Lemma S1 (Structural non-identifiability of the SLE split)**.** *Consider the infinite-dilution SLE observable $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ with $\Delta C_p = 0$, so that $\Phi(T) = (\Delta H_{\mathrm{fus}}/R)(1/T - 1/T_m)$ is affine in $u = 1/T$, and a symmetric two-suffix activity class ($\ln \gamma_2 = \ln \gamma_2^\infty\,(1 - x_2)^2$) in which $\ln \gamma_2^\infty(T)$ is affine in u. Then, for any design with at least two distinct temperatures, μ depends on the parameters only through the two-dimensional statistic (its slope and intercept in u), and the activity intercept and slope already span that plane; hence the crystal parameters ($\Delta H_{\mathrm{fus}}, T_m$) are unidentified, the profiled Fisher information for ΔH_{fus} is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing exter-*

nal constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor $(1-x_2)^2$ *breaks the exact affinity, leaving a non-zero but near-degenerate information of order* x_2^2.

Proof of Lemma S1 (non-identifiability). Write $u = 1/T$ and $h = \Delta H_{\text{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\begin{aligned}\mu(u) &= -\Phi(u) - \ln\gamma_2^\infty(u) \\ &= -h(u-u_m) - (c + d\,u) \\ &= -(h+d)\,u + (hu_m - c),\end{aligned}$$

using $\ln\gamma_2^\infty(u) = c + d\,u$ (affine in u by hypothesis). Thus μ depends on the parameters $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$ only through the pair $(S, K) = (-(h+d),\, hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S,K)/\partial c = (0,-1)$ and $\partial(S,K)/\partial d = (-1,0)$; hence for *any* crystal value (h, u_m) and any target (S,K) there is an activity pair $d = -S-h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h, u_m)\}$, so $(\Delta H_{\text{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S,K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out (c,d) annihilates it and $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$. At finite dilution $\ln\gamma_2(u) = \ln\gamma_2^\infty(u)\,(1-x_2)^2$ adds $\ln\gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in u because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order x_2, so $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$. □

Lemma S2 (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class* $\mathcal{H}$ *with tangent projection* $\Pi_{\mathcal{H}}$*, and let* $v = \partial\mu/\partial\Delta H_{\text{fus}}$ *be the crystal score. The efficient information for* ΔH_{fus} *is* $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}}v\|^2$*. If* $\mathcal{H}$ *contains the crystal score's span* $\{1, 1/T\}$ *(a free intercept and slope) then* $\Pi_{\mathcal{H}}v = v$ *and* $\mathcal{I}_{\text{eff}} = 0$*, recovering Lemma S1; if* $\mathcal{H}$ *is restricted (the slope is known or constrained) then* $\mathcal{I}_{\text{eff}} > 0$ *and the Cramér–Rao lower bound (CRLB) on* ΔH_{fus} *is finite. (The melting point* T_m *is likewise an unknown crystal parameter, but is profiled jointly: its score* $\partial\mu/\partial T_m$ *lies in* $\text{span}\{1\} \subset \text{span}\{1, 1/T\}$*, already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating* T_m *as free.)*

Proof of Lemma S2 (class-dependent information). This is the semiparametric efficient-score construction[S7] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance σ^2, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\text{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the L^2-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}}v$ and the efficient information is $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}}v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u-u_m)/R \in \text{span}\{1, u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \text{span}\{1, u\}$ then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma S1; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing v (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \text{span}\{1\}$) then $\Pi_{\mathcal{H}}v \neq v$ and $\mathcal{I}_{\text{eff}} > 0$, with a finite Cramér–Rao bound. □

Lemma 1, the identity this instrument rests on, is stated and proved in the article at §2.1.1: the cross term vanishes because $g(z^\star)$ is $\sigma(z^\star)$-measurable when $z^\star$ is the closure's whole argument, the Jensen bound needs no smoothness, the law-of-total-variance bound needs only that a binning of $g(z^\star)$ coarsens $\sigma(z^\star)$, and B_{insuff} is a functional of the law of $(m, z^\star)$ alone and therefore the same quantity under both combinatorial conventions—which is why Table S7 carries two valid upper bounds per set. Two consequences of that one-sidedness reconcile the bounding apparatus with the injectivity of the design. First, because every reported B_{insuff} is an *upper* bound and every B_{closure} the matching *lower* one, the bounds are conservative in one direction only: were the superpopulation variance negligible, the finding would be the stronger $B_{\text{closure}} = \text{MSE}$. Nothing reported can therefore be an artifact of having overestimated B_{insuff}, and no entry in any table is the share of the error the inputs contribute. Second, the resampling frequencies are statements about the *instrument* and not about the superpopulation: the clustered-bootstrap frequency P_{boot} is the fraction of cluster resamples in which the *measured bound* still falls below half the total error, MSE/2 being the threshold that orders the two terms. It measures whether this design resolves the ordering, not the probability that an inequality between two population quantities holds, so a low P_{boot} is loss of resolution and never evidence for the inputs.

Theorem S1 (Grounding-gap sandwich)**.** *Assume all risks are finite ($m, g(z^\star) \in L^2$; g and each $h \in \mathcal{E}$ measurable), (A1) the true encoder is representable, $\phi^\star \in \mathcal{E}$ with $z^\star = \phi^\star(x)$, and (A2) the intermediate is a lossless bottleneck, $\mathbb{E}[m \mid x] = \mathbb{E}[m \mid z^\star]$. Then*

$$0 \le \mathcal{G} \le B_{\text{closure}}, \qquad \mathcal{G} = B_{\text{closure}} - \big(R_{\text{e2e}} - R_{\text{free}}\big), \tag{S2}$$

with $\mathcal{G} = B_{\text{closure}}$ iff $R_{\text{e2e}} = R_{\text{free}}$ (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ($B_{\text{closure}} = 0$) gives $\mathcal{G} = 0$; hence under A1–A2, *$\mathcal{G} > 0$ certifies $B_{\text{closure}} > 0$.*

Proof of Theorem S1 (grounding-gap sandwich). Throughout assume $m, g(z^\star) \in L^2$ and g, every $h \in \mathcal{E}$ measurable. *(a) $\mathcal{G} \ge 0$.* By A1, $\phi^\star \in \mathcal{E}$ is a feasible encoder with $g(\phi^\star(x)) = g(z^\star)$, so $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \le \mathbb{E}[(m - g(z^\star))^2] = R_{\text{orc}}$; the infimum need not be attained. Hence $\mathcal{G} = R_{\text{orc}} - R_{\text{e2e}} \ge 0$. *(b) $R_{\text{e2e}} \ge R_{\text{free}}$.* Every $g(h(x))$ is $\sigma(x)$-measurable, and $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$ is the infimum of $\mathbb{E}[(m-p)^2]$ over *all* $\sigma(x)$-measurable p; an infimum over the subclass $\{g(h(\cdot)) : h \in \mathcal{E}\}$ is no smaller. *(c) $R_{\text{free}} = B_{\text{insuff}}$ under A2.* From $\text{Var}(m) = \mathbb{E}[\text{Var}(m \mid x)] + \text{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\text{Var}(m \mid z^\star)] + \text{Var}(\mathbb{E}(m \mid z^\star))$ and A2 ($\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^\star)$, so the two $\text{Var}(\mathbb{E}(\cdot))$ terms agree), the two expected conditional variances agree: $R_{\text{free}} = \mathbb{E}[\text{Var}(m \mid x)] = B_{\text{insuff}}$. *(d) Sandwich and identity.* By Lemma 1, $R_{\text{orc}} = B_{\text{insuff}} + B_{\text{closure}}$. With (c), $\mathcal{G} = R_{\text{orc}} - R_{\text{e2e}} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}})$, and (b) gives $R_{\text{e2e}} - R_{\text{free}} \ge 0$, so $0 \le \mathcal{G} \le B_{\text{closure}}$; equality $\mathcal{G} = B_{\text{closure}}$ holds iff $R_{\text{e2e}} = R_{\text{free}}$. Finally $B_{\text{closure}} = 0$ forces $0 \le \mathcal{G} \le 0$, i.e. $\mathcal{G} = 0$; contrapositively, under A1–A2, $\mathcal{G} > 0 \Rightarrow B_{\text{closure}} > 0$. *(e) A2 is necessary for the certificate.* If A2 fails, take $x \in \{a, b\}$ equiprobable, $z^\star \equiv 0$ (so A1 holds), $m(a) = +1, m(b) = -1$, and g with $g(0) = 0 = \mathbb{E}(m \mid z^\star)$ so $B_{\text{closure}} = 0$; an encoder $h^\dagger(a) = 1, h^\dagger(b) = -1$ with $g(\pm 1) = \pm 1$ gives $R_{\text{e2e}} = 0$ while $R_{\text{orc}} = 1$, hence $\mathcal{G} = 1 > 0 = B_{\text{closure}}$. Thus $\mathcal{G} > 0$ alone does not certify $B_{\text{closure}} > 0$; the decomposition (Lemma 1) does. □

The article states the sign rule in words at §2.1.2, against the asymptote-plus-variance form of the physics penalty $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$ it is written against, and $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$. It is stated and proved here.

Proposition S1 (Sign of the physics penalty)**.** *Write $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$ with $V(n) \downarrow 0$, so that the physics penalty at budget n is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$ with $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$. Then (i) $\mathcal{T}(n) < 0$ (physics helps) iff $V_{\text{direct}}(n) - V_{\text{phys}}(n) > \Delta_\infty$: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If $\Delta_\infty > 0$ and the variance gap vanishes ($V_{\text{direct}} - V_{\text{phys}} \to 0$), there is a critical budget $n^\star$ beyond which $\mathcal{T}(n) > 0$ for all $n > n^\star$.*

Proof of Proposition S1 (sign of the physics penalty). Write $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$ and $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$, so that $\mathcal{T}(n) = \Delta_\infty - D(n)$. *(i)* $\mathcal{T}(n) < 0$ exactly when $D(n) > \Delta_\infty$, immediately. *(ii)* If $\Delta_\infty > 0$ and $D(n) \to 0$, choose $n^\star$ with $|D(n)| < \Delta_\infty$ for all $n > n^\star$ (possible since $D(n) \to 0$); then $\mathcal{T}(n) = \Delta_\infty - D(n) > 0$. Monotonicity of D is *not* required for existence of $n^\star$; it is needed only for the stronger claim "$\mathcal{T}(n) > 0$ at every n," which we treat as an empirical observation. *(iii) A limiting refinement.* Since $V_{\text{phys}}(n) \ge 0$, $D(n) \le V_{\text{direct}}(n)$: the variance a prior can save is bounded by the control's own estimation variance. As the black box approaches the aleatoric floor $V_{\text{direct}}(n) \to 0$, so $\limsup_n D(n) \le 0$; with both estimators consistent, so $V_{\text{phys}}, V_{\text{direct}} \to 0$, one has $D(n) \to 0$ and $\mathcal{T}(n) \to \Delta_\infty$. That limit is ≥ 0 when the control is asymptotically Bayes, so $R_{\text{direct},\infty} = R_{\text{free}}$ (the risk of an unconstrained predictor), and not merely from $V_{\text{direct}} \to 0$; under that condition and A2, $\Delta_\infty = R_{\text{e2e}} - R_{\text{free}} = B_{\text{closure}} - \mathcal{G}$ by (S2). □

The variance term $D(n)$ of Proposition S1 is what carries the semiparametric phenomenon §2.1.2 appeals to, in which an estimated propensity outperforms the known-true one:[S8,S9] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

S3 The claims ledger, per-arm tables and decomposition robustness

S3.1 The claims ledger

Table S2 collects the article's eighteen load-bearing claims in one place: what is claimed, the data set it was measured on and that set's size, and the value with its uncertainty and its scope. It is an index and not the place those scopes live. Every qualification printed in a value cell below—an n, a per-seed triple, "one seed", "one checkpoint", an interval that spans zero, a named exception, the one-sidedness of a margin—is also carried by the article sentence that makes the claim, which is where a reader meets it; nothing in the article depends on this table being open beside it. What the table adds is that the eighteen can be read against one another, and that the five blocks show at a glance which claims share a data set, a confound or an estimator.

S3.2 The scored row set, and why it is not arm-dependent

Every accuracy number of §3.1 is on 5608 rows, and the scaffold partition gives 8103 test rows. The 2495-row difference decomposes with no remainder, identically at all three seeds: 0 rows are lost to duplicate (solute, solvent, T) keys—there are none—and 2495 carry no solubility flag, a melting point and no solubility measurement, with the $\ln x_2$ field a placeholder zero. All 2495 carry a melting point; none is a self-solvation row; they are unscorable for solubility in every arm, the control included.

The clause that could have made this arm-dependent removes nothing. The code that builds the set intersects, over arms, the rows that are both labelled and finite in that arm's $\ln x_2$ prediction, and the second half of that conjunction is outcome-dependent by construction. Audited over all twenty-two per-row prediction files in the deposit—the five intersecting arms and the σ-oracle at each of the three seeds, plus the two co-adaptation controls and the two output-residual arms at seed 42—the count of non-finite $\ln x_2$ predictions is 0 in every file, over every row that file carries and not merely the labelled ones. All twenty-two are full length, covering all 8103 test rows and each carrying exactly 5608 labelled ones. (Twenty-one were full length as first deposited; the seed-44 σ-oracle file was repaired, and the repair is audited below.) The five intersecting arms' labelled row sets are moreover identical as sets, not merely equal in size. Dropping the finiteness clause outright therefore returns the same 5608 rows, so no arm's failures are removed from any arm's score and no number in this paper moves. The set is fixed by which rows carry a label.

Non-finiteness in $\ln x_2$ is not the same event as the saturation that Table 4's clip note reports. That note counts rows whose predicted x_2 approaches 1, where the *unit conversion* to $\log_{10} S$ diverges while $\ln x_2$ itself stays finite; those rows are inside the 5608 and are scored, under the clip that note discloses.

Table S3 gives the exact per-arm metrics behind the six three-seed bars of Fig. 2: three seeds on the labelled test rows (n=5608 of 8103, those carrying a solubility measurement); the two single-seed co-adaptation bars are valued in the paragraph below. Per-seed values are printed to three decimals; every mean, difference and $\pm$ quoted anywhere is computed on the unrounded values, so the printed triples reproduce them only to that precision. Every $\pm$ that spans *seeds* in this paper is a population standard deviation over them; a sample standard deviation would be larger by $\sqrt{3/2}$, and the error bars of Fig. 2 follow the same convention. Three kinds of $\pm$ in this paper are *not* that: the kernel-residual figures of the correctability test ($46 \pm 5\%$, $-5 \pm 16\%$, $+2 \pm 15\%$) carry a standard *error* over five fit seeds; a $\pm$ on a permutation or phase-randomised null (Table S2; §3.3) is that null's

Table S2: One row per load-bearing claim: what is claimed, the data set and its size, and the value measured with its uncertainty and its scope. Notes *a*–*h*, at the foot of the continuation, qualify a row or a column.

Claim	Set (n)[a]	Value, with its uncertainty and scope[b]
The two senses of grounding (§3.1)		
Reference profiles hurt solubility at evaluation (MAE)	labelled test rows (5608 of 8103), 3 seeds	1.85 → 2.25; the per-seed values do not overlap; the gap carries the test-time input swap with it
...and still hurt when injected at training	labelled test rows[c], seed 42	1.80 → 1.98 (own baseline 1.80 → 2.23); one seed and a stream build not certified against its comparator's (§S9), so a sign and not an effect size—free of the test-time swap, which the row above is not
...so part of that gap is test-time distribution shift	labelled test rows[c], seed 42	share 0.59 on one seed: a description of that seed, not a magnitude
Training-time σ supervision *helps* (MAE)	labelled test rows, 3 seeds	2.04 → 1.85; the per-seed values do not overlap; per-seed gains 0.237/0.075/0.281 straddle the co-adapted arm's +0.18, so the two operations are not ordered by magnitude—and the only contrast here a σ-stream leak could inflate, the ungrounded arm carrying no stream (§2.2)
The evaluation substitution degrades the *choice* of solvent, not only the level	labelled test rows grouped by solute and T (823 groups, 107 solutes), 3 seeds	per-solute Spearman -0.197 $[-0.282, -0.117]$ pooled, all three per-seed intervals excluding zero; nDCG@3 -0.084 $[-0.124, -0.043]$; top-1 -0.107 $[-0.192, -0.020]$ pooled only, two of three per-seed intervals spanning zero. Within-checkpoint—one set of weights, $\max\lvert\Delta\Phi\rvert = 4.2 \times 10^{-4}$ at the worst seed—so free of the separate-tuning confound of §3.7, though not of the evaluation-time swap the first row of this block carries; a per-group affine recalibration fitted in sample removes 86% of the *MAE* gap and refitted out of sample removes none of it, and leaves this one untouched either way where the fitted map increases, as it does in 589 of the 823 groups in both arms at all three seeds. The substitution falls on the solvent channel
Closure against inputs on the activity axis (§3.5); margins in squared $\ln\gamma_2^\infty$ units, not the $\ln x_2$ MAE of the block above		
$B_{\text{closure}} > B_{\text{insuff}}$ on *glycol-ether* solvents[d,g]	broad IDAC, 182, 3 sources	the margin, a lower bound on $B_{\text{closure}} - B_{\text{insuff}}$, runs $+2.04$ $[+1.26, +2.87]$ to $+2.94$ over the four cells, $P > 0.95$ in all of them; worst deletion $+1.60$—and the survivor of a 59-fold search, at the ninetieth percentile of a chemistry-blind null on all 477 rows that certifies some row set in 58% of draws, and not testable out of sample, where no rows of that cell exist; the pre-declared cross-fitted $B^{\text{up}}_{\text{insuff}}$, which came back looser and does not headline, puts the same cell at $+1.76$ $[+1.02, +2.49]$ (§S3.4)
... on every other stratum of the map	broad IDAC, 59 strata	§S3.3, Tables S11–S15, each with its margin, interval and the test it fails; none is stated as a finding[h]
... for either set *as a whole*	broad IDAC (477); VT-2005-matched (60)	$+0.51$ (full $+0.95$); -0.38 compounding two cuts ($+0.01$ full); $+0.18$ less one publication; $+0.05$ on the smaller set: an average weighted by composition[f]
$B^{\text{up}}_{\text{insuff}}$ near stratum-independent, MSE not	broad IDAC, 3 boundable classes; 6 with an estimate	$\times 2.0$ against $\times 25$; $\times 6.8$ against $\times 51$; p=0.02 against 2000 null partitions of 9 classes
$B_{\text{closure}} > B_{\text{insuff}}$, 2010 on its *own* typed latent[f]	broad IDAC (477); VT-2005-matched (60)	$+1.29$, $P > 0.95/\approx 0.9/\approx 0.9$[e]; -0.77 at $P \approx 0.3$ on the smaller set, where the ordering is lost: loss of resolution, not a reversal

Table S2: *Continued.* The remaining three blocks of the ledger, and notes *a*–*h*, which qualify rows and columns of both parts.

Claim	Set (n)[a]	Value, with its uncertainty and scope[b]
Whether a better closure moves the ceiling (§3.5.3)		
The 2002 → 2010 step lowers the error	broad IDAC (477)	−0.99 at $P \approx 0.2$: no improvement, and not a measured equality either; under the full convention −0.07, $P \approx 0.6$, and +0.06, $P \approx 0.6$ with dispersion on
…but does on the VT-2005 set's chemistry	57 shared pairs, UD	1.78 → 0.77, $P > 0.95$, at fixed profile database and temperature, full convention; those 57 pairs are also broad-set rows, so not an independent sample
A fitted kernel residual transfers across solvents	VT-2005-matched (60), grouped folds	$-5 \pm 16\%$: spans zero; the ± prices optimiser noise only, and the fold spread that dominates it is not propagated, so no bound follows
Mechanism (§3.3)		
The latent departs from its reference along a shared direction	matched molecules (44), 3 seeds	$51 \pm 2\%$; the magnitude is not stable across analysis configurations, and these are a separate three-seed set of runs, not the arms of Fig. 2, whose latent departure is not measured
…and that departure's top-two mode share exceeds a phase-randomised null	one grounded checkpoint, same 44	0.40 against 0.22 ± 0.01, the ± being the null's spread; one checkpoint, and a wrong-molecule null is not cleared
…and that direction *transfers* to molecules outside the fit	matched molecules (44), 3 seeds	$4.1 \pm 0.5\times$ a zero-drift null, with no constant-offset control on these seeds; the ± spans seeds at one partition; and the held-out half is homologue-saturated, so "outside the fit" is largely within-cluster interpolation
That departure *cancels* closure error	not measured	the one direct check is under-powered and runs against it
Accuracy (§3.7)		
Physics arm vs. DirectGNN accuracy (ΔMAE)	labelled test rows (5608 of 8103)	+0.14 between separately tuned pipelines, so not attributable to the thermodynamic bottleneck

[a] The sets are defined in Table 1. [b] P is a two-way (solute × solvent) cluster-bootstrap frequency that the margin stays positive, reported as a band by the rounding rule at the head of this section. [c] The same 5608 rows: the seed-42 control arms carry the identical labelled set, so their cross-arm intersection with the arms of Table S3 is that set exactly. [d] One estimator cell (§2.5) for both sets, both kernels and every stratum. [e] Bootstrap band over pair / two-way / source clusters. [f] An aggregate margin is a composition-weighted average, and the deletions and cuts recorded here are failures of the headline cell rather than robustness passes. These rows record what the aggregate was and how it fails; the rows above replace it. [g] The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$, a non-positive one is failure to separate and licenses nothing. The margin's size is a lower bound on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate of it, so an interval on a margin bounds that bound and not the difference (Lemma 1, §2.1.1). B_{closure} is the error of the fixed map *read at a computed profile*, so what a positive margin establishes is a property of that composite and not of the kernel alone (§3.5.3). [h] Of fifty-nine strata, fifteen are boundable and six admissible; clause (c) of §2.5 makes those six three distinct row sets, of which one is also positive and not demoted by its residual test—the glycol ethers, the one row set the rule leaves standing.

own spread over draws, not a spread over seeds at all; and the pKa arm errors of §S8 span the 20 within-scaffold splits. Reading the table off the rounded means mis-states differences by up to 0.01: the learned-σ and DirectGNN arms differ by 0.1435 from their unrounded means, which we quote as +0.14, while their rounded values 1.85 and 1.70 differ by 0.15. Both closure families—NRTL and COSMO-SAC over a learned σ-profile—trail the DirectGNN control in the seed mean (the physics penalty), though the arms are separately tuned and that gap is not attributable to the bottleneck (§3.7). Only the COSMO-SAC family trails it at every seed: the per-seed column below puts the NRTL arm at 1.734 against the control's 1.749 at seed 42, so the two arms' per-seed values overlap and that ordering is a property of the seed-mean column and not of the arms. The σ-oracle (reference VT-2005 profile) is the worst arm. Restoring the Staverman–Guggenheim combinatorial term makes the *grounded learned-σ* arm slightly worse (1.8457 → 1.8788)—the signature of a misspecified map (§3.5); no oracle-plus-combinatorial arm was run, so the combinatorial contrast is against the grounded learned-σ row and not against the oracle row. To four decimals the means are 1.7022, 1.7950, 1.8457, 1.8788, 2.0434 and 2.2517, with population standard deviations 0.033, 0.071, 0.053, 0.091, 0.040 and 0.021. The NRTL row is where the two roundings part: averaging its printed per-seed triple gives 1.7953, whose two-decimal form is 1.80, but the arm's mean is 1.794975, so the entry is 1.79. The two senses of grounding are the second and the last of these: training-time supervision moves the arm from 2.0434 to 1.8457 (helps, −0.20), evaluation-time substi-

Table S3: Per-arm scaffold-split metrics (3 seeds, the n=5608 labelled test rows), with per-seed MAE to three decimals. Ordered by MAE. The ± is a population standard deviation over the three seeds. Mean, ± and per-seed column are each rounded once from the unrounded values, the column independently of the mean beside it.

Arm	MAE	R^2	per-seed MAE (42/43/44)
DirectGNN (no closure)	1.70 ± 0.03	$+0.42 \pm 0.03$	1.749/1.674/1.684
NRTL closure	1.79 ± 0.07	$+0.34 \pm 0.03$	1.734/1.758/1.894
COSMO-SAC, learned σ (grounded)	1.85 ± 0.05	$+0.33 \pm 0.05$	1.803/1.921/1.813
+ Staverman–Guggenheim combinatorial	1.88 ± 0.09	$+0.32 \pm 0.05$	1.805/2.007/1.824
COSMO-SAC, ungrounded σ	2.04 ± 0.04	$+0.19 \pm 0.01$	2.040/1.996/2.093
COSMO-SAC, reference σ (σ-oracle)	2.25 ± 0.02	-0.03 ± 0.04	2.232/2.281/2.242

tution from 1.8457 to 2.2517 (hurts, +0.41).

σ-head schedule of these arms. The four COSMO-SAC rows come from three training runs under one pinned configuration — 30 P1, 70 P2 and 10 P3 epochs with the σ-head freeze on, so that head's weights are held from P2 onward. The two grounded rows precede that with a σ warm-up on the grounding pool (up to 40 epochs, early-stopped on a held-out tenth of the pool) and interleave the pool through P1; under the freeze the stream is inactive in P2/P3. The ungrounded row sets warm-up and stream budget to zero, so its σ head is shaped in P1 by the corpus-internal $\ln \gamma_2^\infty$ auxiliary loss alone, never by an external σ label, and is frozen from P2 in the same way. The σ-oracle row is the grounded checkpoint re-evaluated with $z^\star$ injected, so it shares the grounded schedule exactly. Because the freeze holds the head and not the encoder that feeds it, the predicted profile is not pinned at its warm-up value in these arms. We have not run the surrogate isolation (§3.3) on these checkpoints, so how far their latent drifted is not measured. The supervision contrast quoted there ($+0.40 \to +0.14$) is a separate matched pair of runs — one warm-up-plus-short-P1 base resumed twice for SLE, with the freeze off and on, on n=5571 matched rows, so a single pair carrying no seed replicate and no interval — and not these arms.

Oracle-application controls: two of the three are single-seed. All three oracle applications (evaluation-only, training-time injection, channel-swap; §2) are computed on the same 5608 labelled test rows as Table S3, and their per-arm predictions are released with the code. Only the evaluation-only substitution is a three-seed result. Training-time injection (MAE 1.98) and the channel swap (MAE 2.08) were run at seed 42 alone, so their matched comparator is the seed-42 learned-σ value 1.803 on the same n=5608 lock, not the three-seed mean 1.8457 of Table S3. Both exceed the worst of the three learned-σ seeds (1.921), so the direction of both survives a worst-case seed comparison. The channel-swap margin (+0.27) is 2.3 times the learned arm's full between-seed range; the training-time margin (+0.18) is one and a half times it, so that control fixes a sign rather than an effect size. This matters for how the paradox's magnitude should be read. The evaluation-only substitution (+0.41 against the three-seed mean) mixes closure misspecification with the ordinary degradation any pipeline suffers when an input distribution is swapped at test time; the training-time injection, in which the model co-adapts to the reference profile, is the arm that removes that component and it costs +0.18. The co-adapted arm is therefore free of that confound and the evaluation-only +0.41 is not. It is not free of a second: neither this control nor the channel swap is certified to have trained on the same σ-stream build as the seed-42 comparator both are read against, and the file times exclude no history in either direction (§S9). Both are penalties of uncertified stream provenance. How large a share of the gap co-adaptation carries off can be read only on the seed the two arms share: at seed 42 the injection costs $1.981 - 1.803 = 0.177$ where the evaluation-only substitution costs $2.232 - 1.803 = 0.428$, a share of 0.59. We report that share as a description of seed 42 and not as a magnitude, because it is not resolved. Its numerator was measured once, so the injected arm's own between-seed spread is unmeasured, and a spread the size of the learned arm's 0.118 would move the share by about ± 0.27; holding the injected arm at 1.981 and reading it against the seed-43 and seed-44 learned baselines instead gives 0.83 and 0.61. What survives that latitude is the sign—at the matched seed part of the evaluation-only gap does go with co-adaptation, which is what makes the co-adapted arm the one free of the test-time swap—and the sign, not the share, is what the claim rests on. The ratio $(0.406 - 0.177)/0.406 = 0.56$ sets the one-seed injection against the three-seed mean gap; it is a share of nothing and is reported nowhere. Neither $+0.18$ nor $+0.27$ can be ordered against the -0.20 supervision gain either, whose own per-seed values are 0.237/0.075/0.281 and straddle $+0.18$.

The estimator grid. Every MSE, bound and margin on the activity axis is in squared $\ln \gamma_2^{\infty}$ units, and not on the $\ln x_2$ axis the per-arm MAEs above are measured on. Three analyst choices set the separation margin on the VT-2005-matched set, and Table S4 reports every combination of them rather than one setting of each. The margin is positive in 66 of the 72 cells. The six exceptions are the coarsest conditionings—three bins under either combinatorial convention and four bins residual-only—where the law-of-total-variance bound is valid but weakest. The grid is not a set of interchangeable settings, because two opposing biases run across it. Coarser bins condition on less, so the bound is weaker; finer bins condition on more but leave few points per bin, and the within-bin variance is then downward-biased in finite samples, which inflates the margin favourably. Neither end is unbiased. We headline the eight-bin cell under the *unbiased* within-bin variance on an occupancy rule fixed in advance of the grid—the coarsest binning that still leaves fewer than eight rows per bin, here 7.5 at n=60—and we carry the same *bin count* to the broad IDAC set, so the two are conditioned at the same granularity. On 477 rows that is not the same occupancy rule (which would call for about 60 bins) and is deliberately conservative. The broad set's margin grows with bin count over the grid as a whole, though not cell by cell: in the deployed convention under the unbiased variance it falls from $+0.10$ at three bins to $+0.01$ at four, from $+0.57$ at seven to $+0.51$ at eight, and from $+0.83$ at twenty to $+0.82$ at twenty-four. The endpoints are what the choice turns on: $+0.51$ at eight bins against $+0.96$ at forty-eight. Eight bins therefore gives a looser bound and a smaller margin there than the occupancy rule would. It is *not* the least favourable cell of the middle range: seven bins residual-only under the unbiased variance gives $+0.01$ against the eight-bin $+0.05$, and twenty bins gives $+0.08$. Nothing turns on the choice—the whole middle range of the deployed convention lies between $+0.01$ and $+0.47$, all positive—but the reason for it is occupancy and not conservatism, and taking the least favourable middle cell instead gives $+0.01$ and the same verdict of an unresolved margin. The per-cell P holds that cell's bin count fixed across resamples. The main text instead quotes the deposited adaptive bin count $\max(3, \lfloor n/10 \rfloor)$, here in the rows n of the resample, which conditions on less on small resamples and is therefore more conservative: it gives ≈ 0.78 (maximum-likelihood) and ≈ 0.72 (unbiased) residual-only, against ≈ 0.89 and ≈ 0.86 full. The two schemes measure different things and we report both rather than the higher.

A fourth analyst choice: what "equal-count" means when n is not divisible by the bin count. The rule this paper deploys, and which §2.5 names as the fourth analyst choice without restating, is linearly interpolated quantile edges of g at $j/8$, $j = 1, \ldots, 7$, with a value falling exactly on an edge assigned to the upper bin. Neither n divides by eight ($60/8 = 7.5$, $477/8 = 59.625$), and that alone is what makes standard implementations of the same phrase disagree; there are no exact ties in g on either set under either convention, so each of them is a genuine coarsening of $\sigma(g)$ and the bound is valid under all of them. On the VT-2005-matched set at eight bins, the deployed convention and the unbiased variance, $B^{\mathrm{up}}_{\mathrm{insuff}}$ is 0.713 under our rule, under the same edges with ties sent to the lower bin, and under the standard quantile-cut implementation alike; 0.624 under an equal-length split of the g-sorted order (margin +0.225); 0.669 under $\lfloor k\,\mathrm{rank}/n \rfloor$ (+0.134); and 0.736 under order-statistic quantile edges (+0.001). Exact enumeration by dynamic programming over *every* partition of the g-ordered rows into eight contiguous bins of sizes in $\{7, 8\}$ puts the VT-2005-matched bound in $[0.620, 0.742]$ and its margin in $[-0.012, +0.232]$; at seven bins the margin envelope is $[-0.134, +0.157]$, and one such rule realises -0.089 there. The sign of the VT-2005-matched margin is therefore set by the binning rule and not by the data, which is a further reason to read its +0.05 as consistent with zero. The broad IDAC set absorbs the same choice: the six strictly equal-count implementations give 0.694 to 0.700, a spread of 0.006, and the exact envelope over all sizes-$\{59, 60\}$ layouts is $[0.687, 0.701]$, margin $[+0.499, +0.528]$ (at seven bins, $[+0.564, +0.570]$). Our 0.697 sits mid-range in both enumerations. Nothing load-bearing moves, since the VT-2005-matched set's margin is unresolved either way and the claim is carried on the broad set; we name the rule because the phrase "equal-count" does not by itself fix it.

The same grid on the broad IDAC set is Table S5, now with its own per-cell bootstrap frequency rather than a bare range of point margins. None of its 56 cells is negative. Within the *deployed* residual-only convention the margin runs from +0.01 (four bins) and +0.10 (three bins) up to +0.96, with the eight-bin unbiased headline at +0.51; the frequencies at the two coarsest cells, 0.65 and 0.42, say that those conditionings do not resolve the ordering, and they are why the broad set's aggregate margin is not resolved at every conditioning. Under the full convention the same grid runs from +0.23 to +1.25 and reaches $P > 0.95$ from five bins upward.

Table S4: Separation margin MSE $- 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the VT-2005-matched set (n=60), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] Negative margins in bold.

bins	full/ML	full/Be	res/ML	res/Be
3	**−0.06** (.64)	**−0.16** (.59)	**−0.43** (.43)	**−0.53** (.38)
4	+0.18 (.82)	+0.07 (.78)	**−0.14** (.63)	**−0.26** (.57)
5	+0.50 (.90)	+0.38 (.87)	+0.21 (.77)	+0.10 (.70)
6	+0.66 (.94)	+0.54 (.92)	+0.35 (.84)	+0.22 (.77)
7	+0.76 (.96)	+0.62 (.94)	+0.18 (.89)	+0.01 (.83)
8	+0.74 (.97)	+0.58 (.96)	+0.24 (.91)	+0.05 (.86)
9	+0.90 (.98)	+0.75 (.97)	+0.42 (.93)	+0.24 (.88)
10	+1.12 (.98)	+0.99 (.97)	+0.42 (.94)	+0.21 (.89)
11	+1.11 (.99)	+0.95 (.98)	+0.38 (.96)	+0.14 (.91)
12	+0.96 (.99)	+0.75 (.98)	+0.49 (.96)	+0.24 (.93)
13	+1.01 (.99)	+0.80 (.98)	+0.59 (.97)	+0.34 (.94)
14	+1.05 (.99)	+0.81 (.98)	+0.50 (.97)	+0.21 (.94)
15	+1.00 (.99)	+0.73 (.99)	+0.52 (.98)	+0.20 (.95)
16	+1.03 (1.0)	+0.77 (.99)	+0.56 (.98)	+0.22 (.96)
17	+1.25 (1.0)	+1.05 (.99)	+0.56 (.99)	+0.20 (.96)
18	+1.08 (1.0)	+0.76 (.99)	+0.69 (.99)	+0.39 (.97)
19	+1.11 (1.0)	+0.77 (1.0)	+0.79 (.99)	+0.47 (.97)
20	+1.23 (1.0)	+0.94 (.99)	+0.55 (.99)	+0.08 (.97)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive.

The same grid on the stratum the paper states. The two grids above sweep the bin count for the sets taken *whole*, and they are what establishes that the bin count is the analyst choice with the largest leverage here—the broad set's deployed margin runs from +0.01 to +0.96 across Table S5. Every one of the 59 strata of §S3.3 is then computed at the single fixed eight-bin cell, so the one row set the rule leaves standing was reported at one cell of a grid that moves. Table S6 runs the grid of Table S5 on that row set. At n=182 the eight-bin cell holds 22.75 rows per bin, where the occupancy rule that fixed the cell on the n=60 set would ask for 23 bins: the fixed cell is *coarser* than the rule that chose it and therefore conditions on less, which weakens the bound and lowers the margin.

The glycol-ether margin is positive in all 56 cells of that grid at $P \geq 0.99$ throughout, the deployed unbiased column running from +1.06 at three bins to +2.19 at forty-eight, with the fixed cell's +2.04 sitting below the midpoint

Table S5: Separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the broad IDAC set (n=477), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] No cell is negative; the headline is the 8/res/Be cell.

bins	full/ML	full/Be	res/ML	res/Be
3	+0.24 (.60)	+0.23 (.59)	+0.11 (.42)	+0.10 (.42)
4	+0.50 (.93)	+0.49 (.93)	+0.02 (.67)	+0.01 (.65)
5	+0.84 (.98)	+0.82 (.98)	+0.50 (.77)	+0.48 (.76)
6	+0.84 (.99)	+0.82 (.99)	+0.51 (.86)	+0.50 (.85)
7	+0.94 (.99)	+0.93 (.99)	+0.59 (.89)	+0.57 (.88)
8	+0.96 (1.0)	+0.95 (1.0)	+0.53 (.93)	+0.51 (.92)
10	+1.06 (1.0)	+1.04 (1.0)	+0.67 (.96)	+0.64 (.94)
12	+1.05 (1.0)	+1.03 (1.0)	+0.74 (.97)	+0.71 (.97)
16	+1.11 (1.0)	+1.09 (1.0)	+0.76 (.99)	+0.72 (.99)
20	+1.17 (1.0)	+1.14 (1.0)	+0.88 (1.0)	+0.83 (.99)
24	+1.19 (1.0)	+1.15 (1.0)	+0.87 (1.0)	+0.82 (1.0)
32	+1.21 (1.0)	+1.16 (1.0)	+0.93 (1.0)	+0.86 (1.0)
40	+1.30 (1.0)	+1.25 (1.0)	+1.03 (1.0)	+0.95 (1.0)
48	+1.29 (1.0)	+1.22 (1.0)	+1.06 (1.0)	+0.96 (1.0)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. Regenerates from the deposited convention audit.

of that column and the occupancy rule's own 23 bins giving +2.17 deployed and +2.36 full. The pair unit, n=43, supports ten of the fourteen bin counts and is positive in all 40 of the cells it defines, $[+1.41, +2.83]$ deployed and $[+1.69, +3.10]$ full. Across the 36 cells of the sweep where the leave-one-publication-out test can be run at all, the sign survives every deletion in all 36, the worst leaving +1.08. Neither the sign nor the admissibility of that stratum is a property of the bin count. The other two admissible row sets behave differently and are reported here rather than left out: the acceptor-only aprotics are non-positive in every cell of the same grid, $[-0.55, -0.03]$ over the 44 its n=57 defines, which licenses nothing by one-sidedness; and the alkane solutes are positive in all 56 row-unit cells but lose the sign in two of the twenty pair-unit deployed cells and fail the deletion test at eight of the forty cells where it runs, so that set is admissible at the fixed cell and not across the sweep—one more reason the finding is stated on the glycol ethers and not on it.

Table S6: Separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the *glycol-ether stratum* of the broad IDAC set (n=182, row unit), over the same bin-count × variance × convention grid the aggregate is swept on.[a] No cell is negative; the fixed cell of the map is 8/res/Be.

bins	full/ML	full/Be	res/ML	res/Be
3	+1.29 (1.0)	+1.27 (1.0)	+1.08 (.99)	+1.06 (.99)
4	+1.92 (1.0)	+1.91 (1.0)	+1.72 (1.0)	+1.71 (1.0)
5	+1.96 (1.0)	+1.95 (1.0)	+1.77 (1.0)	+1.76 (1.0)
6	+2.02 (1.0)	+2.01 (1.0)	+1.82 (1.0)	+1.80 (1.0)
7	+2.12 (1.0)	+2.11 (1.0)	+1.93 (1.0)	+1.91 (1.0)
8	+2.24 (1.0)	+2.23 (1.0)	+2.05 (1.0)	+2.04 (1.0)
10	+2.24 (1.0)	+2.22 (1.0)	+2.05 (1.0)	+2.04 (1.0)
12	+2.34 (1.0)	+2.33 (1.0)	+2.13 (1.0)	+2.13 (1.0)
16	+2.36 (1.0)	+2.35 (1.0)	+2.16 (1.0)	+2.15 (1.0)
20	+2.37 (1.0)	+2.36 (1.0)	+2.18 (1.0)	+2.17 (1.0)
24	+2.39 (1.0)	+2.38 (1.0)	+2.18 (1.0)	+2.17 (1.0)
32	+2.39 (1.0)	+2.37 (1.0)	+2.19 (1.0)	+2.18 (1.0)
40	+2.40 (1.0)	+2.38 (1.0)	+2.20 (1.0)	+2.19 (1.0)
48	+2.41 (1.0)	+2.39 (1.0)	+2.20 (1.0)	+2.19 (1.0)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. The pair unit, the other five admissible cells, and the leave-one-publication-out curve at every bin count are in the deposited stratum bin sweep.

The decomposition on both sets, and the three kernel arms. Table S7 is the closure–insufficiency decomposition of §3.5.1–§3.5.2 on both sets under both conventions, with the maximum-likelihood variance variant, the constant-offset (Jensen) row and the bootstrap bands. Table S8 is the three COSMO-SAC arms of §3.5.3 on both sets under both conventions, with the bootstrap frequencies, the 90% margins and the block split of the 2002 → 2010 contrast into the 57 pairs shared with the VT-2005-matched set, the complementary 420 and the 80 of those at 298 K. Every claim either table carries has a row in Table S2 with its set, value and its uncertainty.

The bounds inventory, named and valued. Table S9 lists every bound either set supports at the aggregate level, together with the corroborating estimators Table S7 points here for instead of listing them row by row. The inclusion rule is note *a* of the table, and it is a rule for that table and not a claim of

Table S7: Closure–insufficiency decomposition on the reference-input activity path, as bounds rather than a point split. **The headline columns are the deployed residual-only ones on *both* sets**, one rule applied throughout; the full-convention columns are the comparison.

	broad IDAC (477)[a]		VT-2005-matched (60)[a]	
Quantity	**res**[b]	full[b]	**res**[b]	full[b]
$\mathrm{MSE}_{\mathrm{total}} = B_{\mathrm{closure}} + B_{\mathrm{insuff}}$ (exact)	1.90	1.91	1.47	1.80
B_{insuff} upper bound (LOTV)[c]	0.70	0.48	0.71	0.61
B_{closure} lower bound[d]	1.21	1.43	0.76	1.19
B_{closure} lower, constant-offset (Jensen)[e]	0.18	0.71	0.43	0.72
separation margin[f]	**+0.51**	+0.95	**+0.05**	+0.58
… bootstrap band[g]	>0.95/≈0.9/≈0.9	>0.95 (all three)	≈0.7	≈0.9
… maximum-likelihood variance[h]	0.68/+0.53	0.47/+0.96	0.62/+0.24	0.53/+0.74
… tighter of the two conventions[i]	0.48/+0.94		0.61/+0.26	
Further B_{insuff} upper bounds[j]	Table S9			
Label noise σ_η^2 (IDAC)	already inside every row above[k]			

[a] Broad IDAC, n=477 on UD profiles; VT-2005-matched, n=60 on VT-2005 profiles throughout (§3.5.3 scores the same pairs on UD). [b] *full*, with the Staverman–Guggenheim combinatorial term; *res*, residual-only. [c] Law of total variance, at eight equal-count bins of $g(z^\star, T)$ under the unbiased (Bessel-corrected) within-bin variance. It fits no model to the data. [d] $\mathrm{MSE} - B_{\mathrm{insuff}}^{\mathrm{up}}$; this row decides the ordering. [e] $(\mathbb{E}[m] - \mathbb{E}[g])^2$. [f] $\mathrm{MSE} - 2B_{\mathrm{insuff}}^{\mathrm{up}}$, the quantity that orders the two terms. [g] Pair / two-way / source clusters. The VT-2005-matched set supports the two-way clustering only, so it carries one entry per column. [h] Same rows and bins, maximum-likelihood within-bin variance, as B_{insuff} upper / margin. [i] Charged against the deployed MSE, as B_{insuff} upper / margin. Not headlined. [j] The entries of Table S9: out-of-fold RF and ridge (random and blocked folds), kNN-in-$z^\star$, and the binning-free $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$. Not a paper-wide inventory: the bin grids behind the LOTV selection are Tables S4 and S5, and the $\ln\gamma_2^\infty$-stratified bounds, computed on VT-2005-matched subsets, are Table S16. [k] A *component* of B_{insuff}, not a term beside it (§2.1.1).

completeness over the paper: the LOTV half is a selection from two bin grids far too large to reproduce, and the $\ln\gamma_2^\infty$-stratified bounds of Table S16 are computed on subsets of the VT-2005-matched set rather than on either whole set and are not in it. The deposited figure script regenerates it, printing the inventory as it goes, so the table can be re-checked against the artifacts. The regeneration's guard against an artifact changing under the inventory is *not* the same on the two sets, and we state which is which rather than claim one rule. On the VT-2005-matched set it stops both ways: on an estimator the artifact records and the inventory does not name, and on a named one the artifact no longer records. It additionally checks the entries that set's two audit artifacts share against each other. On the broad set it stops only the first way, so a named estimator absent from that set's audit artifact would be dropped rather than raised. $B_{\mathrm{insuff}}^{\mathrm{up}}$ and the B_{closure} lower bound it implies are contiguous shares of one total, so what separates the terms is the margin $\mathrm{MSE} - 2B_{\mathrm{insuff}}^{\mathrm{up}}$ and not a gap between the two. Two rows of the table carry the whole risk on each set. On the broad set the coarsest and the least favourable cell differ—three bins gives 0.901, four bins 0.948 against a threshold of 0.951, a margin of +0.007 at bootstrap frequency 0.65 and the tightest anywhere on that set. On the VT-2005-matched set the two coincide at three bins (1.000), and both it and the four-bin cell (0.866) sit *above* that set's threshold of 0.736: the negative-margin cells Table S4 sets in bold, and both are listed here.

The stratification rule, in full. §3.5.1 reports the decomposition per stratum rather than per set, and the rule is set out here in full because it is the object that has to be checkable against the charge of having been fitted to its own outcome.

Axis 1, solvent class. An ordered cascade over functional groups; the first class a solvent matches is the class it takes, and the order is hydrogen-bonding role first—the axis COSMO-

Table S8: Activity error (MSE, squared $\ln \gamma_2^\infty$ units) of the three COSMO-SAC arms, and the block split that reads the 2002 → 2010 reversal against profile database and temperature held fixed.

(a) Arm[a]	VT-2005-matched (60), UD[b] full[b]	VT-2005-matched (60), UD[b] res[b]	broad IDAC (477), UD[b] full[b]	broad IDAC (477), UD[b] res[b]
fair 2002	1.757	1.436	1.911	1.902
2010, dispersion off	0.765	0.831	1.978	2.893
2010, dispersion on	0.785[e]	—[d]	1.855	—
P (2010 better)[c], dispersion off	>0.95	—	≈0.6	≈0.2
90% margin[c]	[+0.06, +2.18]	—	[−1.95, +0.93]	[−3.06, +0.70]
P (2010 better)[c], dispersion on	—	—	≈0.6	—
90% margin[c]	—	—	[−2.24, +1.31]	—

(b) Block (UD profiles, full convention)	2002 → 2010, dispersion off	P^c	90% margin[c]
shared with the VT-2005-matched set (57)	1.783 → 0.767	>0.95	[+0.06, +2.36]
complementary (420)	1.928 → 2.143	≈0.5	[−2.41, +0.87]
… of these, at 298 K (80)	0.796 → 1.284	≈0.3	—

[a] The three arms, named here and in the text: *fair 2002*, COSMO-SAC-2002 in our NIST-validated layer reading all three profile channels; *2010, dispersion off* and *2010, dispersion on*, COSMO-SAC-2010 without and with the dispersion term (§3.5.3). [b] *full* / *res* as in Table S7. *Both* sets are scored on UD profiles here, the VT-2005-matched set included—Table S7 scores the VT-2005-matched set on VT-2005 instead, and that database difference is the whole difference between its VT-2005/full 1.80 and the 1.757 here (§3.5.3). Within a set, each arm uses the σ-profile averaging its own model prescribes (Mullins for 2002, Hsieh for 2010). [c] P is the two-way cluster-bootstrap frequency that the 2010 arm is the better one, with the 90% margin on the paired squared-error difference beneath it. [d] A dash marks a cell the deposited artifacts do not carry. [e] This one cell is n=57, not 60: `cCOSMO` tabulates no dispersion ϵ for the three bromine- or iodine-containing VT-2005-matched pairs. On those common 57 the *dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n, and no claim rests on that ordering (§3.5.3).

SAC's H-bond term acts on—then dominant functional group. *(1) water* (111 rows / 15 pairs), the only liquid whose entire surface is H-bond active, with σ-profile mass at both extremes and the smallest cavity of any of them, so the Staverman–Guggenheim term is at its most extreme and the two combinatorial conventions disagree most. *(2) glycol ether*, at least one O–H and at least two oxygens: the ethylene-glycol oligomers (182 / 43), donor and acceptor on one flexible chain, so intramolecular H-bonds form and the profile is conformer-dependent—the documented worst case for a model that scores one gas-phase conformer as a bag of independent surface segments with no memory of connectivity. *(3) mono-alcohol*, exactly one O–H, exactly one O, no N, no halogen (35 / 17), the single-donor self-associating chemistry the 2002 H-bond term was fitted on. *(4) N–H protic* (33 / 33): formamide, N-methylformamide, primary and secondary amines, piperidine—nitrogen donors, weaker and differently shaped than O–H and lumped with them under one shared σ_{hb} threshold. *(5) aryl ether* (13 / 2), anisole and phenetole, tested *before* the general acceptor class so as not to be swallowed by it: lone pairs delocalised into the ring put the acceptor strength far below a dialkyl ether's, where a single cut-off is at its crudest. *(6) aprotic acceptor* (57 / 57): carbonyl, nitrile, azine nitrogen, ether oxygen, nitro or tertiary-amine nitrogen, with no donor—pentyl acetate, pyridine and four methylpyridines, acetone, acetonitrile. *(7) halogenated* (30 / 7), any C–F/Cl/Br/I and no donor: large soft polarisable surfaces against a kernel with no dispersion term and one effective contact area. *(8) aromatic hydrocarbon* (14 / 9), a quadrupolar π face with σ near zero over the ring, and *(9) aliphatic hydrocarbon* (2 / 2), pure dispersion, $\sigma \approx 0$ everywhere, so the residual term nearly vanishes and the prediction is almost entirely combinatorial. The nine are exhaustive and disjoint over the set's 36 solvents.

Table S9: **Upper bounds on the input-insufficiency term** B_{insuff}, on both sets taken whole—every bound the inclusion rule keeps, plus one estimator that set cannot support, with the value, what each fits and what qualifies it. What is included, and what is not: note *a*. The per-stratum bounds that replace the aggregate margin are the map tables below.

Estimator[a]	$B^{\text{up}}_{\text{insuff}}$	fits[b]	note[c]
Broad IDAC set (n=477, UD profiles). MSE = 1.902; threshold MSE/2 = 0.951; headline margin +0.51.			
LOTV, 8 bins, unbiased variance [**headline**]	0.697	no model (∘)	
LOTV, 8 bins, maximum-likelihood variance	0.685	no model (∘)	
LOTV, 3 bins (coarsest)	0.901	no model (∘)	
LOTV, 4 bins (least favourable)	0.948	no model (∘)	tightest margin on this set
LOTV, 8 bins, full convention (also valid)[d]	0.482	no model (∘)	
RF out-of-fold, folds blocked on the molecule pair	0.882	out-of-fold (⋄)	
ridge out-of-fold, folds blocked on the molecule pair	1.467	out-of-fold (⋄)	above thr.
RF out-of-fold, folds blocked on the pair, conditioning on $(z^\star, T)$	0.823	out-of-fold (⋄)	
RF out-of-fold, folds blocked on the solute	2.043	out-of-fold (⋄)	vacuous
RF out-of-fold, random folds	0.146	out-of-fold (⋄)	leaky
ridge out-of-fold, random folds	0.801	out-of-fold (⋄)	leaky
$\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$, binning-free, on the 359 rows that support it	0.046	no model (∘)	single-source
VT-2005-matched (n=60, VT-2005 profiles). MSE = 1.472; threshold MSE/2 = 0.736; headline margin +0.05.			
LOTV, 8 bins, unbiased variance [**headline**]	0.713	no model (∘)	
LOTV, 8 bins, maximum-likelihood variance	0.618	no model (∘)	
LOTV, 3 bins (coarsest, least favourable)	1.000	no model (∘)	above thr.
LOTV, 4 bins	0.866	no model (∘)	above thr.
LOTV, 8 bins, full convention (also valid)[d]	0.608	no model (∘)	
kNN-in-$z^\star$ (convention-independent; least conservative)	0.400	out-of-fold (⋄)	deflated
kNN, neighbour from a distinct solute	0.448	out-of-fold (⋄)	
kNN, neighbour from a distinct solvent	1.458	out-of-fold (⋄)	above thr.
kNN, neighbour from a distinct pair	1.831	out-of-fold (⋄)	vacuous
RF out-of-fold, random folds	0.621	out-of-fold (⋄)	deflated
ridge out-of-fold, random folds	0.625	out-of-fold (⋄)	
RF out-of-fold, folds blocked on both molecules	3.653	out-of-fold (⋄)	vacuous
ridge out-of-fold, folds blocked on both molecules	divergent	out-of-fold (⋄)	
ridge out-of-fold, deliberately aggressive	0.877	out-of-fold (⋄)	above thr.
$\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$[e]	—	—	no repeated pairs

[a] One inclusion rule governs both sets, in two halves that differ in kind. The LOTV half is a *selection* from bin grids too large to draw (56 cells on the broad set, 72 on the VT-2005-matched set; Tables S5 and S4): *within the deployed convention and under the unbiased within-bin variance*, the headline cell, the *coarsest*, the *least favourable* and every such cell above the threshold, with the headline cell's maximum-likelihood variant and its full-convention counterpart listed beside it. Cells above the threshold under the other variance or the other convention are not listed; they are in those two grids, where Table S4 sets the negative-margin ones in bold. The estimator half is every out-of-fold estimator that set's audit artifact records, in every fold design it records, vacuous designs included—the per-set guards on that are stated with the regeneration command above. [b] What the estimator fits to the data. *no model* (∘) for the law-of-total-variance (LOTV) cells and for $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$, *out-of-fold* (⋄) for a fitted estimator. The LOTV bound, fitting nothing, is unaffected by any fold design. [c] *above thr.*, the bound exceeds that set's separation threshold MSE/2, so that cell does not separate; *vacuous*, it exceeds the total MSE, which two of the VT-2005-matched set's estimators and one of the broad set's do; *leaky*, random folds leave a held-out row's own molecule pair in the training fold at a neighbouring temperature; *deflated*, biased low by the crossed design, in which two pairs sharing a molecule agree on 51 of the 102 $z^\star$ coordinates; *single-source*, every multi-row pair supporting the bound comes from one publication, so it is valid on those rows but samples no between-laboratory component and is not promoted. [d] Computed under the other combinatorial rule and reported against the deployed-convention threshold beside it. It is valid there, and it is the only entry on either set that is not residual-only. [e] The estimator that set cannot support.

Ether, nitro and tertiary-amine tests sit in the cascade because the same cascade is applied to the *solutes*, where those groups occur; no solvent reaches them.

Two further granularities, so that the granularity is not itself a choice. Coarse: donor-and-acceptor oxygen (classes 1–3, n=328), donor nitrogen (4, 33), acceptor-only (5–6, 70), halogenated non-H-bonding (7, 30), hydrocarbon non-H-bonding (8–9, 16). Fine: amide N–H 25 against amine N–H 8; ester 30, azine 23, ketone or aldehyde 2, nitrile 2; halogenated aliphatic 18 against halogenated aromatic 12. All three levels are in the deposited table, and two of their cells are worth naming. The coarse *acceptor-only* class, which merges the aryl ethers into the aprotic acceptors, carries a *positive* margin (+1.43 deployed at $P \approx 0.7$) where its dominant child is negative—the composition effect of §3.5.1 reappearing one level up, on 13 rows. At the fine level the azine sub-family changes sign between conventions at n=23. Neither overturns a headline-level verdict, and both are why the granularity is reported rather than chosen.

Axis 2, solute role. Water as the dilute solute (37 / 6) against organic solutes (440 / 179), reported alone and crossed with axis 1. Water as the solute is a different physical situation from water as the solvent: an isolated H_2O in an organic continuum has all four H-bond sites unsatisfied by like molecules, a state the segment-activity expression is not fitted near and the one COSMO-SAC-2002's documented weakness is specific to. The same cascade applied to the solute gives a secondary solute-family axis, also deposited, because a practitioner deciding where to ground needs both halves of a pair.

Two mechanical points. The solvent classifier takes one argument, a SMILES string, and the whole cascade is a sequence of SMARTS tests: it never reads m, g, the squared error, MSE, B_{insuff}, the margin, the source DOI or the bootstrap. That is a property of the code and not an assurance. Second, the University of Delaware compound list stores formamide and N-methylformamide as imidic-acid tautomers (`N=CO`, `CN=CO`); RDKit's canonical tautomer restores the amide so that the N–H donor defining their class is visible, and that normalisation is applied to every molecule alike rather than carved for two of them.

What sets the level of the bound, and why it barely moves between strata. From above, $B^{\mathrm{up}}_{\mathrm{insuff}}$ is capped by the within-stratum label variance: the law of total variance on a single bin *is* $\mathrm{Var}(m)$, and refining the bins can only lower it, so under the maximum-likelihood within-bin variance $B^{\mathrm{up}}_{\mathrm{insuff}} \leq \mathrm{Var}(m)$ identically—which the deposited maximum-likelihood column satisfies in every stratum (0.12 against 0.80 on the aprotic acceptors, 0.10 against 3.07 on the glycol ethers, 0.20 against 1.42 on water, 0.22 against 5.61 on the N–H protics, 0.44 against 3.66 on the halogenated, 0.58 against 0.60 on the mono-alcohols). The Bessel-corrected estimate the manuscript headlines can exceed $\mathrm{Var}(m)$ once bins hold only a few rows, and does so exactly once: the mono-alcohols, 0.733 against 0.604 at n=35 and four rows a bin, where the artifact raises its own exceedance flag. That flag fires only on a stratum already declared not boundable, which is the boundability criterion doing its job rather than an unexplained anomaly—and that same value is what sets the $\times 6.8$ span quoted in §3.5.1.

From below the received story needs a correction. $z^\star$ is a deterministic function of the pair, so $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair}, T)]$ *is* B_{insuff} on rows that repeat; but no two of the 477 share a pair *and* a temperature, so pure replicate disagreement is not measurable on this set. The coarser $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$, which additionally absorbs the whole within-pair temperature spread, is 0.0003 on the mono-alcohols, 0.037 on water, 0.044 on the glycol ethers, 0.075 on the halogenated and 0.046 over the 359 rows that support it on the whole set—one to two orders of magnitude below the binning bound of 0.11–0.73. The level of $B^{\mathrm{up}}_{\mathrm{insuff}}$ here is therefore not set by label noise but by the residual spread of m inside a quantile bin of a scalar, which is a property of the estimator rather than of the chemistry. That is the sharper form of the structural claim, and the reason the bound is so nearly stratum-independent while MSE is free to move by a factor of 25 to 51.

The structural claim as a test. 2000 random partitions of the 477 rows into blocks of exactly the observed class sizes, at the same estimator cell and under the same boundability filter. Deployed convention: observed MSE range 2.163 against a null median of 0.405, no draw reaching it; observed $B^{\mathrm{up}}_{\mathrm{insuff}}$ range 0.107 against 0.165, $p = 0.26$; observed ratio 20.2 against 2.46, $p = 0.02$. Full convention: 2.353 against 0.337, again no draw reaching it; 0.058 against 0.133, $p = 0.12$; ratio 40.5 against 2.48, $p = 0.003$. The chemical partition separates the error far beyond chance and separates the bound no more than chance, which is what makes an aggregate margin a composition statistic.

A multiplicity null for the search itself. That test asks whether the chemical partition separates error; it does not ask how often a 59-fold search certifies a stratum by chance, and the admissibility rule is a stability filter rather than a multiplicity correction. The missing null is this one. Each of the 36 solvents keeps its rows and receives the label *triple* (nine-class, coarse, fine family) of a uniformly drawn other solvent, and each of the 78 solutes another solute's (family, role) pair, the triples moving together so that the nesting between granularities is preserved exactly; the stratum family is rebuilt from the permuted labels by the same code and put through the same rule—boundability, leave-one-publication-out in all four unit × convention cells, positivity in all four, and clause (c) of §2.5, the reduction to distinct row sets together with its asymmetric residual test. Preserved: the number of strata, the number of molecules per stratum, the containment lattice, and every row's m, g, source and pair. Destroyed: the association between chemistry and error. Every frequency here is that null on all 477 rows under the coarsening, the row set and the estimator the map itself deploys; §S3.4 re-runs it on 473 and its figures differ for that reason alone. Over 2000 draws the chemical map's six admissible strata are above chance ($p = 0.01$) and its two admissible positive row sets are not ($p = 0.286$): a chemically blind relabelling certifies at least one row set in 58% of draws and two or more in 28.6%. What the search returns is therefore not rare; its *size* is closer to being so. The largest certified margin reaches the observed +2.04 or more in 10.1% of all draws (17% of the draws that certify anything), against a median of +1.57 among those, a 95th percentile of +2.39 and a maximum of +3.14. The consequences for what may be claimed are in §3.5.1 and limitation (vi); in one line, the finding sits at about the ninetieth percentile of what this search returns under a chemistry-blind null, which is not multiplicity control at a conventional level.

Two features of the null are worth stating because they are not what one would guess, and both are measured rather than argued. Permuting the chemistry labels does not leave the *provenance* structure alone—publications measure related sets of solvents—so we recorded what it does: a label-permuted map has *more* boundable cells than the chemical one (62 against 50, median over draws) and each passes the deletion test *less* often (0.56 against 0.70), the two effects running in opposite directions, so the reported frequencies are not signed and are neither an upper nor a lower bound on a chance rate. And because the set taken whole has a positive margin at this cell, a randomly assembled stratum inherits a positive margin on average; a null frequency near one half for "certifies something" is expected and is not evidence that the rule is loose. The exactly matched null would permute chemistry and provenance jointly, which this design cannot do because the two are confounded in these data. The draw-by-draw record is deposited.

The out-of-sample test, declared before it was run. A selection objection is settled by rows the search never saw. The only such rows this work holds are the PGL 6th-edition IDAC database scored by the deployed closure on VT-2005 profiles (Table 2): 7889 records over 2279 pairs, another group's compilation, sixteen times the broad set's rows on a different profile database. Two other candidates were considered and rejected as not out of sample: the 14900-row pool the broad set was drawn from contains 184 neutral glycol-ether-solvent rows of which 182 are already in the broad set, its remaining glycol-ether-classified rows being hydroxy-functionalised ionic liquids with no reference profile and no scoreable closure; and the VT-2005-matched set is inside the search. Before any margin was computed on the PGL rows we fixed the row set (the same solvent-classifier cascade, unmodified, minus any pair shared with the broad set), the estimator cell (the headline one: row unit, deployed convention, eight equal-count bins, Bessel within-bin variance), the admissibility rule (the manuscript's, with the source axis declared to be the record file refined by the compilation label, which is the finest provenance these files carry and is coarser than the broad set's DOI), and the three verdicts available— confirmed, not confirmed, and not testable, the last counting as not confirmed for what the title may say. The verdict was **not testable, hence not confirmed**. The cell the finding is about does not exist there: no glycol-ether solvent carries an organic solute in that database, its non-aqueous half listing 88 solvent molecules and no glycol ether. That emptiness needed no margin to see. The geometry of the candidate row sets—how many rows exist, which solvents and solutes they carry, which files they come from—was established before the declaration was written, because a test cannot be designed without knowing whether its data exist, and the declaration records the zero-row count for the exact analogue cell along with the two mismatches of the nearest one. What the pre-registration delivered on this cell is therefore the unavailability of confirming data and not a replication that ran and failed; what it protected is the rule and the row set that would have decided the verdict had rows existed, fixed and staged before anything was scored. The nearest cell is 52 rows of water in seven glycol-ether and polyol solvents over seven pairs, drawn 50–2 from the two compilations of one record file, so the one deletion that matters leaves two rows, the fixed cell undefined and the sign untested—the same reason the rule refuses water-as-solute in sample. Its margin, which licenses nothing under the rule declared in advance, is $+0.35$ $[+0.28, +0.58]$, against $+2.04$ in sample. Of the four solvent molecules in the in-sample cell only ethylene glycol appears; the three oligomers carrying 141 of its 182 rows do not. Descriptively, and declared in advance as description rather than as a test, the same cell over that database's ten solvent classes is Table S10, printed whole because a ranking quoted in part is a different statement: it runs from -2.53 (water, 3074 rows, the largest stratum there) to $+1.73$ (aliphatic hydrocarbons, 875 rows). The same admissibility rule admits two of the ten—the monoalcohols, whose $+0.09$ carries an interval covering zero, and water at -2.53—so no out-of-sample class both passes the rule and separates the two terms, and the classes with the larger positive margins are ones the rule refuses. The glycol ethers are not where this closure fails hardest out of sample, and the instrument behaves out of sample as §3.5.1 says it behaves in sample—a map, not a verdict. The deposited artifact carries the pre-declaration in full.

The leave-one-out curves. At the headline cell under the deployed convention, baseline $+0.509$: leaving out `10.1021/acs.jced.7b00114` gives $+0.184$ and leaving out `10.1021/je020196d` (105 rows) $+0.419$, while the remaining thirteen deletions run from $+0.503$ to $+0.875$. Under the full convention the same curve is flat, $[+0.821, +1.073]$ against a baseline of $+0.947$, so the single-publication dependence is specific to the deployed convention. All 185 leave-one-pair-out margins lie in $[+0.422, +0.721]$ with median $+0.507$ and none non-positive, the seven most damaging deletions being *five* of the six water-as-solute pairs interleaved with two glycol-ether ones; the sixth water pair ranks 178th of 185, so those six rows do not move the aggregate in one direction. Leave-one-solvent-class-out spans $[+0.352, +1.156]$—removing water raises it most, by $+0.647$, and removing the halogenated solvents lowers it most, by -0.157—so exactly one class of the nine moves the aggregate by more than the aggregate's own distance from zero, and six of the other eight move it by less than 0.38. The span is wide because one class is; it is not evidence that the aggregate is broadly composition-sensitive, and the composition argument of §3.5.1 rests on the compounded cut and the null-partition test

Table S10: **The out-of-sample set, class by class**: the descriptive companion to the pre-declared test above (item 7 of the declaration, fixed in advance as description and *not* as a test—no verdict is read from this table and no row of it is stated as a finding), over every solvent class of the PGL 6th-edition set at the map's own headline cell: eight equal-count bins of $g(z^\star)$, unbiased within-bin variance, row unit, deployed residual-only convention. Rows in descending margin; the ten classes carry all 7889 records, so nothing is dropped for being small or for being negative. The manuscript's admissibility rule, applied here unchanged, admits two of the ten: the mono-alcohols, whose +0.09 carries a 90% interval covering zero, and water, whose margin is negative. So no class of this set both passes the rule and separates the two terms, and the three largest positive margins sit on classes the rule refuses. The glycol-ether row is the nearest cell of the test above; the exact analogue cell—a glycol-ether solvent with an organic solute—has no rows here at all.

Solvent class	n	pairs	sources[a]	margin	90% margin[b]	boundable[c]	deletions[d]	admissible	what it shows, or the test it fails
aliphatic hydrocarbon	875	358	3	+1.73	[−0.18, +5.26]	yes	3/3	no	the margin changes sign when one source is deleted: WinOrg:J takes +1.726 → −0.151
halogenated	428	211	3	+0.67	[−0.07, +2.23]	yes	3/3	no	the margin changes sign when one source is deleted: WinOrg:J takes +0.670 → −0.275
N–H protic	368	119	3	+0.60	[+0.22, +1.50]	yes	2/3	no	1 of 3 deletions leave fewer than 16 rows and the fixed cell is then undefined (Lazzaroni), so the sign is not verifiable
glycol ether	52	7	2	+0.35	[+0.29, +0.57]	yes	1/2	no	1 of 2 deletions leave fewer than 16 rows and the fixed cell is then undefined (WinOrg:J), so the sign is not verifiable
unclassified	143	53	2	+0.10	[−0.13, +1.15]	yes	1/2	no	1 of 2 deletions leave fewer than 16 rows and the fixed cell is then undefined (Lazzaroni), so the sign is not verifiable
mono-alcohol	1027	344	3	+0.09	[−0.06, +0.32]	yes	3/3	yes	boundable, and the sign holds when any one source is deleted, so the rule admits it; the margin is positive but its interval covers zero, so nothing is separated at this cell
aprotic acceptor	1577	714	3	+0.02	[−0.19, +0.31]	yes	3/3	no	the margin changes sign when one source is deleted: WinOrg:J takes +0.019 → −0.005; WinOrg:T takes +0.019 → −0.114
aromatic hydrocarbon	344	133	3	−0.13	[−0.28, +2.00]	yes	3/3	no	the margin changes sign when one source is deleted: Lazzaroni takes −0.130 → +10.249
water	3074	339	3	−2.53	[−3.09, −1.77]	yes	3/3	yes	boundable, and the sign holds when any one source is deleted, so the rule admits it; the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault
aryl ether	1	1	1	—	—	no	0/1	no	the fixed cell is undefined at $n = 1$: no estimate

[a] Source labels under the provenance axis declared before the run: the record file, refined by the compilation label inside the two Jaubert/TDE files (J Jaubert, T NIST TDE, Y Yaws). *WinOrg* is the water-in-organic file, *OrgInW* the organic-in-water one, *Lazzaroni* the third file, which carries no label and counts as one source. Six labels in all, against the broad set's fifteen source DOIs: this axis is coarser than the in-sample one and the test it supports is therefore weaker, which the declaration states in advance.
[b] Two-way solute × solvent cluster bootstrap, 3000 draws, drawn separately at each class: the glycol-ether row is the same 52 rows as the test above and repeats its margin exactly, while its endpoints differ from that row's in the last digits because the draw is a different one. [c] $n \geq 40$, so eight equal-count bins hold at least five rows. [d] How many of the class's leave-one-source-out deletions can be run: a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and a sign that cannot be checked reads as untested rather than as a pass.

rather than on this curve. Per-stratum leave-one-source-out, and the VT-2005-matched set's own curves, are in the artifact and summarised in §3.5.1–§3.5.2.

S3.3 The map, stratum by stratum

What is measured and not established. Water carried as the dilute solute is 37 rows and six pairs from `10.1021/acs.jced.7b00114` alone, which supplies 100% of its rows and 100% of its squared error: at n=37 it is not boundable, and with one publication there is no deletion to run, so its +6.33 was never put at risk and is not evidence about water solutes. The halogenated solvents (30 rows, +3.39) are not boundable either, and 83% of that cell's squared error is the same publication; deleting it leaves 12 rows, fewer than the fixed cell needs, so the sign cannot be verified. Neither may be used to explain a reversal. The coarse acceptor-only class, whose +1.43 this table names, is 96% carried by that publication's 13 aryl-ether rows and goes to −0.19 when it is removed. Water as a *solvent* (+0.05 at $P \approx 0.8$ deployed, +0.57 full) flips under two of its three deletions; the coarse donor-and-acceptor-oxygen class—"the hydroxylic solvents"—flips to −0.42 when `10.1021/je020196d` is removed, so it is the glycol-ether sub-class and not the hydroxylic solvents as a class that carries the finding; the organic-solute subset flips to −0.09 under the same deletion; the mono-alcohols and the N–H protics are unboundable and each has a deletion that cannot be run; the aryl ethers are 13 rows from that same publication and carry no estimate at all; and the ester (30 rows, one publication, −0.06) and azine (23 rows, one deletion unverifiable) sub-families, which the acceptor-only reading would lean on, are inadmissible on the same grounds. Of the 59 strata only 15 are boundable at all, and 24 carry no estimate at any n this set supplies. The whole-set row is admissible at the headline cell and fails once the pair unit is required, where deleting `10.1021/acs.jced.7b00114` takes +0.270 to −0.119; the rule therefore does not by itself retire the aggregate at the headline cell, and its retirement rests on the composition argument of §3.5.1 rather than on this rule.

The two axes cross, and the map measures the crossing. 362 of the 1711 stratum pairs share rows, so cells drawn from the two axes are not independent evidence. All 37 water-as-solute rows and all six of its pairs sit inside three solvent classes and all come from one publication: 18 rows in the halogenated class (60% of it), 13 in the aryl ethers (all of it) and 6 in the aromatic hydrocarbons. Naming water-as-solute and the halogenated solvents as two findings would count those 18 rows twice, and adding the aryl ethers or the coarse acceptor-only class a further 13. The two positive admissible cells overlap the same way: alkane ∩ glycol-ether is 108 rows over 24 pairs and carries the whole effect (+2.51 on those rows), while the glycol ethers minus the alkanes remain admissible—74 rows at +1.40 deployed and +1.69 full, under all three deletions—and the alkanes minus the glycol ethers, 21 rows, do not, being below the boundability threshold at the fixed cell and returning −0.22 there (−0.01 full). That is clause (c) of §2.5 run on these two cells: the glycol ethers hold the more rows and hold 84% of the alkanes', and the alkane residual is inadmissible, so the alkane solutes are demoted to the glycol ethers and not the reverse. The overlap by itself decides nothing—had the 21 rows carried the margin, the alkanes would have stood. Those rows are counted once.

Concentration, inside the map as well as under it. Three solvent classes holding 47% of the rows carry 74% of the squared error, the 12% of rows that are acceptor-only aprotics carry 0.6%, and three of fifteen publications carry 76%. A map cell can be as source-carried as the aggregate was, so each is put through the same deletion (Tables S11–S14). Eleven multi-source cells have every deletion runnable in all four cells and six keep the sign throughout—the admissible ones. What singles the glycol ethers out among them is positivity and clause (c), not a deletion nobody else survives. On the halogenated solvents and the mono-alcohols only one of the two deletions can be run at all—in

Table S11: **The map, solvent axis, in full**: every solvent stratum of the broad IDAC set, including the ones the estimator cannot bound, at the pre-declared cell—eight equal-count bins of $g(z^\star)$, unbiased within-bin variance, row unit, deployed residual-only convention—applied identically to all of them. *full* is the same cell under the full combinatorial convention. Every margin here and in the three map tables that follow, and the MSE and bound behind it, is in squared $\ln\gamma_2^\infty$ units. The map runs over 59 strata in all: the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole, over 477 rows, 185 pairs, 78 solutes, 36 solvents and 15 source publications. Together the four tables carry all 59, so nothing is dropped for being small, and each row states either what it shows or the test it fails.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent class								
glycol ether	182	43	3 (P8, 0.53)	+2.04	$[+1.26, +2.87]$	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	$[-0.15, +0.39]$	+0.57	1	the margin changes sign when one publication is deleted: P2 takes $+0.046 \to -0.075$; P14 takes $+0.046 \to -0.194$
aprotic acceptor	57	57	4 (P7, 0.32)	−0.19	$[-0.44, +0.12]$	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
mono-alcohol	35	17	2 (P3, 0.55)	+0.78	$[+0.18, +3.24]$	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
N–H protic	33	33	3 (P11, 0.62)	+0.61	$[-0.27, +1.96]$	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated	30	7	2 (P13, 0.83)	+3.39	$[+0.86, +8.12]$	+1.95	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there
aromatic hydrocarbon	14	9	2 (P13, 0.88)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aryl ether	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
Solvent class, coarse								
H-bond donor and acceptor (O)	328	75	8 (P8, 0.40)	+0.53	$[-0.21, +2.18]$	+1.34	3	the margin changes sign when one publication is deleted: P15 takes $+0.527 \to -0.417$
H-bond acceptor only	70	59	5 (P13, 0.96)	+1.43	$[-0.28, +6.19]$	+0.77	0	the margin changes sign when one publication is deleted: P13 takes $+1.434 \to -0.192$
H-bond donor (N)	33	33	3 (P11, 0.62)	+0.61	$[-0.27, +1.96]$	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated, non-H-bonding	30	7	2 (P13, 0.83)	+3.39	$[+0.86, +8.12]$	+1.95	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there
hydrocarbon, non-H-bonding	16	11	3 (P13, 0.88)	+2.99	$[-0.07, +8.86]$	+1.84	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P10, P11, P13 (3 of 3 deletions), so the sign is untested there

[a] Number of source publications, then the largest one and its share of that stratum's squared error. The keys, used in all five map tables: P1 `10.1016/j.fluid.2013.06.028`, P2 `10.1016/j.fluid.2015.01.010`, P3 `10.1016/j.fluid.2015.11.037`, P4 `10.1016/j.fluid.2016.08.030`, P5 `10.1016/j.fluid.2016.10.009`, P6 `10.1016/j.jct.2013.05.006`, P7 `10.1016/j.jct.2015.12.034`, P8 `10.1016/j.jct.2016.10.013`, P9 `10.1016/j.jct.2018.05.003`, P10 `10.1016/j.jct.2019.05.011`, P11 `10.1016/j.tca.2012.02.005`, P12 `10.1016/j.tca.2015.11.010`, P13 `10.1021/acs.jced.7b00114`, P14 `10.1021/acs.jced.9b00726`, P15 `10.1021/je020196d`. [b] Two-way solute × solvent cluster bootstrap, 3000 draws; each endpoint carries ±0.02 of Monte-Carlo spread. Row-identical strata print one interval, under the row set's first name above, and the glycol ethers print it converged. [c] In how many of the four unit × convention cells the stratum is both boundable ($n \geq 40$, so eight bins hold five rows) and keeps its margin's sign under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 strata—three distinct row sets—satisfy. [d] The condition the row is in, not a label for it. A one-source cell cannot be tested and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive puts $B_{\text{closure}} > B_{\text{insuff}}$ on those rows at that cell, non-positive is failure to separate and licenses nothing (Table S2, note *g*). Every row here is a cell of a 59-fold search, and the row that passes is not separated from it: a chemistry-blind relabelling of the same molecules, on all 477 rows, certifies at least one row set in 58% of 2000 draws and a margin of +2.04 or more in 10.1% (§S3).

Table S12: **The map, solvent axis at the fine granularity**—the third solvent granularity, at the same estimator cell and under the same rule as the solvent-axis map above, whose notes apply here unchanged. A finer class is a subset of a coarser one, so these rows are the same measurements re-partitioned and not new ones.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent family, fine								
glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.26, +2.87]	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
mono-alcohol	35	17	2 (P3, 0.55)	+0.78	[+0.18, +3.24]	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
ester	30	30	1 (P9, 1.00)	−0.06	[−0.08, +0.03]	−0.05	0	$n < 40$: too few rows for eight bins of five; one source (P9), so there is no deletion to run and the sign was never put at risk
amide N–H	25	25	2 (P12, 0.96)	−0.24	[−0.34, +0.76]	−0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 2 deletions), so the sign is untested there
azine	23	23	2 (P11, 0.58)	−0.05	[−0.15, +0.14]	+0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P5 (1 of 2 deletions), so the sign is untested there
halogenated aliphatic	18	5	2 (P3, 0.87)	+0.11	[−0.18, +2.33]	−0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3, P13 (2 of 2 deletions), so the sign is untested there
aromatic hydrocarbon	14	9	2 (P13, 0.88)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aryl ether	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated aromatic	12	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
amine N–H	8	8	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ketone/aldehyde	2	2	1 (P7, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitrile	2	2	1 (P7, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n

Table S13: **The map, solute axis**—the solute half of the map, at the same estimator cell and under the same rule as the solvent axis above, whose notes apply here unchanged. Rows at different granularities, and rows on the two axes, can be the same measurements: 362 of the 1711 stratum pairs share rows, and every row that keeps its sign carries its overlap with the others.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solute role								
organic solute	440	179	14 (P8, 0.35)	+0.18	[−0.23, +1.36]	+0.82	2	the margin changes sign when one publication is deleted: P15 takes +0.184 → −0.094
water as solute	37	6	1 (P13, 1.00)	+6.33	[+3.97, +9.49]	+3.54	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
Solute family								
alkane	129	45	7 (P15, 0.53)	+1.38	[+0.79, +2.33]	+1.72	4	the sign holds under every deletion, but 84% of these rows are the glycol ethers' and the 21 outside give −0.22, so this margin does not stand outside the glycol ethers: demoted to them by clause (c); 10% shared with the aprotic acceptors
aromatic hydrocarbon	71	32	7 (P15, 0.53)	−0.19	[−1.01, +0.75]	−0.07	0	the margin changes sign when one publication is deleted: P6 takes −0.188 → +0.081
ketone	63	11	3 (P14, 1.00)	−0.53	[−0.85, +0.64]	−0.33	0	too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there
mono-alcohol	39	21	5 (P11, 0.46)	+0.27	[+0.09, +2.10]	+0.44	0	$n < 40$: too few rows for eight bins of five; the sign holds, but all 5 deletions leave $n < 40$
water	37	6	1 (P13, 1.00)	+6.33	[+3.97, +9.49]	+3.54	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
alkene	32	16	4 (P8, 0.98)	+2.29	[+0.12, +3.43]	+2.76	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P8 (1 of 4 deletions), so the sign is untested there
aldehyde	29	3	1 (P2, 1.00)	−0.20	[−0.51, +0.13]	+0.68	0	$n < 40$: too few rows for eight bins of five; one source (P2), so there is no deletion to run and the sign was never put at risk
halogenated aliphatic	27	9	3 (P3, 0.99)	+1.06	[+0.15, +1.99]	+1.36	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 3 deletions), so the sign is untested there
amine N–H	12	12	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ether	12	4	3 (P2, 0.98)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated aromatic	8	8	4 (P1, 0.75)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
azine	6	6	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
tertiary amine	4	4	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ester	3	3	2 (P5, 0.56)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
alkyne	2	2	2 (P5, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitrile	2	2	2 (P9, 0.98)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitro	1	1	1 (P9, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n

Table S14: **The map, the solvent × solute cross-tab and the set as a whole**—the last two granularities, at the same estimator cell and under the same rule as the two axes above, whose notes apply here unchanged. The whole-set row is the aggregate this work declines to state, printed with the rest.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent class × solute role								
glycol ether | organic solute	182	43	3 (P8, 0.53)	+2.04	[+1.26, +2.87]	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water | organic solute	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
aprotic acceptor | organic solute	57	57	4 (P7, 0.32)	−0.19	[−0.44, +0.12]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
mono-alcohol | organic solute	35	17	2 (P3, 0.55)	+0.78	[+0.18, +3.24]	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
N–H protic | organic solute	33	33	3 (P11, 0.62)	+0.61	[−0.27, +1.96]	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated | water as solute	18	3	1 (P13, 1.00)	+5.84	[+2.70, +9.43]	+3.39	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
aryl ether | water as solute	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated | organic solute	12	4	1 (P3, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aromatic hydrocarbon | organic solute	8	8	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aromatic hydrocarbon | water as solute	6	1	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon | organic solute	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
The set as a whole								
whole set	477	185	15 (P13, 0.32)	+0.51	[−0.09, +2.16]	+0.95	3	passes here and fails at the pair unit (deployed convention), where deleting P13 takes +0.270 → −0.119

Table S15: **The map's error, bound and source curve**, stratum by stratum, at the same headline cell as the four map tables above—eight equal-count bins of $g(z^\star)$, unbiased within-bin variance, row unit, deployed residual-only convention. These are the columns those tables do not carry, on the same 59 strata, whose margins, intervals and admissibility are printed there.

Stratum	n	MSE	$B^{\mathrm{up}}_{\mathrm{insuff}}$ [a]	P [b]	share [c]	leave-one-source-out [d]
Solvent class						
glycol ether	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water	111	0.477	0.215	$\approx$0.8	5.8%	$[-0.19, +0.25]$ (3)
aprotic acceptor	57	0.089	0.140	< 0.2	0.6%	$[-0.41, -0.13]$ (4)
mono-alcohol	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
N–H protic	33	1.175	0.281	$\approx$0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated	30	4.569	0.592	> 0.95	15.1%	$+5.84$ (one deletion)
aromatic hydrocarbon	14	4.297	—	—	6.6%	none possible
aryl ether	13	9.582	—	—	13.7%	none possible
aliphatic hydrocarbon	2	0.011	—	—	0.0%	none possible
Solvent class, coarse						
H-bond donor and acceptor (O)	328	1.651	0.562	$\approx$0.9	59.7%	$[-0.42, +0.97]$ (8)
H-bond acceptor only	70	1.852	0.209	$\approx$0.7	14.3%	$[-0.19, +2.94]$ (5)
H-bond donor (N)	33	1.175	0.281	$\approx$0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated, non-H-bonding	30	4.569	0.592	> 0.95	15.1%	$+5.84$ (one deletion)
hydrocarbon, non-H-bonding	16	3.761	0.385	$\approx$0.9	6.6%	none possible
Solvent family, fine						
glycol ether	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water	111	0.477	0.215	$\approx$0.8	5.8%	$[-0.19, +0.25]$ (3)
mono-alcohol	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
ester	30	0.043	0.049	$\approx$0.2	0.1%	none possible
amide N–H	25	0.591	0.414	$\approx$0.8	1.6%	$+0.47$ (one deletion)
azine	23	0.096	0.072	$\approx$0.5	0.2%	-0.00 (one deletion)
halogenated aliphatic	18	1.526	0.709	$\approx$0.9	3.0%	none possible
aromatic hydrocarbon	14	4.297	—	—	6.6%	none possible
aryl ether	13	9.582	—	—	13.7%	none possible
halogenated aromatic	12	9.135	—	—	12.1%	none possible
amine N–H	8	2.999	—	—	2.6%	none possible
aliphatic hydrocarbon	2	0.011	—	—	0.0%	none possible
ketone/aldehyde	2	0.404	—	—	0.1%	none possible
nitrile	2	0.397	—	—	0.1%	none possible
Solute role						
organic solute	440	1.400	0.608	$\approx$0.8	67.9%	$[-0.09, +0.41]$ (14)
water as solute	37	7.866	0.770	> 0.95	32.1%	none possible
Solute family						
alkane	129	2.315	0.467	> 0.95	32.9%	$[+1.39, +1.67]$ (7)
aromatic hydrocarbon	71	0.534	0.361	$\approx$0.5	4.2%	$[-0.86, +0.08]$ (7)
ketone	63	0.628	0.577	$\approx$0.5	4.4%	$[-0.14, -0.14]$ (2)
mono-alcohol	39	1.394	0.563	> 0.95	6.0%	$[+0.17, +1.37]$ (5)
water	37	7.866	0.770	> 0.95	32.1%	none possible
alkene	32	2.661	0.184	> 0.95	9.4%	$[+2.00, +2.68]$ (3)
aldehyde	29	0.091	0.144	$\approx$0.4	0.3%	none possible
halogenated aliphatic	27	1.631	0.286	> 0.95	4.9%	$[+1.23, +1.71]$ (2)
amine N–H	12	3.362	—	—	4.5%	none possible
ether	12	0.469	—	—	0.6%	none possible
halogenated aromatic	8	0.619	—	—	0.5%	none possible
azine	6	0.135	—	—	0.1%	none possible
tertiary amine	4	0.278	—	—	0.1%	none possible
ester	3	0.013	—	—	0.0%	none possible
alkyne	2	0.350	—	—	0.1%	none possible
nitrile	2	0.124	—	—	0.0%	none possible
nitro	1	0.068	—	—	0.0%	none possible
Solvent class × solute role						
glycol ether | organic solute	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water | organic solute	111	0.477	0.215	$\approx$0.8	5.8%	$[-0.19, +0.25]$ (3)
aprotic acceptor | organic solute	57	0.089	0.140	< 0.2	0.6%	$[-0.41, -0.13]$ (4)
mono-alcohol | organic solute	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
N–H protic | organic solute	33	1.175	0.281	$\approx$0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated | water as solute	18	6.294	0.228	> 0.95	12.5%	none possible
aryl ether | water as solute	13	9.582	—	—	13.7%	none possible
halogenated | organic solute	12	1.982	—	—	2.6%	none possible
aromatic hydrocarbon | organic solute	8	0.873	—	—	0.8%	none possible
aromatic hydrocarbon | water as solute	6	8.862	—	—	5.9%	none possible
aliphatic hydrocarbon | organic solute	2	0.011	—	—	0.0%	none possible
The set as a whole						
whole set	477	1.902	0.697	$\approx$0.9	100.0%	$[+0.18, +0.87]$ (15)

[a] A dash is a stratum at which the fixed cell is undefined. Strata with $n \geq 40$ are *boundable*: eight bins hold at least five rows each. The others are reported anyway, and additionally carry the adaptive bin count $\max(3, \lfloor n/10 \rfloor)$, here in the stratum's own rows n, in the deposited table. Every stratum here is at the one fixed bin count; the sweep over bin count for the strata the rule passes is Table S6. On the mono-alcohols $B^{\mathrm{up}}_{\mathrm{insuff}} = 0.733$ exceeds that stratum's own $\mathrm{Var}(m) = 0.604$, which the artifact flags; it is the value that sets the ×6.8 span of §3.5.1, and it occurs only on a stratum already not boundable. [b] Two-way solute × solvent cluster bootstrap, 3000 draws, quoted as a band by the rounding rule of §3. [c] That stratum's share of the set's squared error. Shares do not sum to 100% down the column: the granularities and the two axes overlap, which §3.5.1 measures. [d] Range of the margin over the deletions that can be run, with how many that is in parentheses; a deletion leaving fewer than 16 rows is not runnable, and a stratum with one source has none. Whether the sign survives—which is what admissibility turns on—is in Table S11.

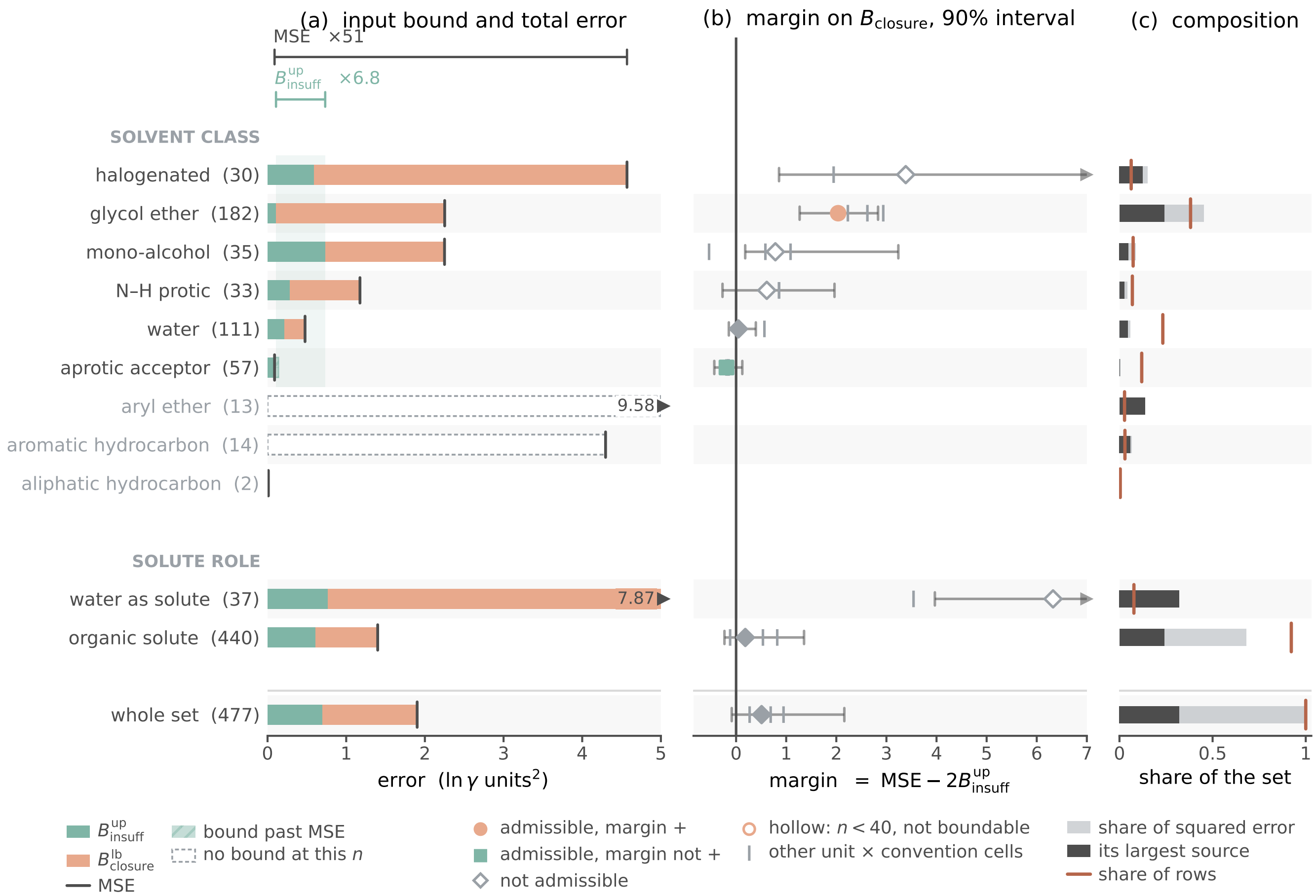

Figure S2: **Where the closure binds, and where the two terms do not separate**, by solvent class and by solute role, on the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Bars, points and shares are at the row unit under the deployed residual-only convention, at one estimator cell for every stratum—eight equal-count bins of $g(z^\star)$ with the Bessel within-bin variance. **(a)** each stratum's input-insufficiency bound and its total error; **(b)** the margin between them with its 90% two-way cluster-bootstrap interval, the other three unit × convention cells as grey ticks; **(c)** each stratum's share of the set's squared error, the part carried by its largest source publication drawn dark. This is the drawn form of the twelve cells at the reported granularity; the map itself, all 59 strata with their sources, intervals and admissibility, is the four map tables below. The marker in (b) is the admissibility verdict, read off the same deposited columns the tables are built from. The one row set both positive and left standing by clause (c) of the rule is the glycol ethers, here at $P > 0.95$; on no other row, the aggregate included, do the two terms separate. *Caveat:* a hollow marker is a stratum below n=40—four of the drawn cells, with margins +0.78, +0.61, +3.39 and +6.33, none of them admissible—and three classes carry no bound at any n. The one cell the rule leaves standing and the one admissible cell whose margin is not positive are both filled; panel (c) shows how much of each cell one publication carries. The bars on the two axes overlap: all 37 water-as-solute rows lie inside three solvent classes, which is why panel (c)'s shares do not sum down the column.

both, the one that cannot is the one removing the dominant source—and it leaves +5.84 and +1.95, so those two cells are unverified rather than shown to fail. At the level of the individual pair the broad set is stable, all 185 leave-one-pair-out margins falling in [+0.42, +0.72], so what the aggregate is fragile to is the source and the stratum, which is what a composition effect looks like. The chromatographic-only cut that *inverts* the margin on the enlarged 298 K

VT-2005-matched set (§3.7, limitation (i)) does not invert here under either convention.

The broad set: which chemistry does not separate, cell by cell. Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets, and neither establishes anything. The alkane solutes (129 rows, 45 pairs, +1.38, [+0.79, +2.33]) are admissible and positive, and clause (c) demotes them to the glycol ethers, which hold more rows and hold 108 of these 129: the alkane residual outside them is 21 rows, too few to bound at the fixed cell, and returns −0.22 there, so the alkanes' margin does not stand outside the glycol ethers—where the glycol ethers' does stand outside the alkanes, on the 74 rows at the +1.40 above. What demotes the cell is that residual, not the 84% overlap and not the −0.22, which is itself subthreshold. The acceptor-only aprotics (57 rows, −0.19, negative in all four cells, $P < 0.2$ at both units under the deployed convention and $P \approx 0.3$ under the full one, at an MSE of 0.089 that is at the scale of the measurements) pass the rule and license nothing, the instrument being one-sided; their deletion test is additionally *anti*-conservative, two of the four deletions leaving $n < 40$, where the Bessel within-bin variance inflates—0.140 to 0.258 on the n=38 remainder—and drives a negative margin further negative. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and left standing by clause (c)—*not* the only admissible stratum and not the only one sign-stable under every deletion. Of the rest, water carried as the dilute solute (37 rows, +6.33) is too small to bound and draws on one publication, so its deletion cannot be run and that margin was never put at risk; the halogenated solvents (30 rows, +3.39) are too small to bound and only one of their two deletions runs, leaving +5.84; water as a solvent, the coarse hydroxylic class and the organic-solute subset each lose their sign under a deletion, which is why the finding belongs to the glycol-ether sub-class and not to the hydroxylic solvents as a class; the mono-alcohols, the N–H protics and the ester and azine subfamilies, which an acceptor-only reading would lean on, are unboundable. None may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

The VT-2005-matched set, stratum by stratum and deletion by deletion. §3.5.2 states what this set is and that its aggregate does not stand; here is the arithmetic. The two worst-fit pairs each rest on a single record and carry 34% of the squared error, a concentration we do not read as established chemistry; the separation does not survive their deletion at the headline cell (margin −0.002), a margin of +0.05 being unable to absorb 34% of the squared error leaving the sample, and that deletion is a failure of the headline cell rather than a robustness pass. Four neighbouring cells do survive it, and the full composition audit is above. These are not 60 exchangeable observations, and the folds hinge on the same molecules. The stratification rule of §2.5 carries the composition half of §3.5.1 here and not the bound half: the 17 solvents span six of the nine classes—no water, no halogenated solvent, no glycol ether beyond ethylene glycol—and the largest reaches 24, so the set's composition is missing two of the three classes that carry most of the broad set's error, and of its 38 strata not one is admissible in every unit × convention cell. Within it, scored on UD profiles so that the two sets are commensurable, the ordering of the error is the broad set's—2.71 on the mono-alcohols, 3.00 on the N–H protics and 2.53 on the glycol ethers against 0.092 on the acceptor-only aprotics, that last cell again returning a negative margin, −0.05 at $P \approx 0.5$—but 57 of the 60 pairs are the broad set's own rows, so that agreement is a consistency check and not independent corroboration. What is this set's own is the granularity of its fragility: deleting the publication supplying 33 of the pairs takes the margin to −0.23. Scored against VT-2005 profiles under the deployed residual-only convention and the *unbiased* within-bin variance, the LOTV bound gives $B_{\mathrm{insuff}} \lesssim 0.713$ and the separation margin of +0.05; the maximum-likelihood variant biases the within-bin variance downward and so the margin upward to +0.24. That number stands with the broad set's aggregate, and on weaker evidence, four analyst choices setting it (§2.5).

Which defect the broad set points at, and why it states neither. §3.5.3 places the closure's failure on the VT-2005-matched set in a *chemistry*—90% of that set's squared error falls on strong hydrogen-bond-donor solvents, the chemistry the single-parameter H-bond kernel governs—and not in one component of the composite. On the broad set that reading does not reproduce, and neither does its replacement. The donor classes carry 64% of the squared error on 76% of the rows, which is no enrichment, while the halogenated and aryl-ether solvents—which the 2002 kernel has no dispersion term for at all—carry 29% on 9%, which is. But 31 of those 43 rows are `10.1021/acs.jced.7b00114`, and they are the same water-as-solute rows the map already declines to state; they carry 91% of that 29%. Put those two shares through the deletion the map applies to every cell and both invert: without that publication the halogenated and aryl-ether rows fall to 2.7% of the rows and 3.9% of the error, and the donor classes rise to 82% and 94%, which *is* the VT-2005-matched set's localisation. So one publication decides the reading in either direction, which is why the article states neither.

The constant-offset (Jensen) bound on both sets. §2.5 promises these here. On the broad IDAC set $\mathbb{E}[m] = 3.424$, $\mathbb{E}[g_{\text{res}}] = 2.998$ and $\mathbb{E}[g_{\text{full}}] = 2.579$, so the constant-offset bound is $B_{\text{closure}} \geq 0.182$ under the deployed residual-only convention and $B_{\text{closure}} \geq 0.715$ under the full one. Against the law-of-total-variance ceiling on B_{insuff} (0.697 residual-only, 0.482 full) the assumption-free bound therefore separates under the full convention and inverts under the deployed one, exactly as it does on the VT-2005-matched set (0.426 against 0.713) and by a wider margin—the deployed ratio is 0.26 here against 0.60 there. On this set it is in addition strictly slacker than the bound the decomposition already supplies, $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$ (0.182 against 1.205 residual-only, 0.715 against 1.429 full), so it adds nothing to the ordering under either convention; the deployed-convention separation rests on the law-of-total-variance bound alone, which is what §2.5 says.

The broad set's matching rule, and two bounds we decline to headline. Three details of §3.5.1 follow. *(a) The matching asymmetry.* The VT-2005-matched set is matched to VT-2005 by canonical SMILES, a rule we defend there on the ground that a profile imported across a tautomer or protonation variant is the wrong $z^\star$. The broad IDAC set is matched to UD by InChIKey with a first-block fallback, which is looser, and a wrong $z^\star$ inflates MSE and so B_{closure}—that is, it runs against the ordering we claim. The exposure is small and we measure it rather than argue it: of the 90 distinct molecules there, 86 resolve by full InChIKey and four only through the first block, touching 12 of the 477 rows. On the 465 rows matched exactly the verdict is unchanged—deployed convention MSE 1.930, $B_{\text{insuff}} \lesssim 0.70$, margin +0.53 at $P \approx 0.9$; full convention 1.926, 0.49, +0.94 at $P > 0.95$—so the looser rule is not carrying the result. *(b) The tighter of the two conventions' bounds.* B_{insuff} is the same population quantity under both conventions (step (d) of the proof of Lemma 1, §S2); what differs is the statistic the coarsening bins on, $g_{\text{full}}(z^\star, T)$ or $g_{\text{res}}(z^\star, T)$, each a deterministic function of the same argument and each therefore giving a separately valid upper bound. The smaller of the two is consequently also a valid upper bound in population, and charging it against the deployed error gives 1.902 against 2×0.48, a margin of +0.94 (on the VT-2005-matched set, 1.472 against 2×0.61, +0.26). We report this but do not headline it: a minimum over two finite-sample estimates is anti-conservative even when each is valid in population, and the same-convention pairing is the conservative reading. *(c) The binning-free pair bound.* $\mathbb{E}[\text{Var}(m \mid \text{pair})] = 0.046$ is thirteen times tighter than the binning bound on the same 359 rows (0.61), and would put the margin *there* at +2.16 against the +1.02 the binning bound gives on those rows. It is not promoted because all 67 multi-row pairs draw their rows from a single publication, so it samples no between-laboratory spread (§3.5.1).

Whether the closure defect is correctable: within a solvent, not across one. Where B_{closure} dominates—on this set, the one row set of §3.5.1 the rule leaves standing—the ceiling is in the map *as deployed*, kernel and computed profile together (§2.1.1); that does not say the composite is *fixably* wrong. We tested this with a small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel ($\Delta = B\,\mathrm{diag}(a)\,B^{\top}$ over the σ-grid, symmetric, zero-initialised), fit on the *reference* VT-2005 profiles and cross-validated five-fold over five fit seeds. The answer depends on the fold design, and only one design is free of leakage. With random folds over the 60 matched pairs, a rank-one residual (52 parameters) removes $46 \pm 5\%$ of the reference-input error out of sample (activity MSE $1.47 \to 0.79$). The same solvent, however, then appears in training and test, and the closure error is predominantly a between-solvent systematic. Under folds grouped so that no solvent is shared, the reduction is $-5 \pm 16\%$ at rank one and $+2 \pm 15\%$ at rank two. Each $\pm$ is a standard error over the five fit seeds on one fixed 60-pair set and prices optimiser noise only; the held-out spread across folds is an order of magnitude larger (sd 1.02 in MSE under the grouped design, 0.54 under random folds) and is not propagated into it. That spread is itself about 70% of the reference-input MSE the reduction is measured against, so the grouped design yields a null and no upper bound: an interval wide enough to price it would exclude nothing about a cross-solvent reduction, and the narrow one quoted prices the wrong thing. What the two designs do support is the contrast between them—the 46% appears only when the same solvent may sit on both sides of the split, so it is a within-solvent effect.

That fold spread also exceeds the gaps between ranks (≤ 0.06) by an order of magnitude, so these data do not resolve a rank ordering; and the 60 pairs form a single connected component under shared molecule identity, so a molecule-disjoint holdout does not exist at any fold design. B_{closure} is therefore an error that a handful of kernel parameters can absorb *within* a solvent, and that we have not shown to be correctable across solvents—weaker than a general repair, and a statement about what a correction of that shape absorbs rather than about which component of the composite the error came from. The fitted correction also gains nothing over a free one: the parameter-free 2010 kernel reaches 0.765 on the same molecule pairs (§3.5.3) against the 52-parameter residual's random-fold 0.79. The residual's leakage-free number, however, is the grouped-fold one, a $-5 \pm 16\%$ reduction, i.e. no reduction at all, so the comparison that holds is a free 2002 $\to$ 2010 step against a fitted correction that does not transfer. The two also run on different profile sources and conventions (residual-only on VT-2005 here, full on the University of Delaware (UD) profiles there[S10]), so this is indicative rather than a head-to-head; but each stays above its own convention's insufficiency ceiling (0.62–0.71 residual-only, 0.53–0.61 full), so neither erases B_{closure}.

Where the residual error sits chemically. The composition audit behind §3.5: 90% of the VT-2005-matched set's squared error falls on the strong hydrogen-bond-donor solvents, 84% once the two doubted amine/methanol pairs are deleted, and the acceptor-only solvents are near-exact. That localisation places B_{closure} on the chemistry COSMO-SAC-2002's single-parameter hydrogen-bonding term governs. It does not place it in the kernel rather than in the profile: those solvents' reference profiles are single gas-phase conformers as well, and the glycol ethers—the one stratum the rule leaves standing, and the class the cascade above names conformer-dependent—are where that displacement is expected to be largest, so a displaced reference would load onto the rows that carry the error. Neither set separates the two (§S9). A counter-test on the same closure points the same way, and is the one §3.5 cites. The Staverman–Guggenheim combinatorial term is the physically *more* complete treatment and acts only where the two molecules differ in size; restoring it *increases* the VT-2005-matched set's reference-input error, MSE = 1.47 under the deployed residual-only convention against 1.80 with the term in place (Table S7), the gap between the two being carried by the size-asymmetric pairs. A term that adds physics and costs accuracy is what a misspecified map does and not what missing information about the inputs would do. The same signature appears on the solubility axis, where restoring the term makes the grounded learned-σ arm slightly worse (Table S3).

The closure-fidelity lever: the arithmetic behind §3.5.3. *(a′) The input-fixed kernel control.* The prescribed reduction of the typed Hsieh profile to a single 2002 profile is its sum $p = p_{\mathrm{nhb}} + p_{\mathrm{OH}} + p_{\mathrm{OT}}$ (Eq. 9 of Ref.[S1]), which holds database, cavity area, volume and averaging fixed and varies only the kernel. On the VT-2005-matched set that contrast is $2.047 \to 0.765$, two-way cluster bootstrap $P > 0.95$ with 90% margin $[+0.05, +2.90]$; the averaging step taken alone moves 2002 by +0.29 in the opposite direction and is not resolved ($P < 0.2$, $[-0.79, +0.11]$). That averaging term is why the control's 2002 endpoint is 2.047 and not the 1.757 of the *fair 2002* arm, and why the input-fixed step (1.28) is *larger* than the native-input one (0.99): the control is favourable to the conclusion it supports. The control cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction. What fixes the *fair 2002* arm at 1.757 rather than lower is an implementation detail: the NIST reference implementation, handed a typed three-profile database, silently reads the nonpolar channel only, an arm scoring 1.26 on the VT-2005-matched set (§S1 gives it unrounded), which is not the reference closure, so the arm is taken from our own NIST-validated layer instead. *(a) The* 2010, dispersion on *arm's coverage.* `cCOSMO` returns no dispersion value for the three bromine- or iodine-containing VT-2005-matched pairs—bromobenzene and iodobenzene in 1,2-ethanediol, and 2-bromopropane in pyridine—so that arm's VT-2005-matched run scores on n=57. On those common 57 the *2010, dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n. They are the *same* three pairs the broad set's construction excludes, its ladder also requiring a tabulated ϵ, so the dispersion-defined 57 and the 57 shared between the two sets are one set, not two. *(b) A rounding coincidence.* The 1.783 quoted for the shared 57 and the 1.783 quoted for the 334 chromatographic rows are distinct numbers that round alike (unrounded 1.78257 against 1.78245), the smaller set lying wholly inside the larger. *(c) The 2010 kernel on its own typed latent.* That latent is three 51-bin profiles per molecule, 153 in all, so the conditioning variable is 306-dimensional and the resolution ratios are $307/477 \approx 0.64$ per row and $307/185 \approx 1.66$ per distinct pair on the broad IDAC set, and $306/60 \approx 5.1$ on the VT-2005-matched set, where all rows sit at 298 K and T adds no dimension. Under the deployed residual-only convention the VT-2005-matched set gives 0.831 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 0.03$ and a margin of -0.77 at $P \approx 0.3$; the broad IDAC set gives 2.893 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 2.09$ and a margin of +1.29 at $P \approx 0.9$. Under the literature-standard full convention the same rows give 1.978 against $B_{\mathrm{insuff}} \lesssim 0.60$, margin +0.78 at $P \approx 0.8$, and the VT-2005-matched set 0.765 against 0.69, margin -0.61. The switch to the deployed convention raises the 2010 kernel's broad-set error from 1.978 to 2.893 and its floor bound from 0.60 to 0.80, so the wider margin comes from a worse closure and not from a tighter floor, and the reading does not move. On the broad set the binding clustering is the source-DOI bootstrap on 15 clusters, whose 90% interval $[-0.41, +4.53]$ still admits zero, as does the two-way one $[-0.24, +4.71]$; only the pair clustering is strictly positive, and it was so before the convention switch as well. *(d) The published comparison.* The comprehensive assessment[S10] covers 29,173 γ^{∞} points for the 2010 and dsp kernels only. Its largest documented infinite-dilution step—adding the whole dispersion term—yields a ~10-point mean-absolute-deviation gain (95% → 86%) with a *reversal* on aqueous systems (203% vs. 194%); 2010's gains are otherwise concentrated in liquid–liquid equilibria and in the temperature dependence of the electrostatic term, where an infinite-dilution design has little leverage. *The scope of the search behind the null of §3.5.3*: the papers introducing each kernel,[S2,S3,S11] the benchmark implementation,[S1] this comprehensive assessment[S10] and the works citing them. It was not exhaustive, which is why the main text reads the result as an absence of evidence rather than as evidence of absence. *(e) The floor checks.* Each is same-database, same-latent and same-convention, stated under the deployed residual-only rule with the full-convention reading beside it. *(1)* Broad IDAC set: the fair 2002 error 1.902 against $2B_{\text{insuff}} \lesssim 1.39$ on the *same* UD profiles, margin +0.51 at $P \approx 0.9$ (1.911 against 0.96, +0.95, $P > 0.95$ full). *(2)* VT-2005-matched, VT-2005 against VT-2005: 1.472 against $2B_{\text{insuff}} \lesssim 1.427$, so $B_{\text{closure}} \gtrsim 0.759$ against $B_{\text{insuff}} \lesssim 0.713$ and a margin of +0.05 (1.80 against 1.22, +0.58 full; Table S7). The 1.757 of (a′) is those same 60 pairs scored on *UD* profiles, which is the whole difference between 1.80 and 1.757. Charged against a UD-native floor recomputed for those pairs, the deployed convention gives 1.436 against $B_{\text{insuff}} \lesssim 0.71$, a margin of +0.01 at $P \approx 0.8$, against 1.757 versus a UD-native $B_{\text{insuff}} \lesssim 0.54$ and a margin of +0.69 full—so on UD profiles the VT-2005-matched set's own floor check is, under the rule we deploy, at zero.

Robustness of the closure/insufficiency separation. We stress-test the deployed-convention verdict $B_{\text{closure}} > B_{\text{insuff}}$ (§3.5)—equivalently $B_{\text{insuff}} < \text{MSE}/2 = 0.74$ at $\text{MSE}_{\text{res}} = 1.47$—four ways on the n=60 set.

(i) Estimator sensitivity. The four random-fold B_{insuff} upper bounds of Table S9 (the LOTV one is also the headline row of Table S7)—0.40 (k-NN), 0.71 (binning/LOTV at the headline unbiased variance; 0.62 at the maximum-likelihood one), 0.62 (random forest), 0.62 (ridge out-of-fold)—all fall below the 0.74 threshold, the binning bound only just. The forest is the one estimator our two VT-2005-matched audit artifacts disagree on: the standalone decomposition run's own forest gives 0.565 on these rows against the audit's 0.621, a spread of the size a forest's own randomisation produces. The audit value is the one Table S9 carries, because the audit artifact is what the inclusion rule names; both are below the threshold and neither is the headline. The k-NN value is the smallest and least conservative input to $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$, so we take the leakage-immune LOTV bound as the headline instead. The LOTV bound bins $g(z^\star)$ and is therefore finite-sample downward-biased when points per bin are few, which would *inflate* B_{closure} favourably. The two non-binning bounds (RF 0.62, ridge 0.62) are immune to that particular per-bin sparsity, but they do not provide independent corroboration: under random folds every held-out row shares its solvent or its solute with a training row, so both can borrow strength from a near-twin, anti-conservatively and in the same direction as the binning (§3.5). Blocking folds on both molecules leaves no informative estimator on this set. The broad IDAC set behaves the same way once its own leak—a held-out row's molecule pair recurring at a neighbouring temperature—is blocked: the random forest goes from 0.15 to 0.88 and the ridge from 0.80 to 1.47, past its threshold. The headline should therefore be read as resting on the leakage-immune LOTV bound alone, on either set, not on agreement between three estimators. The sole non-certifying configuration is a deliberately aggressive ridge, which lifts B_{insuff} to 0.88 (then $B_{\text{closure}} \geq 0.59 < 0.88$): a sensitivity edge, not a refutation.

(0) The VT-2005-matched set's design. The 60 pairs span 41 solutes and 17 solvents; two formamide pairs are excluded for a canonical-SMILES tautomer mismatch. Methanol occupies 25 of the 60 rows (12 as solute, 13 as solvent) and pyridine is the largest solvent stratum (20 rows); $\text{Var}(m) = 2.47$ across the set. The two worst-fit pairs, 1-hexanamine and cyclohexylamine in methanol, carry 34% of the squared error; both rest on a single record, and 1-hexanamine sits far off its own homologous series ($\ln \gamma_2^\infty$=2.62 against −0.78 for 1-butanamine in the same solvent), which

is why we do not read that concentration as established chemistry. Deleting both leaves MSE 1.01; charged against this the headline cell of the deployed convention—eight equal-count bins under the *unbiased* within-bin variance—gives $B^{\mathrm{up}}_{\mathrm{insuff}}$ 0.5056 and margin -0.0015, so the separation does *not* survive the deletion where the paper headlines it. The four neighbouring cells do: 0.4370 (margin +0.14) at eight bins under the maximum-likelihood variance, 0.4717 (+0.07) at six bins unbiased, 0.4241 (+0.16) at six bins maximum-likelihood, and, under the full convention on the same 58 rows, MSE 1.20 against 0.3309 (+0.53) at eight bins unbiased. The reading we take from the cell that fails is that the VT-2005-matched set's +0.05 headline margin has no room to absorb the loss of 34% of the squared error, which is the conclusion the leverage folds below reach by a different route.

(i bis) Leverage folds on the VT-2005-matched set. The separation survives all 17 leave-one-solvent and all 60 leave-one-pair folds and deletion of the eight highest-error pairs, but only 40 of 41 leave-one-solute folds (six equal-count bins): the exception drops methanol, the maximum-leverage solute (12 pairs), and loses the separation (-0.26), as does dropping methanol as solute with pyridine as solvent (n=28; -0.36). Deleting methanol from both roles (n=35) holds only marginally—the margin falls from 0.35 to 0.11 at six bins and from 0.24 to 0.003 at eight, and inverts under the Bessel-corrected variance (-0.03, -0.21). Those fold counts and the bootstraps use the maximum-likelihood variance at six bins. Repeated at the headline cell—eight bins, unbiased variance—the leave-one-solvent count is unchanged at 17/17 (minimum margin +0.03), the leave-one-pair count falls to 56/60 (minimum -0.03) and the leave-one-solute count to 36/41 (minimum -0.22); deletion of the eight highest-error pairs still holds there (+0.04).

(ii) Leave-one-solvent-out. Dropping each of the 17 solvents in turn, the VT-2005-matched set's aggregate margin stays positive in all 17 folds under the LOTV bound—a fold curve on a quantity §3.5.1 retires, so it records what that aggregate did and not a robustness pass. Under the random-forest estimator it holds in only 2/17, because an RF over-fits conditional variance in 102 dimensions at n=60 (its out-of-fold MSE approaches $\mathrm{Var}(m)$), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip.

(iii) Label noise. Injecting zero-mean Gaussian noise $\sigma_\eta \in \{0.2, \ldots, 0.8\}$ into m leaves the B_{closure} lower bound stable ($0.84 \to 0.70$), as expected since $B_{\mathrm{closure}} = \mathrm{MSE} - \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ is invariant to zero-mean noise.

(iv) Clustered-bootstrap interval. A solvent-clustered bootstrap (3000 resamples of the 17 solvents, LOTV bound) makes the confidence explicit: the B_{closure} lower bound has median 0.93 (CI_{90} [0.40, 1.62]), and the margin $B_{\mathrm{closure}} - B_{\mathrm{insuff}}$ has median 0.36 (CI_{90} [-0.22, +0.97]) with $P(B_{\mathrm{closure}} > B_{\mathrm{insuff}}) \approx 0.89$ (single-axis; the conservative two-way solute×solvent bootstrap, which also charges for repeated solutes, gives ≈ 0.78, the value reported in the main text). With $G = 17$ solvent and 41 solute clusters this small-G regime has optimistic interval coverage, and the finite-cluster correction acts in the direction of less confidence and non-certification. We therefore treat the P values as a coarse band, not two-decimal quantities, and the main text quotes them as bands (rounded to one decimal, with > 0.95 and < 0.2 closing the ends) throughout; the two-decimal values are in the deposited artifacts and in the tables here, where the estimator that produced each is named. The separation on the VT-2005-matched set does not clear the 90% level under either clustering, which is why the article treats it as supporting rather than decisive. On the broad IDAC set, with 78 solute and 36 solvent clusters and a third clustering over 15 source publications, the bands mean something; but they do not all clear. Under the deployed residual-only convention the margin bootstraps to $P > 0.95$ over the 185 molecule pairs (90% margin $[+0.23, +1.16]$) and only $\approx$0.9 two-way over the 78×36 solute/solvent clusters ($[-0.08, +2.16]$) and over the 15 sources ($[-0.07, +1.74]$), with two of the three 90% intervals extending just below zero, so that set is not resolved either. Read as a ratio, the broad set's +0.51 is 1.1 times its own 90% bootstrap half-width under pair clustering

and 0.46 times it under the two-way one, which is the sense in which §3.5 calls that margin unresolved. It is only under the full convention that all three clear ($P > 0.95$), in the same order: $[+0.71, +1.27]$ over pairs, $[+0.48, +1.85]$ two-way, $[+0.50, +1.57]$ over sources.

(v) A label systematic tracking $z^\star$, and what the VT-2005-matched set's fold designs leave. Zero-mean noise cannot manufacture the separation: it raises MSE and $B^{\mathrm{up}}_{\mathrm{insuff}}$ alike, leaves the B_{closure} lower bound unchanged, and shrinks the margin by σ^2_η, so it can understate the separation but not create it. What the margin does bound is a *systematic* label error correlated with $z^\star$, which enters B_{closure} rather than B_{insuff}: on the VT-2005-matched set it would close the separation at root-mean-square $0.49 \ln\gamma$ under the maximum-likelihood binning but at only ≈ 0.21 under the Bessel-corrected one. The matched IDAC is essentially single-method (158/165 gas chromatography), which makes a within-method systematic of 0.5 implausible; one of 0.21 is not, so on the corrected binning that threat is *not* excluded. The VT-2005-matched set's out-of-fold estimators cannot police it either: under random folds every held-out row shares its solvent ($47/60 = 78\%$ of rows) or its solute ($27/60 = 45\%$) with a training row, and under folds blocked on both molecules three of the four estimates (3.65, 1.83, and the divergent ridge) exceed the total MSE = 1.47, beyond which an upper bound on B_{insuff} constrains B_{closure} vacuously.

In-regime (γ-stratified) separation. The n=60 measurement is matched at low γ, while the paradox spans all γ. Stratifying the n=60 set by $\ln\gamma_2^\infty$ tests whether closure-dominance is a regime artifact. It is not: the ordering $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ (leakage-immune LOTV binning bound) holds in every γ stratum and *strengthens* toward higher γ (Table S16)—in the highest stratum ($\overline{\ln\gamma_2^\infty} = 2.94$, approaching the non-ideal regime) $B_{\mathrm{closure}} \geq 2.68$ exceeds $B_{\mathrm{insuff}} \leq 0.67$ several-fold. The high-γ strata are small (n = 17–30 in 102 dimensions): the law-of-total-variance bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the several-fold separations rather than the precise values that carry the qualitative reading. The bound here bins each stratum into three equal-count bins (against eight in the main text) and uses the maximum-likelihood within-bin variance, as in the headline row of Table S7; recomputed with the unbiased (Bessel) one it rises to 0.08/0.95/0.81 across the three strata, so each still separates, but the middle stratum only marginally (0.95 against its own MSE/2 = 0.96). The $m >$ med. stratum separates through its upper (high-γ) half, and the *isolated* moderate band (med.$< m \leq 1.5$, n=13) does *not* separate. This is not a counter-example. The total error there is only ≈ 0.06, so $B_{\mathrm{closure}} \leq \mathrm{MSE} \approx 0.06$ assumption-free (since $B_{\mathrm{insuff}} \geq 0$), and there is nothing to attribute—near ideality the target is small and the split is uninformative. We read no closure-quality claim from it in either direction. This supports—but does not prove—the conjecture that the same closure misspecification drives the all-γ paradox.

Table S16: Closure/insufficiency verdict across $\ln\gamma_2^\infty$ strata of the matched set (deployed residual-only convention; $B^{\mathrm{up}}_{\mathrm{insuff}}$ is the leakage-immune LOTV binning bound throughout, at the maximum-likelihood within-bin variance).

Stratum	n	$\overline{\ln\gamma_2^\infty}$	$B_{\mathrm{insuff}} \leq$	$B_{\mathrm{closure}} \geq$
lower γ ($m \leq$ med.)	30	-0.40	0.07	0.95
higher γ ($m >$ med.)	30	$+1.90$	0.85	1.07
high γ ($m > 1.5$)	17	$+2.94$	0.67	2.68

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §3.5. Figure S3 shows the shape of that error on the VT-2005-matched set: the reference-σ closure's parity plot against the measured $\ln\gamma_2^\infty$.

Four qualifications on the decomposition (full). Four qualifications bear on the decomposition of §3.5. None of them inverts the sign of the VT-2005-matched set's aggregate margin; each sharpens its scope. That aggregate

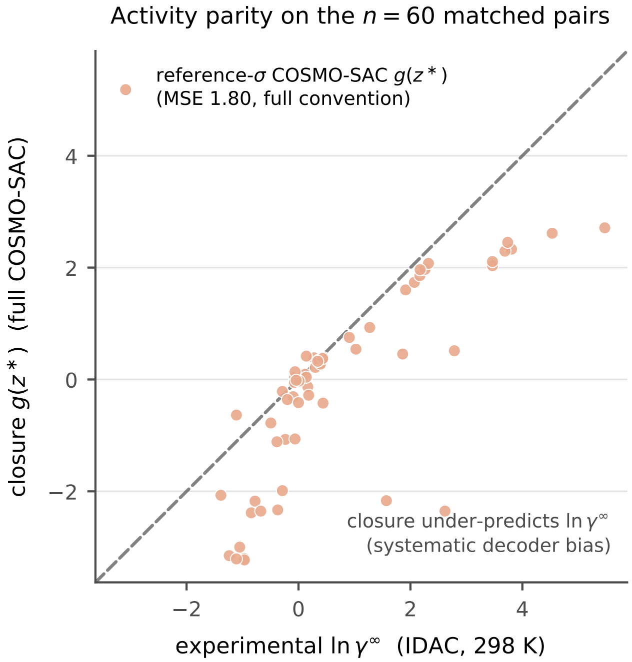

Figure S3: Activity-coefficient parity on the n=60 matched pairs. The reference-σ COSMO-SAC closure $g(z^\star)$ (salmon) falls below the diagonal at larger $\ln\gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention).

is retired on other grounds (§3.5.1–§3.5.2), so these are recorded as what the four choices do and not as robustness passes for a live verdict.

1. The deployed combinatorial default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.71$; it separates only under the full convention, $B_{\text{closure}} \geq 0.72$ against $B_{\text{insuff}} \leq 0.61$), so that assumption-free bound does not separate under deployment; Table S7 carries that row for both sets, and the full-convention margin (the sharpened MSE$-$$B_{\text{insuff}}$ bound, 1.19 vs 0.61) is a full-COSMO-SAC-only result. What separates under the deployed convention is the leakage-immune bound $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B^{\text{LOTV}}_{\text{insuff}} \approx 0.76 > 0.71$, which carries the headline claim, together with the combinatorial argument above. Every B_{insuff} ceiling quoted in this item is the unbiased within-bin variance, the headline setting of Table S7; the maximum-likelihood pair (0.85 against 0.62) is the demoted variant and is not quoted here.

2. The closure error is predominantly a *between-solvent* systematic bias ($\sim$64% of total error); within the largest single-solvent stratum (pyridine, n=20) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of $z^\star$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).

3. Confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The leakage-immune margin (0.76 vs 0.71) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. The 60 matchable pairs are the low-to-moderate γ regime (matched mean $\ln\gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln\gamma_2^\infty$ at 298 K, so the numeric ceiling does not necessarily transfer unchanged to the finite-composition solubility regime.

S3.4 A pre-declared attempt to tighten the insufficiency bound, and its failure

What was declared, and when. In the solubility domain $B^{\text{up}}_{\text{insuff}}$ is a coarsening throughout: the within-bin variance of m over eight equal-count bins of the scalar $g(z^\star)$, whose slack is the information the binning destroys and is not estimable from these data (§2.5). The other domain does not coarsen—in the pK_a closure the same quantity is the out-of-fold residual mean square of a cross-fitted forest of m on $z^\star$ (§S8)—and because every margin in the map is $\mathrm{MSE}-2B^{\mathrm{up}}_{\mathrm{insuff}}$, a regressor that predicts m from $z^\star$ better than an eight-bin histogram of a scalar summary of $z^\star$ does would raise all of them at once. The experiment that asks whether it does was pre-declared, in the header of the estimator itself (sha256 `d158989fe9cfdc79…`, the full digest deposited beside it), written and staged in version control before any cross-fitted margin existed. It fixed in advance: the feature set ($z^\star$ and nothing else—the two σ-profiles, their areas and T, 105 dimensions—with v_{cosmo} excluded because it is an argument of the full convention only, so conditioning on it would not bound the deployed convention's B_{insuff}); the regressor, the pK_a domain's own, untuned, with a ridge sensitivity that never headlines; the folds, five-fold grouped on the molecule pair, with a source-grouped variant reported beside every cell; the row set, 473 of the 477, the four rows whose solutes carry no VT-2005 profile dropped; an outcome-blind validity gate; a ban on forming the per-cell minimum of the two bounds; the rule for which estimator the paper would headline; and three verdicts. Nothing above the declaration was edited afterwards, and both bounds are deposited for every one of the 1164 cells.

It did not tighten. On the same 473 rows, broad set, row unit, primary folds, the cross-fitted bound is $B^{\mathrm{cf}}_{\mathrm{insuff}} = 0.856$ against the coarsening's 0.704 (deployed residual-only convention) and 0.489 (full)—looser by 22% and by 75%. It is looser having *passed* its own gate: no pair straddles a fold, the global out-of-fold R^2 is +0.823, and the value sits well above the deposited replicate floor $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})] = 0.046$. This is the estimator working and losing, not failing. The arithmetic of the loss is one comparison: with $\mathrm{Var}(m) = 4.85$ on these rows, a fitted bound beats the coarsening only above $R^2 = 1 - B^{\mathrm{bin}}_{\mathrm{insuff}}/\mathrm{Var}(m)$, which is 0.855 under the deployed convention and 0.899 under the full one. The forest reached 0.823.

Every consequence runs against this work, and the numbers are printed here for that reason. The glycol-ether margin—the paper's one surviving stratified finding—falls from +2.036 to +1.757 at the headline cell, its 90% two-way cluster interval from $[+1.26, +2.87]$ to $[+1.02, +2.49]$ and its leave-one-publication-out envelope from $[+1.597, +2.866]$ to $[+1.342, +2.339]$, on the same 182 rows either side—no dropped row carries a glycol-ether solvent. The envelopes are exact deletion extrema and print to three decimals; a bootstrap endpoint carries ± 0.02 of Monte-Carlo spread and prints to two (§S3.3, note *b*). The map is poorer in every count that reads a margin: admissible cells $35 \to 32$, strata admissible in all four unit $\times$ convention cells $6 \to 5$, and distinct admissible positive row sets after the maximality reduction $2 \to 1$, the alkanes being the row set that drops. The count of *boundable* strata does not move, 15 of 59 under both estimators, because boundability is the $n \geq 40$ test and the declaration froze it. The retired whole-set aggregate does not return: its margin falls from +0.944 to +0.211 (full) and from +0.506 to +0.203 (deployed), and it now fails the deletion test outright, two publications flipping its sign. Nor does the chemistry-blind multiplicity null improve where it matters. *Both* arms of that null are re-run on the 473 rows, so the comparison is like for like and its coarsening column is not the one the rest of the submission prints: $p(C)$ moves from 0.297 to 0.255, but $p(D)$, the null p of the largest certified margin, moves the wrong way, $0.094 \to 0.137$; conditioned on drawing a map that certifies anything at all, a chemistry-blind relabelling reaches the observed maximum in 54% of draws against 16% under the coarsening. The declared verdict is therefore LOSS on branch L1, with L2 firing independently. Restricting $477 \to 473$ rows leaves the map itself untouched—the four dropped rows leave the 477-row baseline's 35/6/2 and its +2.036 exactly where they were—but it does move the null: on all 477 the same coarsening gives $p(C) = 0.286$ and $p(D) = 0.1005$, that is 17% of certifying draws. Every statement of this null outside this subsection is that 477-row one.

A deviation from the declaration, named as one. The declaration's Sec. 3 fixed an unconditional rule—"THE RULE. The cross-fitted estimator headlines UNCONDITIONALLY … provided it passes the validity gate of Sec. 3.1, which never reads a margin, an MSE, a g or a stratum label"—and spelled out what it committed to: "If the cross-fit is LOOSER than the coarsening, the headline margins SHRINK, cells that are certified today lose their certification, and the map gets poorer. That outcome is reported as the result, in the paper, with the same prominence a gain would have had." **We did not follow that rule.** The coarsening remains the headline estimator, and every margin printed in this paper is the coarsening's. The reason is that the rule was written against one hazard—choosing the smaller bound cell by cell after seeing the cells, which is biased low and which the paper disowns elsewhere—and it did not anticipate the other, that one valid bound would be uniformly *looser* than the other. Where that happens, obeying the rule prints a bound that is valid and strictly less informative: every claim weakens and none becomes more correct. What we did instead is what the declaration's Sec. 2 already required independently—both bounds, both margins and their difference are printed for every cell of the deposited map, the cells that favour this paper and the cells that do not, so that the poorer map is legible in the deposit and every number in the paragraph above rederivable from it. Two things about this deviation should be said rather than absorbed. It runs in this paper's favour, which is exactly why it is named here instead of being folded into a methods sentence. And it is a choice made *after* seeing a number, which is what the declaration existed to prevent; what limits it is that the number is one global comparison of two bounds on one row set, not a per-cell selection, and that the alternative is on the page.

Why the coarsening wins here, and where it would not. The mechanism is measurable, and it is not the obvious one. It was established on synthetic data matched to the real design—477 rows in 185 pairs from 15 sources, the real rows-per-pair multiplicities, real σ-profiles drawn from the deposited 1432-row artifact so the regressor faces the real collinear geometry, and B_{insuff} known in closed form by construction. The coarsening is handed g: a physics-built statistic that was designed to predict m and is close to sufficient for $\mathbb{E}[m \mid z^\star]$, so binning it starts from the answer. The forest is handed $z^\star$ and must rediscover g from about 145 of the 181 pair groups in each training fold. *It is not a dimensionality effect*: coarsening the regressor's features to the leading unsupervised principal components of $z^\star$—a legitimate operation on the bound, since a smaller σ-field can only raise $\mathbb{E}[\mathrm{Var}(m \mid \cdot)]$—never helps. On the synthetic design the raw 105 dimensions give the tightest cross-fitted bound of the six feature sets tried (median 3.39, against 3.71 at eight components, 4.04 at sixteen, 4.59 at four, 5.00 at two and 6.10 at one), and all six sit above the eight-bin coarsening's 2.3–2.6 on the same draws. What separates the two estimators is sample size alone: the ratio of the two bounds falls from 3.2 at n=40 to 1.24 at n=960, and a log–log extrapolation of the two curves crosses near $n \approx 2 \times 10^3$ rows, three to four times the 473 actually scored (1.7×10^3 fitting all eight sizes, 2.0×10^3 omitting the two smallest; it is an extrapolation and is quoted as one). The same account explains why the cross-fitted estimator is the right one in the pK_a domain and is not an inconsistency between the two: there $z^\star$ is four series labels and a scalar, $\mathbb{E}[m \mid z^\star]$ is estimable directly within a scaffold at the sample sizes available, and the closure confers no head start because there is no high-dimensional statistic left for it to summarise (§S8). One further asymmetry is worth recording because a maximum-over-strata statistic inherits it: across the 15 boundable strata the coarsening's bound has mean 0.315 and standard deviation 0.213, the cross-fitted one 0.658 and 0.519, and the same ordering holds under a chemistry-blind relabelling (0.241 against 0.553, median over draws), so the extra spread is a property of the estimator and not of the chemistry—which is why $p(D)$ worsened while the bound merely loosened.

The design rule this yields. For anyone reusing this instrument on another closure, the choice of $B^{\text{up}}_{\text{insuff}}$ is not a choice between an old estimator and a modern one. Both are valid upper bounds; the choice is about power, and it turns on three things that are known before any margin is formed. *(i)* If the closure's own output $g(z^\star)$ is available as a feature and is near-sufficient for $\mathbb{E}[m \mid z^\star]$—the normal case when g is a physical model built to predict m—coarsening g begins from a statistic the regressor would have to learn, and it wins at every sample size that does not let the regressor learn it. *(ii)* If $\dim(z^\star)$ is small enough that $\mathbb{E}[m \mid z^\star]$ is directly estimable—a handful of labels and a scalar, as in the pK_a domain—condition directly and do not coarsen, since the coarsening would then discard information for no return. *(iii)* Between those poles the crossing is a sample-size question with a computable answer: fit both and prefer the regressor only if its out-of-fold R^2 exceeds $1 - B^{\text{bin}}_{\text{insuff}}/\text{Var}(m)$, the requirement being a function of $(m, z^\star, g)$ that reads no MSE and no margin. Here that requirement was 0.855 and the forest returned 0.823, which is the whole of the result. The rule must be declared before the margins are read; choosing per cell between two valid bounds after seeing them is biased low, and choosing once and globally after seeing them, as we did above, is a disclosed deviation and not a method.

The fold warning, which is the part that can invalidate a bound rather than merely loosen it. The cross-fitted quantity is an upper bound only if the regressor is independent of the held-out row's label, and on clustered data that is a property of the fold scheme, not of the estimator. This was measured before the real rows were touched, against known truth. With pair-grouped or source-grouped folds the estimator is a genuine upper bound: 0 violations in 636 cells over the declared headline scope—fit once globally, evaluate on label-blind sub-strata, both grouped schemes, random and chemically coherent sub-strata alike—including all 100 cells at n=40, the map's own boundability threshold. With row-level folds it is not a bound at all: 9.4% of cells (18 of 192) fall below truth in the realistic regime, $B_{\text{insuff}}/\text{Var}(m) \leq 0.30$, and 71.9% (46 of 64) at 0.60. In the null design, where m is independent of $z^\star$ and the truth is $\text{Var}(m) = 4.85$, row-level folds return 1.73 under a pair shock and 1.42 under a source shock—64% and 71% *below* truth. The numerical gates do not catch this. The leaky version returns out-of-fold $R^2 = +0.61$ and $+0.64$ in exactly those two cells and clears the replicate-floor check by a factor of thirty; the only gate that fires is the assertion that no group straddles a fold, which is a statement about the folds and not a number. **One deposited number is compromised by this and must not be cited as a bound**: the deposited decomposition artifact records an out-of-fold forest upper bound of 0.5646, fitted on row-level folds, with no grouping. Nothing in this paper rests on it: Table S9 carries the audit artifact's 0.621 instead, and the disagreement between the two forests is reported above. The deposit now carries the warning in the file itself, beside the number. The scheme's exposure differs by set and both readings reach the same verdict: on the VT-2005-matched set where that number lives no two rows share a pair, but 60 rows carry 41 solutes and 17 solvents, so every held-out row has a near-twin in training; on the broad set the pair channel is open outright, 359 of the 477 rows sharing a molecule pair with another row (§S9), and blocking it moves the same forest from 0.15 to 0.88. The cross-fitted map, its per-cell table, its multiplicity null, its within-stratum sensitivity, its verdict and its synthetic validation are all deposited, together with the declaration above, the scoring pass and the validation run that produced them.

S4 External baselines and ablations

Reading Table 4 across its three blocks, in full. The external block is not like-for-like and the table says so in its own rows. FastSolv and SolProp are deposited here on a *by-solute* split, this work's arms on the harder solute-scaffold split; retraining them on the scaffold split needs a GPU run we did not perform, and scoring released FastSolv weights on our test rows is invalid because FastSolv is trained on BigSolDB, which is this corpus. The third block is as published, on other corpora, with $\log_{10} S$ in molarity against mole fraction here, so the dimension matches—through the conversion the caption defines, which puts our S per litre of *solvent* where the quoted block's basis is unstated—and the task does not. One sentence survives all three caveats, and it is not a favourable one: this work's control, at $\log_{10} S$ RMSE 1.00 on held-out scaffolds, is the same order as published new-solute extrapolation rather than an order away. The published leaders run 0.83 to 0.99 and the physics-informed cycle 1.43 to 2.16, so the control's seed mean sits just past the worst leader—at one of its three seeds (0.97) it is under it; the physics arms, at 1.39 to 1.60, are worse than every published leader at every seed and every clip. The NRTL and grounded arms also trail their control by more here—an RMSE deficit—than in MAE, at every clip of the caption's sweep and at every seed; the σ-oracle does not, and §S4 says where. Where they sit *within* the cycle's range is not a reading this column supports: it moves with the saturation clip the caption defines—the control's figure does not move with that clip and all three of theirs do—and at the tight end of that sweep the NRTL and grounded arms fall below the cycle's better end at every seed, the σ-oracle in its seed mean and at two of its three seeds but not at seed 43. They are also worse, in this column, than predicting one number for the entire test set: the scale reference reads 1.33 on the same rows, below all three printed figures. That is a reading of the seed-mean column at the deposited clip and not of the models—at seed 42 the grounded arm is below the reference at every clip, 1.13 against 1.33 at the deposited one, and seed-averaged the grounded and NRTL arms fall under it at the tight end as well; the caption counts the cells. What may be read from the three figures is what survives the sweep—all three above every published leader and above their own control—and not the two decimals. The 0.75 aleatoric estimate[S12]—itself a different functional on a different base from this paper's own floor (§2.2)—sits below every row of the table but two, the closest published figure being the leaders' 0.83; both exceptions, at 0.61 and 0.45, are on the random-pair split, where the same solute appears in training and the task is not extrapolation. The two arms run on both splits move the way the split predicts: the FastSolv retrainings score MAE 1.44 in $\ln x_2$ by solute against 1.02 on random pairs, the retrained SolProp encoder 1.62 against 0.64. That FastSolv pair is re-scored, and it replaces one that read the other way round. Both runs had been scored through a prediction path of ours that standardises the model's inputs twice, which saturates the network's input activation on temperature and leaves the predictions exactly constant in T; the fitted checkpoints were unaffected, so the repair is an inference pass on those same checkpoints and not a retraining (§S4). Re-scored, the by-solute row reads $\log_{10} S$ RMSE 0.84, inside the published 0.83 to 0.95 but on an easier split than this work's arms are held to.

We evaluated the direct state-of-the-art model (FastSolv[S12]) and the physics-informed state-of-the-art model (SolProp[S13]) on a by-solute test split ($n \approx$ 10–12k). The two arms are not obtained the same way, and the difference matters below: FastSolv is retrained on our corpus, whereas SolProp is the released pretrained thermodynamic cycle—the arm that instantiates the physics—mapped onto our target by a three-parameter linear recalibration fitted on the training rows, not by retraining. That cycle is run at 298 K on every row, so the temperature of a measurement reaches the SolProp arm only through the calibrator's temperature term. In $\ln x_2$ they reach MAE 1.44 (FastSolv, R^2 0.55) and 2.34 (SolProp, calibrated, R^2 −0.18); in $\log_{10} S$, MAE 0.65 and 1.04, the same rows reading RMSE 0.84 and 1.34 in Table 4's column. These are on a *by-solute* split that is *not comparable* to our scaffold numbers (Table S3): they bound the task's difficulty, not a ranking of the models against one another, so this does *not* establish that our scaffold-split DirectGNN (1.70) outperforms them. A like-for-like scaffold evaluation of the external models was not run, the deposited external runs being by-solute and pair-random only. The internal ordering—the physics-informed SolProp trailing the direct FastSolv, the same ordering we find between our physics closures and DirectGNN—is therefore a comparison of the *cycle* against a direct model. Discarding the cycle and retraining

SolProp's encoder directly on $\ln x_2$ closes most of that gap without closing it: that arm reaches MAE 1.62 (R^2 0.39) in $\ln x_2$ on the n=10,570 rows it predicts, and MAE 0.76 in $\log_{10} S$ on the 10,287 of those the conversion is finite on, against FastSolv's MAE 1.44 and 0.65 on 10,287 rows in both units, and both direct arms are well ahead of the cycle's 2.34. The two arms' $\ln x_2$ figures are therefore not on quite the same rows; Table 4 prints both sample sizes for the two rows of that block whose two metric columns disagree, and one figure for the four where they coincide. We quote the cycle as the physics-informed arm because only it assembles the separately learned components; the retrained arm measures what discarding the closure achieves on this corpus, not a better physics-informed model. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §3.7. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table S3.

The two FastSolv rows are re-scored, and what was wrong with the first pair. The bundles first written for these two runs were read through a prediction path that standardises the model's inputs *twice*. Our own inference wrapper scales solute descriptors, solvent descriptors and temperature with the checkpoint's own buffers and then hands them to the released package's prediction step, which scales the same tensors again; the training and validation steps do not re-scale, so the fitted models were never what was wrong. The network's sigmoid input activation then annihilates the twice-scaled temperature, which reaches it at about −19.3 rather than at the order −1 a single scaling gives, a sigmoid of 4×10^{-9}. The signature was unmistakable: those predictions were *exactly* constant in temperature—span 0 to machine precision—for all 1102 by-solute and 1149 random-pair (solute, solvent) pairs that span more than 5 K, where the measurements move a median 0.62 in $\log_{10} S$ over a median 40 K. What survived the double scaling was a systematic offset, and it was the whole of the anomaly: the by-solute run's mean $\log_{10} S$ error was −0.36 and the random-pair run's −1.12, and once each was centred the two runs' RMSEs were 1.04 and 1.07—the same dispersion, on both splits. The offset was present in validation too (−1.10), so checkpoint selection did not catch it.

Because the fault was in the scoring and not in the fit, the repair is one inference pass per saved checkpoint. A re-scoring pass forwards the same two checkpoints over the same rows with the single scaling the models were trained under, and re-runs the doubly scaled path beside it as a control. That control reproduces the withdrawn bundles to every digit they printed—by-solute MAE 1.940994 in $\ln x_2$ and RMSE 1.096639 in $\log_{10} S$ on 10,287 rows, random-pair 2.910154 and 1.544042 on 10,696—which is what establishes that the scaling is the only thing that changed. Re-scored, the by-solute run reads MAE 1.44 in $\ln x_2$ (R^2 0.55) where the withdrawn bundle read 1.94, and the random-pair run 1.02 where it read 2.91; the temperature response returns, the predictions spanning a median 0.500 (by solute) and 0.631 (random pair) in $\log_{10} S$ over those same pairs against a measured median 0.623, and the offset falls to −0.048 and −0.024. Both re-scores are on the deposited runs' own row sets, which the block's other rows use; the current data preparation resolves a solvent molarity for 100 by-solute rows more, and scoring those as well moves that row to 1.43 and 0.84. Nothing is retrained: the checkpoints, the descriptors and the row sets are those of the original runs, and the withdrawn bundle is kept beside the re-scored one. The SolProp arms go through a different path and always carried a temperature response: on the by-solute run their predictions span a median 1.31 in $\ln x_2$ across the temperatures of a pair, against a measured 1.35.

What the $\log_{10} S$ column owes to the saturation clip. Converting $\ln x_2$ to $\log_{10} S$ sends a prediction at $x_2 = 1$ to $+\infty$, so predictions are clipped at $x_2 = 0.999999$ first, the standing prediction-side constant of the released code. The constant is a choice, and three of the four block-1 cells of Table 4 are a function of it, because a saturating row's error is whatever the clip makes it. Sweeping x_2 over $1-10^{-3}$, $1-10^{-4}$, $1-10^{-5}$, $1-10^{-6}$ (used), $1-10^{-7}$, $1-10^{-8}$ and $1-10^{-10}$ gives, in RMSE $\log_{10} S$: DirectGNN 0.995 at every one of the seven; NRTL 1.233, 1.324, 1.429, 1.544, 1.667, 1.797, 2.072; COSMO-SAC-grounded 1.183, 1.242, 1.311, 1.387, 1.470, 1.558, 1.746; σ-oracle 1.421, 1.471, 1.531, 1.598, 1.672, 1.752, 1.925. Two readings hold across the whole span, and separately at each of the three seeds: the control is below every physics arm, and every physics arm is above every published leader (0.83 to 0.99). Four do not, and each of the four fails on the seed axis as well as the clip one. The NRTL and σ-oracle rows exchange places between $1-10^{-7}$ and $1-10^{-8}$ in the seed-averaged sweep, between $1-10^{-4}$ and $1-10^{-5}$ at seed 42, and not at all at seeds 43 and 44, where NRTL is the lower at all seven; their printed order is not a fact about the models. The scale reference predicts one number for the whole test set, does not saturate, and reads 1.326 at every clip and at every seed, the locked row set being the same 5440 rows at all three; the physics arms are under it in 15 of the 63 arm–seed–clip cells. Seed-averaged, the grounded arm is below it at the three tightest clips (1.183, 1.242, 1.311) and the NRTL arm at the two tightest (1.233, 1.324); the σ-oracle is above it at all seven. Per seed, the grounded arm is below it at all seven clips at seed 42 (at the deposited clip 1.125 against 1.326, the one cell of the printed column that beats the reference), at the tightest clip at seed 43 and at the two tightest at seed 44; the NRTL arm at the tightest clip at seed 42 and at the two tightest at seeds 43 and 44; the σ-oracle at no clip and no seed. So whether a physics arm beats that reference depends on the constant *and* on the seed: at the value we deposit the seed-mean column does not beat it, and one arm at one seed does. In MAE $\ln x_2$, by contrast, every arm including the σ-oracle is below the same reference (2.325) at every seed, so the reading is confined to this column. "Near the cycle's better end" (1.43) fails at the tight end, where the NRTL and grounded arms sit below it at every seed and the σ-oracle's seed mean (1.421, printed as 1.42) does too, though not its seed-43 value (1.479, printed as 1.48); §3.7 does not place the physics arms within that range. And the physics arms do not *uniformly* trail their control by more in this column than in MAE (a comparison of an RMSE deficit against an MAE one). The NRTL and COSMO-SAC-grounded arms do, at all seven clips and separately at each of the three seeds: the smallest such excess anywhere is +0.011, the grounded arm at seed 42 and the tightest clip. The σ-oracle does not. Its seed-averaged $\log_{10} S$ deficit over the control runs 0.425, 0.476, 0.536, 0.603, 0.677, 0.757, 0.930 across the seven clips against an MAE deficit of 0.549, so the reading fails at the three tightest clips and holds at the deposited one by 0.054; per seed it fails at all seven clips at seed 42 and at the two tightest at seeds 43 and 44. §3.7 therefore states that reading of the other two arms only. A fifth reading fails on the seed axis alone, and neither document carries it: the grounded arm is the lowest of the three physics arms at every clip in the seed-averaged sweep above, but not at every seed. At the deposited clip the per-seed values are, at seeds 42, 43 and 44, grounded 1.125, 1.558, 1.478 against NRTL 1.643, 1.472, 1.515; at seed 43 the NRTL arm is lower at all seven clip values. It is a property of the seed-mean column and not of the models. What may be carried away from that column is what survives the sweep, and that is the two readings above and not the printed decimals.

The unit bridge behind the $\log_{10} S$ column. $\log_{10} S$ is obtained from $\ln x_2$ as $S = x_2 c_{\mathrm{solv}}/(1-x_2)$, the solvent's molarity c_{solv} in $\mathrm{mol\,L^{-1}}$ being $1000\rho/M$ from a tabulated *mass* density ρ ($\mathrm{g\,cm^{-3}}$) and the molar mass M ($\mathrm{g\,mol^{-1}}$) where the solvent has one, and $1000/V_{\mathrm{m}}$ from the molar volume V_{m} ($\mathrm{cm^3\,mol^{-1}}$) of one generated conformer where it has none (this ρ is unrelated to the Hammett reaction constant of §3.6). Since $x_2/(1-x_2)$ is a mole ratio, S is moles of solute per litre of *solvent* rather than of solution, and the two conventions diverge as $x_2 \to 1$. The same converter runs on every row of the first two blocks of Table 4. It carries *no* temperature dependence—one ambient c_{solv} at every T—and it needs a molarity, so the 168 of the 5608

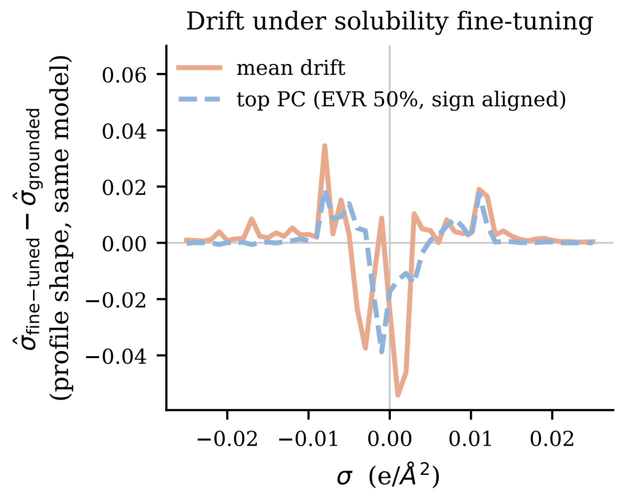

Figure S4: The drift solubility fine-tuning imposes on the learned σ-profile (n=44 VT-2005-matched molecules): $\hat{\sigma}$ after fine-tuning minus the *same model's grounded profile*, not minus the external reference. Shown are the mean drift and its dominant PCA mode, defined only up to sign and drawn in the mean drift's direction; the legend's EVR is that mode's explained variance ratio.

labelled test rows whose solvent has none carry no true $\log_{10} S$ and leave that column for *every* arm, which is why that table's rows carry two sample sizes. Predicting the mean of the true $\log_{10} S$, rather than running the constant-ln x_2 scale reference through this bridge, would read 1.28.

S5 Supplementary mechanism results

The latent's departure from its reference, in full.

The drift under fine-tuning. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning (the σ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; it is repeated over *three seeds* and reported mean±sd. Those three runs are not the arms that produce the paradox: they are a separate set, trained with the σ head left unfrozen through fine-tuning, where the four COSMO-SAC arms of §3.1 hold the head from P2 onward (Table S3). The freeze holds the head and not the encoder feeding it, so those arms' profiles are not pinned at their warm-up value either; how far *their* latent drifted is not measured, and the deposited per-arm predictions carry no σ-profile, so it cannot be recovered from them. What follows is therefore measured on runs of its own and is not established of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference profile to within 36±1%, so the intermediate *can* carry the physical shape, though only as an in-sample fit: all 44 probe molecules lie inside the σ-grounding pool. Solubility fine-tuning then drives the profile 41 ± 3% off the grounded one, for a total departure from the reference of 51 ± 2%. The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry 72 ± 1% of the variance. The quantity read against a null is a different one, the top-two share of the learned-versus-reference deviation, 0.399; its null is *phase-randomised*, matching each deviation's own smoothness and destroying only the between-molecule alignment at issue, and the observed share clears it (0.399 against 0.221, $p < 0.001$), at roughly 1.8×. The 72±1% mean-centred share has no matched null of its own, a wrong-molecule null built from real reference profiles does *not* clear (0.772, p = 1.00), and if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting. To test whether the structure transfers across molecules we fit one distortion operator D on half the 44 molecules and score it on the other half, where it predicts the drift 4.1±0.5× better than the zero-drift (identity) null out of sample. That null is deliberately weak, nothing controls it on these three seeds, whose checkpoints were not retained, and a molecule-independent constant offset reaches 2.1× on the one companion run where that control can be computed, so the ratio stands against the identity null alone; the held-out half is homologue-saturated. Structure *beyond* an offset therefore rests on the mean-centred spectrum, not on the ratio—and that spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape. On the reference ladder of §S5 the drift consumes essentially all the molecular specificity grounding had provided, as such a collapse would also look.

Dominant mode, and the direct cancellation test. The dominant drift mode (Fig. S4) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings—the mode being defined only up to sign and drawn there in the mean drift's direction: fine-tuning makes the network represent molecules as more polar than its own grounded profile does, that profile and not the external reference being what the plotted drift is taken against. Whether that shift *cancels* part of B_{closure}—the input compensating the map, as the non-identifiability of the split allows (Lemma S1)—is the comparison $\text{MSE}(m, g(\hat{\sigma}))$ against $\text{MSE}(m, g(z^{\star}))$ on the same pairs. It does not support the cancellation reading. Holding the cavity area at its reference value so that only the profile shape varies, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint gives 18.75 against 1.47: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure, not better. Three limits sit on that number. The checkpoint is the uncontrolled one, whose departure (82%) is half again the three-seed 51 ± 2%. The axis is wrong: the drift is produced by the finite-composition solubility objective and scored here at infinite dilution. And the check has almost no power in the direction that matters, so it separates "no cancellation" from "large cancellation" but not from "some cancellation". It is the only direct evidence available, and it is evidence against rather than a clean pass or fail. A second, independent check on the closure's own axis points the same way: a *more* polar profile moves g further from m (§S5). Both sit on that same infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift; they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. What is established is weaker and still consequential: the model's "physical" intermediate is not the molecule's charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,[S14,S15] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[S16] Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above.

The activity map's Fisher geometry, and why σ is under-determined. At infinite dilution the aggregate activity Fisher $J^{\top}J$ ($J = \partial \ln \gamma_2^{\infty}/\partial\sigma$) is rank-one (participation ratio 1.0 by autograd; finite differences decorrelate the redundant flat-profile coordinates and inflate it, §S6.8, so we read PR against a permuted control and not absolutely). The per-molecule Fisher that governs the drift has participation ratio ≈1.4 of 51 even at full solvent coverage. The participation ratio is a concentration statistic and not an exact count. It is measured at infinite dilution, so its transfer to the finite-composition regime that generates the drift is not established, the same scope gap that qualifies B_{closure} (§3.7). The external reference is itself *computed* (VT-2005 quantum chemistry) rather than experimental, so none of this establishes that recovery of the physical profile is impossible in principle.

Why the top-two share is read against a phase-randomised null, and how unstable the magnitudes are. A top-two explained-variance share is not interpretable against an isotropic null, because smoothness alone concentrates a spectrum: smoothing isotropic noise reaches a top-two share of 0.25 to 0.83 as the correlation length grows from one bin to eight, spanning the observed value without any between-molecule alignment at all. The *phase-randomised* null of §3.3 preserves each deviation's amplitude spectrum exactly—and so its smoothness—and destroys only the alignment across molecules that a shared compensating direction asserts; the observed 0.399 against its 0.221 ± 0.011 is therefore a statement about alignment and not about smoothness. The wrong-molecule null (0.772 ± 0.023), which the observed share does *not* clear, is a difference between two points on the reference manifold and inherits that manifold's own concentration, so it demands more alignment than a shared direction implies. That is the argument for preferring the phase-randomised one, and §3.3 states that if it is not accepted the spectrum is uninformative rather than supporting. In each case the $\pm$ is the null's spread over resamples, not an uncertainty on the observed value. The three magnitudes §3.3 reports but does not carry move in both directions across analysis configurations: a shortened run gives only $\sim$16% departure and a transfer of 1.7×, below what a constant offset alone reaches on the one run where that control can be computed, while uncontrolled two-model comparisons over-state both the departure ($\sim$82%) and the transfer (16–26×). With n=3 the $\pm$ on the three-seed figures is indicative and not a precision. Nor does mean-centred low-rank structure exclude a shrinkage of each profile toward the corpus-mean shape: the reference profiles themselves sit at 0.77 mean-centred, so any drift linear in the profile inherits that concentration.

The drift spectrum, and the null it lacks. Figure S5 is the spectrum of the drift whose mean and dominant mode are Fig. S4. Its top two modes carry 72.9% in the companion run plotted there and 71.8 ± 1.3% over the three seeds (quoted as 72 ± 1% in §3.3): the same statistic on different runs. The companion run's spectrum and the three-seed aggregate are both deposited; the aggregate's top-two share has mean 0.71819 and standard deviation 0.01265 over seeds 0–2. Both this figure and Fig. S4 are drawn by one deposited script. That share has no matched null, so the panel is titled by what it plots and not by an inference. The phase-randomised and wrong-molecule nulls of §3.3 are run on the $\hat{\sigma}$-versus-reference deviation instead, a different quantity; the isolation drift has not been re-tested that way, because the two locally available checkpoints share a σ head and so have identically zero drift between them.

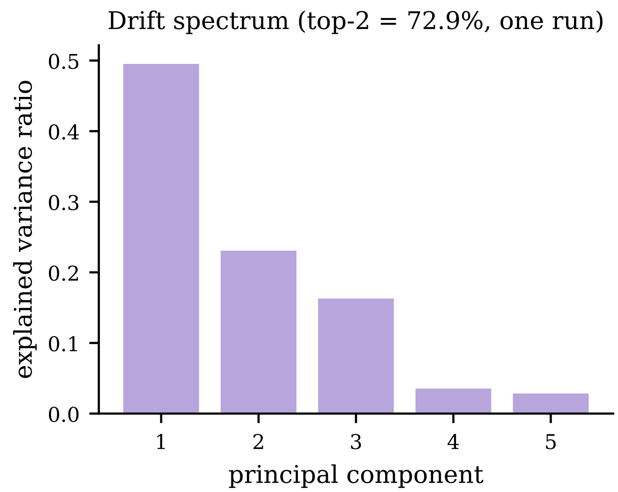

Figure S5: Explained-variance spectrum of the fine-tuning drift on the 44 matched molecules (companion run). *Caveat:* this share carries no matched null (text).

The grounded warm-up checkpoint's 0.73, and the power of the two-MSE check. The direct cancellation check of §3.3 has almost no power in the direction that matters. On the VT-2005-matched set, 18.75 against $\mathrm{Var}(m) = 2.47$ is 7.6× worse than predicting the mean, the signature of a broken profile rather than of a partly compensating one, so the comparison separates "no cancellation" from "large cancellation" and not from "some cancellation". The same comparison run on a grounded warm-up checkpoint, whose profile sits 28% from the reference, scores 0.73 where the end-to-end SLE-trained checkpoint scores 18.75. That is a statement about the closure's sensitivity to profile roughness and about an in-sample fit to $z^{\star}$ on molecules inside the grounding pool, not evidence that the encoder carries information the reference lacks, and we do not read it as a both-channel reproduction of the paradox.

The closure's own-axis check on polarity. The second check of §3.3: on the n=60 VT-2005-matched set the reference-input closure *under*-predicts, its mean output being +0.10 under the residual-only convention and −0.10 under the full one against a mean target m = +0.75. Sweeping its hydrogen-bond coefficient at zero, one and two times the standard value gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs: raising the coefficient makes them worse and zeroing it makes them better, so a *more* polar profile moves g further from m, which is the direction opposite to a compensating drift.

The reference ladder. A relative deviation carries no absolute scale, which is what the ladder supplies. Its rungs are built from the reference profiles the grounded arm was supervised on, so it calibrates that deviation rather than competing with it. The rungs calibrating the 51% departure of §3.3, all on the same 44 molecules and the same relative-L_2 metric: a leave-one-out corpus-mean profile (one fixed shape reused for every molecule, carrying no molecular information) 0.493; each molecule assigned another molecule's reference profile, 0.651–0.684 over five derangements; a uniform profile 0.889. Against these, the grounded base sits at 0.364 and the fine-tuned latent at 0.507. The grounded base is the σ-supervised recovery §3.3 reports as reproducing the reference profile far more closely: it is $36 \pm 1\%$ over the three seeds in the norm ratio $\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$, and it is an *in-sample* fit—all 44 probe molecules lie in the σ-grounding pool and the warm-up early-stops on it—so no out-of-sample recovery estimate exists on this set.

The distortion operator D. The transfer test of §3.3 fits a dense affine map on the 51-bin grid: a 51×51 linear part plus a 51-bin offset, 2652 parameters, each output bin a ridge regression of 52 coefficients (ℓ_2 penalty 10^{-3}) on 22 of the 44 molecules. The constant-offset control replaces D by a single molecule-independent shift of the whole profile, fitted on the same 22 molecules and applied unchanged to the held-out 22. It can be computed on one companion run only—the two-model comparison whose two checkpoints are both still on disk—and there it is 2.1× better than the identity null on both of that run's arms. That run is the uncontrolled one, whose own transfer of 16–26× §3.3 reports as over-stated, so its residual factor is not read across. The three seed checkpoints behind the $4.1 \pm 0.5\times$ were not retained, and the remaining companion run records only an ephemeral checkpoint path, so no offset control exists for either and the 4.1× is a ratio against the identity null with no constant-offset control of its own. The homologue saturation that makes "held out" largely within-cluster interpolation is six n-alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44; the median held-out molecule sits 0.048 in L_2 from its nearest training-half neighbour, about 16% of its own profile norm.

Non-identifiability, empirically (the corrupted twin). Training a twin model on data whose crystal labels are corrupted (every T_m shifted by +273 K) leaves the *solubility* fit essentially unchanged—$\ln x_2$ MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin (T_m MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma S1 made concrete.

Crystal grounding on physical labels. When the crystal head is supervised on measured melting points it predicts T_m to 45 K MAE, about 4× better than a Joback group-contribution reference—so the flat downstream result of §S6.3 is not an inability to learn crystal properties. That experiment's external pool was 97.76% melting points the model was already trained on.

The $\Delta C_p = 0$ approximation. The ideal term omits the constant-heat- capacity correction; a group-contribution audit finds a nonzero ΔC_p for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ($\ln \gamma_2^\infty$, at 298 K on the VT-2005-matched set and at each row's own temperature over 273–403 K on the broad IDAC set), where ΔC_p does not enter at all on either set: the ideal term is not evaluated in the decomposition, so no temperature range of it is at issue. The grounding paradox is an arm-to-arm difference (learned-σ vs. oracle-σ) whose two arms share the *identical* crystal term, so any ΔC_p error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

Fusion-supervision scarcity. Direct ΔH_{fus} labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative basis of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

The NRTL branch trains (a trainability check). The asymptotic physics penalty is not an under-training artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters—τ_{12} over the n=8103 test pairs has mean 0.97 and std 2.5 (range $[-2.6, 22]$), far from collapsed—yet the arm still trails the closure-free DirectGNN in the seed mean (1.79 vs 1.70; Table S3), at two of the three seeds and not at seed 42, where it is ahead (1.734 against 1.749). A trained, non-degenerate activity branch that the control outperforms at two seeds of three is evidence that the penalty is structural rather than a failure to train the branch, and weaker evidence than a clean sweep of the seeds would be. (The core closure-misspecification result—the σ-oracle and the decomposition—is in any case training-*independent*: it feeds the reference VT-2005 profiles through the fixed closure with no encoder optimisation, so it cannot be a training pathology.)

Structured temperature extrapolation. Withholding *temperature* rather than structure (train at $T \leq 309.75\,\text{K}$, test at higher T; $n = 3343$), the affine-in-$1/T$ hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table S17); the van't Hoff class also predicts the *direction* of the high-T shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class and not an accuracy claim for our model.

It is also not a claim in the other direction, and the reason is a defect in the evidence rather than a result. The one TGNN checkpoint ever scored on this split predicts, statistically, a constant: prediction standard deviation 0.58 against a measured 2.56, and an MAE of 1.95 against 2.00 for the best global constant and 1.46 for the per-pair constant already in Table S17. Its activity and correction outputs never left initialisation—$\ln\gamma_2$, τ_{12}, α, the gate and the correction magnitude have standard deviations between 4×10^{-6} and 6×10^{-4} over the test rows—and it was trained at a one-epoch first phase, on the pre-repair crystal labels of the corrupted-twin paragraph above. What it measures is the trainability of the bottleneck at that budget, not the temperature behaviour of a trained physics path, so no trained arm appears in Table S17 and none should be inferred from its absence. Settling the question needs the closure-carrying and closure-free arms retrained on this split at the budget used everywhere else here, read on the *bias* of the fitted per-pair slope as well as its dispersion; that is a GPU run we have not made.

Table S17: Same-pair high-temperature extrapolation ($n = 3343$, train $T \leq 309.75\,\text{K}$). "dir." is the fraction of pairs whose high-T shift direction is predicted correctly.

Model	MAE	R^2	dir.
per-pair van't Hoff ($1/T$)	0.37	0.89	0.99
per-pair linear in T	0.41	0.85	0.99
per-pair last-low-T hold	1.20	0.69	0.01
RF (Morgan $+T$)	1.29	0.66	0.40
per-pair mean	1.46	0.56	0.01

S6 Supplementary diagnostics and broad negatives

S6.1 Supporting diagnostics

The interpretive claims of the preceding sections are based on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that are unfavourable to the physics-informed hypothesis.

Aleatoric noise floor. The derivation, the two aggregating functionals and their values are in the article at §2.2. One property of the per-group distribution is not: it is strongly right-skewed, which is why the mean and the median differ by a factor of five. On the groups backed by two sources the 90th percentile of the per-group disagreement is 0.42 $\ln x_2$ units and the 99th is 1.41; on those backed by three, 0.83 and 3.55. The floor's mean is therefore set by a minority of badly disagreeing systems rather than by a broad band, and on either reading it lies far below the scaffold-split error.

A linear probe on the encoder, and what it does not establish. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a ridge probe from the encoder representation onto fifteen RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.310. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23), so the train–test gap partly reflects covariate shift in the descriptor *targets* across scaffolds rather than an encoder property alone, and the probe measures descriptor-representability and not task-relevant expressivity.

That gap does not establish that the encoder is transfer-limited rather than expressivity-limited. The probe carries no control that would separate the two, and it does not survive one. An untrained, randomly initialised encoder of the same architecture scores 0.707/0.415/0.291 on the same three quantities and is awarded the *identical* verdict by the same decision rule; trained-minus-random bootstrap intervals cross zero on all three; a second random initialisation moves the gap to 0.194 and flips the rule's verdict string, so the 0.25 threshold that rule turns on sits inside its own null's spread. Nor is a high train R^2 evidence that the embedding is expressive: isotropic Gaussian features at the same $d = 768$ and $n = 700$ reach a train R^2 of 0.979, above the trained encoder's, which is ridge geometry at $d \approx n$ and not chemistry. What the probe establishes, once the random-encoder arm is subtracted, is only that training adds about 0.11 in R^2 of linear descriptor decodability on train solutes and 0.00 ± 0.07 on test. Whether a new graph encoder would move out-of-distribution accuracy is therefore *unmeasured* here rather than settled; the expectation that pretraining and coverage are the more likely levers is cited to the arm comparisons and not to this probe.

Temperature extrapolation. A structured van't Hoff activity class (log-solubility affine in $1/T$) encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this is apparent when the split withholds temperature rather than structure. Training on measurements at $T \leq 309.75$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in $\ln x_2$ units), consistent with the temperature-aware $\ln \gamma_2^\infty$ modelling of.[S17] This is a property of the structured hypothesis class, not a performance claim for the trained model: the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not. Neither is it a claim that our trained arms fail to carry it. The one checkpoint scored on this split is a near-constant predictor whose activity branch never left initialisation, so it separates nothing (§S5); what would separate it is a retraining of both arms on this split at the budget used elsewhere here.

Ranking and calibration. Absolute error and decision quality diverge, and within one checkpoint the divergence has a direction. Table S18 ranks the solvents of each solute–temperature group of the labelled test rows under the two σ-profiles of §3.2: the same weights, the same rows, and only the profile fed to the closure differing ($\max |\Delta\Phi| = 4.2 \times 10^{-4}$ at the worst of the three seeds, against 4.4 to 5.9 between separately trained arms). With its own learned profile the model orders solvents above two solute-blind references, at every one of the twenty-four margins that comparison has—four metrics, three seeds, two floors. With the reference profile substituted, none of its own twenty-four margins separates it from those references: twenty-four of the twenty-four include zero, or twenty-three, one cell sitting on the boundary and changing side with the bootstrap draw. That cell is named in the caption below and discussed after it. Both arms remain far above the within-group permutation floor, so the substitution does not reduce the ordering to chance; what it leaves cannot be told apart from what a ranker never shown the solute produces, which is a failure to separate the two and not a demonstration that they are the same. Groups are formed on solute and temperature rounded to 1 K, kept when they hold at least three distinct solvents, with duplicate measurements averaged per solvent. The effective sample is the 107 solute clusters, not the 823 groups, and every interval below is a 95% bootstrap over those clusters.

The shuffled-profile floor gives each group another randomly drawn solute's predictions over the same solvents, 200 to 300 draws; the solute-blind label floor ranks the group's solvents by their leave-one-solute-out mean measured solubility, which is label-derived and therefore an optimistic reference. Both are paired per group against the arm they are built from. The reading is of the intervals and not of the point estimates: the reference arm's twenty-four margins are all small and all but one positive (the exception is top-1 at seed 43 against its own shuffled-profile floor, -0.028), and what the intervals say is that none of the twenty-four separates that arm from its floor—twenty-four of them, or twenty-three, the one cell on the boundary changing side with the bootstrap draw. Two are marginal rather than comfortable—seed 44 against its own shuffled-profile floor, ρ $[-0.005, +0.165]$ and nDCG@3 $[-0.0008, +0.0762]$, the second of which is that boundary cell and is discussed in the caption above—so *indistinguishable from its floor* is the description all twenty-four support, and *below* it is not. The contrast survives excluding the 16 groups where some solvent has no reference profile and the arm therefore runs on a mixture (807 groups: $-0.212/-0.158/-0.172$), survives a leave-one-solute-out jackknife over the 107 clusters (ranges $[-0.239, -0.211]$, $[-0.190, -0.157]$, $[-0.203, -0.171]$), and exceeds at every seed the largest difference between seed replicates of one arm ($|\Delta\rho| \leq 0.060$ over nine seed pairs, every interval spanning zero). Two further readings are not available from it. The floors are built at the prediction level, not by a forward pass on a permuted profile, so they are faithful to *which* profile the closure sees and not to how its nonlinearity would answer a profile no molecule has; and the closure-carrying arm cannot be ranked against DirectGNN here, that difference—DirectGNN minus the learned-σ arm—being $-0.056/+0.053/-0.040$ by seed, sign-flipping, each interval spanning zero and each smaller than the seed-replicate spread.

Per-group recalibration of the MAE gap, in sample and out. The reading on which the substitution merely miscalibrates is granted in §3.2 at three strengths, each fitted inside each group, on exactly the row set the ranking uses: the 823 groups, 4912 solute–solvent–temperature cells, 107 solute clusters. No recalibration; an offset only, each group's own mean subtracted from prediction and from truth; and an affine map, slope and intercept fitted per group by ordinary least squares of measured on predicted $\ln x_2$. What is reported is the mean of the per-group MAEs and the paired reference-minus-learned difference, bootstrapped over solute clusters as everywhere here. In sample, pooled over the three seeds, the ladder is $+0.431$ $[+0.308, +0.553]$ raw, $+0.371$ $[+0.269, +0.474]$ offset-only and $+0.060$ $[+0.021, +0.101]$ affine; per seed the affine residual is $+0.069/+0.030/+0.081$, the middle one spanning zero, which is why it is quoted pooled.

Those three maps are fitted on the rows they are then scored on, and the groups are small: sizes run from 3 to 18 solvents with a median

Table S18: Solvent ranking under the two σ-profiles, on the labelled test rows grouped by solute and temperature (823 groups, 107 solute clusters, ≥ 3 solvents per group; ρ and τ are undefined on a group whose predictions are all tied and so run on 823/807/808 by seed, as §S6.1 sets out). Both columns are one checkpoint per seed; only the profile injected into the closure differs. Per-seed values are 42/43/44. The last column is the paired per-group difference averaged over the three seeds, with a 95% cluster bootstrap over solutes; all three per-seed intervals exclude zero for ρ, τ and nDCG@3, and only seed 42's does for top-1, which is why that row is quoted pooled. The lower block gives each arm's margin over its *own* solute-blind floors—a floor built from the other arm is not a comparison—and counts, over four metrics × three seeds × two floors, how many of those margins have an interval excluding zero. Each floor is rebuilt from the predictions of the arm it is set against. One reference-σ cell decides that count on a knife edge and is named so nobody has to rediscover it: nDCG@3 at seed 44 against its own shuffled-profile floor is $[-0.0008, +0.0762]$, inside the resampling noise of the boundary itself, and an independent recomputation places it on the other side. The reference-σ count is therefore 0 or 1 according to the bootstrap draw; nothing here turns on which, and the margins themselves, printed above, are what the reading rests on.

	learned σ	reference σ	reference − learned
Spearman ρ	0.592/0.546/0.598	0.365/0.369/0.410	-0.197 $[-0.282, -0.117]$
Kendall τ	0.518/0.469/0.521	0.303/0.306/0.333	-0.188 $[-0.263, -0.118]$
nDCG@3	0.796/0.748/0.792	0.697/0.690/0.698	-0.084 $[-0.124, -0.043]$
Top-1 accuracy	0.586/0.515/0.548	0.447/0.428/0.454	-0.107 $[-0.192, -0.020]$
Margin over its own solute-blind floor (ρ)			
shuffled-profile floor	+0.120/+0.209/+0.264	+0.039/+0.033/+0.079	
solute-blind label floor	+0.240/+0.207/+0.273	+0.038/+0.037/+0.080	
of 24, excluding zero	24	0	
permutation floor, ρ	0.00 ± 0.02, both arms		

of 5 and a mean of 6.0, 103 groups holding exactly three and 273 four or fewer, so in one group in eight two parameters are being fitted to three points. The out-of-sample variant refits the same map leaving out the row it scores—for each solvent of a group the map is fitted on that group's other solvents and the held-out solvent scored under it. Leave-one-out rather than a held-out half, because a half-split is undefined at three solvents, where a two-parameter fit consumes two points and leaves one; and rather than fitting on one seed and applying to another, which reuses the same rows and would test seed stability instead of overfitting. Out of sample neither map absorbs the gap. The offset-only map, one parameter, transfers as a level correction—it takes the learned-σ arm's mean per-group MAE from 1.83 to 1.03 and the reference-σ arm's from 2.26 to 1.51—but not on the quantity printed here, which is the gap between them: that stands at $+0.481$ $[+0.349, +0.619]$ pooled, no smaller than the $+0.431$ raw gap and larger than the $+0.371$ in sample. The affine map is worse still: the gap becomes $+1.09$ $[+0.3, +2.2]$ pooled—one decimal, and more draws do not buy a second, the leave-one-out affine statistic being heavy-tailed on the three-solvent groups: at 200,000 draws the upper endpoint still runs $[+2.193, +2.202]$, astride the rounding boundary—$+1.63/+0.49/+1.14$ by seed, each interval excluding zero, though seed 44 only just: its lower bound is $+0.009$. Over 200 independent 4000-draw bootstraps that bound falls below zero in 31% of them, and does so at 40,000 draws too, as the boundary cell of Table S18 does; seed 43's bound of $+0.013$ crosses in 2.5% and seed 42's of $+0.221$ in none. Its point estimate is larger than the raw gap's, though not separably so—the paired difference is $+0.66$ $[-0.12, +1.75]$ and spans zero—and the direction is what the two arms do separately: the refit *improves* the learned-σ arm, 1.83 to 1.68 pooled, and *worsens* the reference-

σ arm, 2.26 to 2.77—one arm better and one worse, not both worse, which is what opens the gap. That divergence is carried by the small groups. On the 550 groups with five solvents or more the out-of-sample affine gap is $+0.277$ $[+0.065, +0.503]$, $+0.074$ $[-0.082, +0.241]$ and $+0.035$ $[-0.172, +0.208]$ by seed, against an in-sample $+0.089/+0.029/+0.080$ there—the same small residual, measured too imprecisely at two of three seeds to be distinguished from zero. The in-sample 86% is therefore a bound on how much of the MAE gap a per-group affine map *can* absorb when it is fitted on the answer, not an estimate of how much a recalibration a user could apply would absorb, and §3.2 states it in that direction. Both fits, the ranking, the floors and every per-group fitted slope are deposited.

The fitted slopes, and where the rank invariance holds. Rank statistics are invariant under a per-group affine map only when that map is *increasing*; a fitted slope that comes out negative reverses the group's order, and the invariance argument of §3.2 does not cover those groups. Of the 823 groups the fitted slope is negative in 98/94/84 of the learned-σ arm's (11.9/11.4/10.2% by seed) and in 146/153/142 of the reference-σ arm's (17.7/18.6/17.3%); the median slopes are 1.22/1.19/1.10 and 0.50/0.58/0.53, the reference arm's the smaller because its within-group spread is the wider—median $\mathrm{sd}(p)/\mathrm{sd}(t)$ of 1.05 to 1.31 against 0.64 to 0.69—while its alignment with the measurement is the weaker. Both arms' maps increase in 652/633/650 groups by seed and in 589 at all three seeds together. Restricted to those 589, the reference-minus-learned ranking gap is ρ -0.148 $[-0.218, -0.079]$ pooled; per seed it is $-0.178/-0.112/-0.152$, and those three are on each seed's *own* both-increasing set—the 652/633/650 above, a different row set at each seed—not on the 589 the pooled value is on. Also on the 589: Kendall -0.155 $[-0.227, -0.084]$, nDCG@3 -0.068 $[-0.110, -0.026]$ and top-1 -0.087 $[-0.187, +0.015]$: about a quarter smaller than on the full set and of the same sign, with top-1—already quoted pooled only—now spanning zero there too. Groups whose predictions are constant have no fitted slope—there are 0/16/15 of them—so the 589 excludes them by construction. They leave two of the four metrics and not the other two: Spearman and Kendall are undefined where every prediction in the group is tied, which is why those two contrasts run on 823/807/808 groups by seed rather than 823, while top-1 and nDCG@3 are defined there and score all 823. That seed-43 count of 807 is a coincidence of arithmetic and not the 807 of the mixed-coverage check above, which is a different subset.

Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. Those two figures come from an earlier proxy checkpoint on a superseded split and are a property of an uncalibrated regressor of this family rather than of any arm of Fig. 2; the repair is the generic split-conformal guarantee and does not depend on which arm is wrapped. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

S6.2 Identifiability and compensation

Lemma S1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and compute the conditioning $\mathrm{cond}(I)$ and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes.

Every entry below is conditional on a design that must be stated, because a Cramér–Rao bound in J/mol scales directly with the assumed measurement noise. The operating point is $\Delta H_{\mathrm{fus}} = 25{,}000\,\mathrm{J/mol}$, $T_m = 400\,\mathrm{K}$, and an activity term $\ln \gamma_2^{\infty}(T) = a + b\,(T_{\mathrm{ref}}/T - 1)$

Table S19: Fisher audit of the SLE design at $\Delta T = 40\,\mathrm{K}$ (synthetic, closure-independent).

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ $T_m + \Delta H_{\mathrm{fus}}$	1.6×10^3	1,689

with $a = 1$, $b = 200\,\mathrm{K}$ and $T_{\mathrm{ref}} = 298.15\,\mathrm{K}$; x_2 is not a free input but the SLE solution at each temperature, running from 3×10^{-8} to 0.15 across the window. The design is $n = 9$ equally spaced temperatures spanning $\Delta T = 40\,\mathrm{K}$ from 280 K, one measurement each. The noise levels are $\sigma_{\ln x_2} = 0.3$ on each log-solubility observation, $\sigma_{T_m} = 10\,\mathrm{K}$ on a melting-point label and $\sigma_{\Delta H_{\mathrm{fus}}} = 2{,}000\,\mathrm{J/mol}$ on an enthalpy label. Parameters are non-dimensionalised by the scales used in the model's own configuration (5,000 J/mol, 50 K, 1, 300 K), so the condition number reports rank structure rather than unit choice. Table S19 reports the result at that window. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label reduces the conditioning by nearly two orders of magnitude and lowers the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together lowers it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which the grounding streams supply.

Widening or narrowing the temperature window does not remove the ill-conditioning of the solubility-only design. Varying ΔT over $\{120, 80, 40, 20, 10\}$ K moves the solubility-only conditioning through 1.7×10^3, 5.6×10^3, 2.3×10^5, 6.5×10^9 and 1.9×10^{16}, and the CRLB on ΔH_{fus} through 1,348, 2,249, 19,895, 3.2×10^6 and 8.0×10^6 J/mol. The point value quoted in the main text is the $\Delta T = 40$ K entry of that sweep and is not a design-independent constant. Multi-temperature data alone therefore does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps only partially: it reduces sd(ΔH_{fus}) to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic is not part of the primary argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated ($\mathrm{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$). This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative) $\ln x_2$ fit residual, so their correlation tends to -1 as the fit improves, a permutation null that destroys any learned coupling still reproduces most of it, and the model's own factors barely co-vary. The large apparent crystal offset is likewise not a usable misgrounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics, and capping predicted T_m at a physical bound collapses the offset from ≈ -7.1 to ≈ -1.5.[1] We therefore base the non-identifiability claim on the reference-free Fisher/CRLB audit above, and present the compensation figures in the SI (§S5) as a cautionary diagnostic rather than a decomposition-quality measurement.

S6.3 Grounding the crystal factor: a null measured on labels the model already had

Whether the ceiling lies in the inputs rather than the closure can be tested by grounding a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points (15,427 molecules), added through the grounding-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. Neither the property the pool supplies nor solubility improved on the held-out solutes: melting-point MAE 45.0 $\rightarrow$ 46.6 K (paired +1.59 K over the $n = 2518$ test solutes

[1] Permutation null (learned coupling destroyed) -0.90 to -0.93; model-factor $\mathrm{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$; the Joback reference predicts $T_m > 1000\,\mathrm{K}$ for a substantial fraction of these polycyclics.

carrying a T_m label) and $\ln x_2$ MAE 1.81 $\to$ 1.88.

That comparison did not test external crystal grounding, because the pool was barely external. Of its 15,427 molecules, 15,082—97.76%—already carried a T_m label in the training corpus, 98.2% of them agreeing with the in-corpus value to within a millikelvin (mean $|\Delta|$ 0.07 K); the net-new contribution is 345 molecules, 2.24%. The two label sets are drawn from the same open melting-point tables, and nothing in the pool builder counts the overlap. Where the pool did carry something new, the crystal head moved. Scored on the pool itself, T_m MAE falls 35.85 $\to$ 35.21 K (paired -0.636 K, 95% CI $[-1.09, -0.20]$), splitting into -0.371 K $[-0.81, +0.05]$ on the 15,082 duplicated molecules and -12.233 K $[-18.25, -6.27]$ on the 345 new ones—thirty times the movement per molecule on the fraction that supplied information.

Dose was not the constraint. The exported state is the one entering phase 3, after 79 epochs, by which point the stream had run 632 optimiser steps—2.62 passes over the pool, 93.0% of it seen at least once, the loader reshuffling each epoch—and 240 of those steps ran at loss weight 1.0, the phase-1 melting-point weight that an unset stream weight inherits, not at the phase-2 value 0.05. The downstream half of the null is in turn weaker than its row count suggests: the $+0.065$ $\ln x_2$ MAE gap is 5608 rows from 147 test solutes, and clustered by solute it is $+0.088$ with CI $[-0.127, +0.284]$, crossing zero. Nor is it the activity branch absorbing the crystal branch. Between the two arms the predicted Φ differ by 1.25 ln units per row on average; regressing the predicted $\ln \gamma_2$ on that difference gives a slope of -0.08 where full compensation would be -1, so 91% of the crystal head's disagreement passes into $\ln x_2$. The activity-path $B_{\text{closure}}/B_{\text{insuff}}$ decomposition does not apply on this path in any case: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$, not a misspecified map.

What the experiment licenses is narrower than a null on external crystal labels, and more useful: a grounding pool has to be measured by its net-new content before it can test anything. This one was scaffold-guarded correctly—none of the 2518 evaluation solutes appears in it—and was redundant all the same, because the guard protects the estimand and says nothing about what the model already holds. A real test declares the net-new count in advance, the builder reporting it and refusing to run below a floor of a few thousand molecules, which in turn needs a melting-point source disjoint from the one the training corpus itself drew on. Compute is not the obstacle: the grounded arm here was an eight-hour CPU run, so three seeds by two arms is on the order of 50 CPU-hours. Sourcing the labels is.

S6.4 Chemistry-resolved structure of the asymptotic physics penalty

Scope. The two arms compared in this section differ from each other in capacity and in the per-arm optimisation settings of §2.4, and no hyperparameter-matched control was run, so nothing here is an accuracy claim about the thermodynamic bottleneck: Table S2 records that ΔMAE as a difference between separately tuned pipelines, and what follows is a hypothesis-generating description of where two particular pipelines differ (§3.7). "Physics penalty" names that measured difference throughout, not an effect of the bottleneck.

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.03, while the grounded physics arm (learned-σ COSMO-SAC closure) reaches 1.85 ± 0.05, for a global $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)} = +0.14$ (positive meaning the closure adds cost).[2] At this granularity the difference between the two pipelines is small and near label noise. Once the error is resolved along chemical axes, that difference is an average over regimes

[2]The per-seed ΔMAE spans $+0.05$ to $+0.25$, so the point estimate is an imprecise 3-seed mean; being computed on unrounded per-seed means, the rounded 1.70/1.85 differ by 0.15 rather than 0.14.

that disagree in sign.

The solvent axis is the most pronounced. Binning test pairs by solvent class and recomputing ΔMAE per seed (Table S20) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln\gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class weakens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the seed-robust stratum in which the penalty reverses sign ($\Delta\text{MAE} = -0.22 \pm 0.11$ over seeds). This is exploratory rather than a confirmed improvement: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the asymptotic physics penalty shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved counterpart of the synthetic closure-fidelity sweep (§S11).

The solute axis shows a compatible but weaker pattern (Table S21). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while the charged, high-$|\ln\gamma_2|$ stratum of salts is neutral. On this axis the penalty therefore does not track $|\ln\gamma_2|$: it falls on neutral organics, not on the ionic tail.

The novelty axis inverts the usual coverage expectation: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto $\leq$ 0.3) and hurts most on the most familiar ($>$ 0.8; Table S21). This is exploratory rather than a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Taken as a direction rather than a tested quantity, the inversion suggests that the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit. That misfit is masked on novel solutes by the large errors both arms already incur there, and exposed on well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual, which is the same closure-misspecification reading as the

Table S20: Per-solvent-class asymptotic physics penalty $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)}$ on the scaffold split (mean $\pm$ sd over 3 seeds; positive means the closure adds cost).

Solvent class	ΔMAE (mean $\pm$ sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

Table S21: Per-solute-class and novelty-stratum asymptotic physics penalty $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)}$ on the scaffold split (positive means the closure adds cost). Entries carry the 3-seed spread where it was computed: the two Tanimoto strata and the charged/salt class. The other four solute classes are point estimates.

Solute stratum	ΔMAE
Oxygenated	$+0.32$
Polyaromatic	$+0.27$
Heterocycle	$+0.17$
Halogenated-aromatic	≈ 0
Charged / salt	-0.04 ± 0.19
Tanimoto ≤ 0.3 (novel)	-0.04 ± 0.18
Tanimoto > 0.8 (familiar)	$+0.51 \pm 0.09$

rest of the paper. No multiplicity-controlled, within-bin-bootstrapped test of it was run, and the separate-tuning confound stands over both maps, so neither is an accuracy claim in either direction.

S6.5 The data-efficiency sweep, and what survives its confounds

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct factorisation should outperform a data-hungry black box. We test it directly, subsampling the training set *by solute* at fractions 0.05–1.0 and training both the physics-grounded model and DirectGNN on the identical subsample, evaluated on the fixed scaffold test (Table S22, seed 42). The physics arm is *worse at every fraction*, and against the hypothesis the gap is *smallest* at the least data (ΔMAE $= +0.08$ at 5%); its R^2 is negative throughout. We do not read the growth of the gap with data (+1.0 at full) as a data-scale effect, because the sweep's endpoint does not reproduce the headline comparison. Its physics arm carries the NRTL closure, and at fraction 1.0 it gives 2.98 against 1.97 (control) on the same scaffold test where the three-seed runs of Table S3 give 1.7950 against 1.7022: ΔMAE an order of magnitude apart (+1.01 here, +0.09 there—0.0928 before rounding, and that table's rounded column gives the same +0.09). Both arms are worse here than their headline counterparts and the physics arm much the more so (+1.18 against +0.27 MAE), so the trend measures how the light per-fraction budget interacts with training-set size at least as much as how the physics prior does. The sweep is also single-seed, its physics-arm MAEs are non-monotone, and its two arms differ from each other in capacity and in the per-arm optimisation settings of §2.4, and both are wider than the headline arms (hidden width 256 and 128 here, 64 there). What survives is the level and not the trend (Table S22, plotted in Fig. S6)—the physics arm is worse at every fraction under a matched light budget—which is *consistent with* the sign rule (Prop. S1) rather than a sharp confirmation of it. The one inverting metric is top-1 solvent ranking (0.42 vs 0.23 at 5%; the ranking-versus-accuracy split of §S6.1), which does not translate into lower error. This is the last regime in which a physics prior should plausibly help; that it does not is consistent with the broad accuracy null and with the structural account—a non-identifiable split through a misspecified closure—as the operative explanation.

Table S22: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(direct)}$.

train fraction	MAE		ΔMAE	R^2	
(by solute)	physics	direct	(phys−dir)	physics	direct
0.05	2.28	2.20	+0.08	−0.09	+0.07
0.10	2.44	2.07	+0.37	−0.25	+0.15
0.25	2.93	1.81	+1.12	−0.75	+0.38
0.50	3.12	1.98	+1.14	−0.98	+0.22
1.00	2.98	1.97	+1.01	−0.79	+0.22

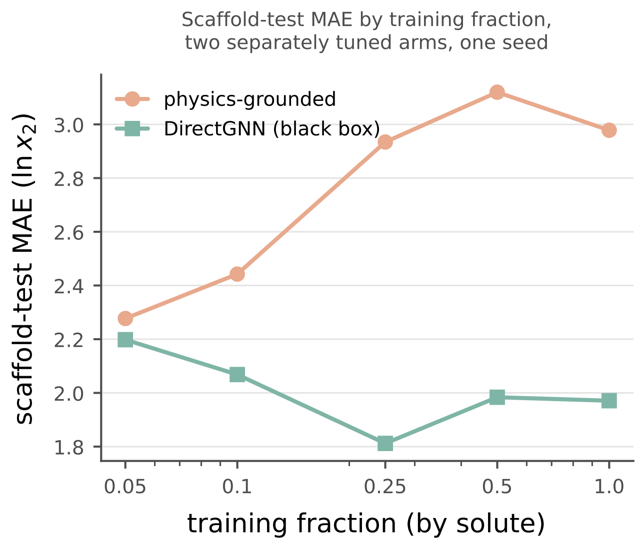

Figure S6: Data-efficiency curve: physics-grounded arm (salmon) and DirectGNN (teal) as a function of training-data fraction. *Scope:* one seed, five fractions, under a light per-fraction budget whose endpoint does not reproduce the headline comparison; the physics arm's MAEs are non-monotone. *Caveat:* the two arms differ in capacity and in per-arm optimisation, so nothing drawn here is attributable to the thermodynamic bottleneck, and this paper makes no accuracy claim in either direction.

S6.6 The fidelity lever and the chemical placement of the closure error, in full

The article's §3.5.3 states two things—that the fidelity lever it can demonstrate is confined to the VT-2005-matched set's chemistry, and that the closure error is chemically placed and placed differently on the two sets. This section is the working behind both.

The 2002 $\rightarrow$ 2010 **step, in full.** If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 $\rightarrow$ 2010 step delivers one on the VT-2005-matched set and not on the broad IDAC set, and a block split identifies the factor that carries that reversal.

The two kernels have no *prescribed* common input, σ-profile averaging being part of each model's definition, so evaluating each on its native profile is the specified use. An input-fixed *control*—profile database, cavity area, volume and averaging held fixed, kernel alone varied—gives 2.047 $\rightarrow$ 0.765 on the smaller set ($P > 0.95$), under three limits: it cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction; its step is larger than the native-input one, so it is favourable to the conclusion it supports; and its re-averaging term is unresolved. The NIST implementation detail fixing the *fair 2002* arm at 1.757, and the three arms on both sets under both conventions, are in Table S8.

The step more than halves the smaller set's error under the full convention ($P > 0.95$), the convention under which the two kernels' published error figures are reported, and cuts it 42% under the deployed one. On the broad IDAC set it is not an improvement under either rule, and the 3% the dispersion term recovers there is a null whose interval is an order of magnitude wider than the effect. That arm's run on the smaller set is on n=57, three halogenated pairs having no tabulated dispersion value—a coverage limit of the model offered as the lever; no claim rests on the 0.765-versus-0.785 ordering, which is not an artifact of the differing n.

The reversal is the *chemistry* and neither the profile database nor the temperature range: with UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set, does not win on the complementary 420, and still does not win on the 80 complementary rows that are *also* at 298 K—that last block at $P \approx 0.3$ and no interval, the three blocks not having been tested against one another. The VT-2005-matched set is a low-γ, hydrogen-bond-dominated sub-design, the regime the donor/acceptor-typed 2010 kernel was built for. The map does not corroborate the reversal from the other side and we do not ask it to: the typed kernel is not an improvement on the broad set, of whose squared error the glycol ethers carry 45%, and the classes that would explain a reversal—the halogenated solvents, the aryl ethers and water as the solute, largely one row set, all 13 aryl-ether rows and 18 of the 30 halogenated being water-as-solute rows from one publication—are the cells the rule excludes. The block split is the whole of the evidence for the reversal being chemical.

The floor checks are same-database, same-latent *and* same-convention. The *UD-native* check on the smaller set, the one commensurable with the broad set, is at zero under the deployed rule and +0.69 under the full convention. Under the 2010 kernel's own typed latent, 306-dimensional and so several times worse resolved, the ordering is *lost* on the smaller set (−0.77) and holds on the broad set (+1.29)—both aggregates, and read as such. That is loss of resolution, not a demonstrated flip to the inputs, which would need a *lower* bound on B_{insuff}. We know of no large-scale infinite-dilution benchmark of the 2002 $\rightarrow$ 2010 step, and our search was not exhaustive, so the broad-set result is an absence of evidence rather than evidence of absence; it lies within the range of magnitudes the comprehensive assessment[S10] reports. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent.

Where the closure fails chemically, in full. Where B_{closure} dominates—on this set, the one row set the rule leaves standing—the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads, which is the composite B_{closure} is the error of (§2.1.1) and the object the pipeline runs on. Which of the two carries the error is a question this section does not answer; nor does dominance say the composite is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit on the reference VT-2005 profiles, removes $46\pm5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. It gives a null and not a bound: the $\pm$ prices only optimiser noise over five fit seeds, the held-out fold spread that dominates it going unpropagated (§S3). B_{closure} is therefore an error a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents—a statement about what a correction of that shape can absorb, not about which component of the composite the error came from.

Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones. That is the chemistry COSMO-SAC's documented single-parameter H-bond term governs—the term the donor/acceptor-typed 2010 closure was built to replace. Half of that reading carries to the published evaluation of Table 2, 130 times as many rows: the acceptor-only solvents take 10% of its squared error at MSE 0.60, the lowest of its three donor classes. The donor half does not carry intact—strong donors take 56% there rather than 90%, and its inert solvents, on which the smaller set's reading is silent, take 34% at the highest per-row error of the three. It places the error in a *chemistry* and not in one component of the composite: the reference profile for each of those solvents is one gas-phase conformer too, and on the glycol ethers—donor and acceptor on one flexible chain, the class this paper's own cascade names conformer-dependent (§S3)—a displaced reference would load onto exactly the rows that carry the error. Neither set separates that from the kernel. On the broad set that reading does not reproduce, and neither does its replacement, an absent dispersion term: which of the two the broad set points at is decided in either direction by the same publication the map already declines to state, so we state neither, and the H-bond reading remains the VT-2005-matched set's, neither corroborated nor replaced by the broad set. The residual's fold design and rank ordering, its failure to gain over the free 2002 $\rightarrow$ 2010 step above, and the Staverman–Guggenheim counter-test in which the physically *more* complete combinatorial term *increases* error on size-asymmetric pairs are in §S3.

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure's fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound. One arithmetic sizes that exposure. The ground truth charging B_{closure} is quantum chemistry and not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state, and the successor Delaware database revises those conformations for about a third of its compounds;[S10] where the reference is displaced, even a perfectly specified g carries a term in B_{closure} through its curvature (§S9). Table 2 measures that axis directly—one closure, two profile databases, HF-TZVP against BP-TZVPD-FINE, a factor of 2.6 in AAD, 6.6 in squared error if the errors scale uniformly—while the glycol-ether stratum's margin, read on the UD profiles the broad set is matched to and not on the VT-2005 profiles the solubility pipeline deploys, closes only once its squared error falls below twice its 0.108 bound, a more than ten-fold reduction (Table S15). A profile source better than UD by the whole of that published bracket would leave that sign standing. No side of that arithmetic is on the same database as another: the bracket enters as the size of a database-to-database effect, transported to the stratum, and not as a comparison run on it. The arithmetic holds the insufficiency bound fixed, which a new profile source would not, so it sizes the exposure and does not bound the share.

S6.7 Limitations (ii), (iii), (v), (vii)–(xi) and (xiii), in full

(ii), (iii) and (v): the convention rule, the other admissible cells, and the constant-offset bound. (ii) Under the deployed convention no *aggregate* margin is resolved on either set and only the literature-standard convention resolves one, while the headline convention is the deployed one; the aggregate falls for the stronger reason of §3.5.1 rather than for want of resolution. (iii) Of the map's six admissible cells the glycol ethers are the only row set also positive and left standing by clause (c), and not the only cell whose deletions all run; the halogenated solvents and the monoalcohols keep their sign under the one deletion each admits, so they are unverified rather than shown to fail. (v) The constant-offset margin is convention-specific and inverts under the deployed default on both sets; there the separation rests on the law-of-total-variance bound and the more-physics-worse effect. That bound is not itself convention-independent—only the estimand B_{insuff} is, Lemma 1(d)—so the deployed claim rests on it *computed under the deployed convention.*

The article's §3.7 carries limitations (i), (iv) and (vi), which scope what the two data sets and the estimator can deliver, and (xii), which the leak-free re-run of §2.2 rewrites. The remaining nine are here, keeping those numerals.

(vii) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor of §2.2, so label noise does not explain the ceiling. (viii) The latent-departure result is three-seed mean±sd (n=44): what it establishes is the *sign* and a transferable shared direction, the magnitudes (51±2%/72±1%/4.1±0.5×) being reported but not stable across analysis configurations. Which null each of the two distinct top-two shares clears, and that none of it is measured on the arms of Fig. 2, whose own latent departure is not measured, are in §3.3. (ix) The probe on the unconstrained control is bounded by the same reference scarcity: its crystal-term and temperature tests rest on 13 held-out solutes and cannot be enlarged, so the negative there is a bound and not a demonstration that no crystal information is present; the crystal-slope test is additionally limited by the corpus, whose empirical slope carries $-\Delta H_{\mathrm{fus}}/R$ at $r = +0.24$. The scaffold-organisation result needs no external reference, but its two partitions nearly coincide by construction, so a ratio near one is weak evidence of within-scaffold indistinguishability (§S10). (x) The closure-fidelity contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself is not measured under a modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested; both need a typed σ-profile head trained end to end through the 2010 kernel, and the solubility test a UD$\cap$BigSolDB matched set. (xi) No difference in accuracy between the physics and control arms is attributable to the thermodynamic bottleneck; the confound and what survives it are stated above. (xiii) Several of the numbers the verdict rests on are checkable only by re-running the released code, whose input artifacts are larger than the repository carries; the two decomposition sets are therefore deposited as row-level tables (Data Sets S1 and S2), so the sets and their margins can be recomputed without code, which does not deliver the provenance of the closure evaluations in them or of the trained arms.

S6.8 Diagnostics that do not work, and why

The drift is not the closure's signed first-order inverse—but the test is under-

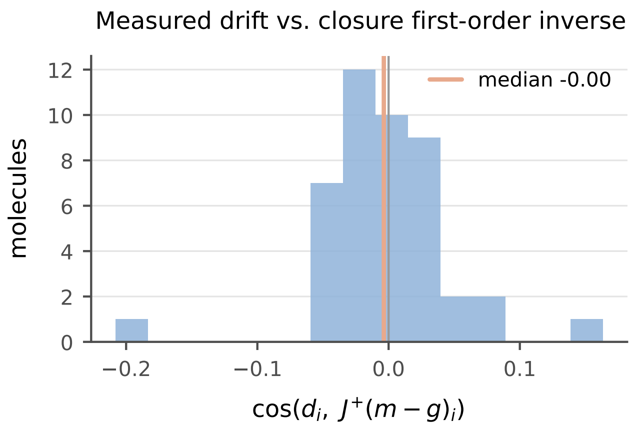

Figure S7: Per-molecule cosine between the measured drift $\hat{\sigma} - \sigma$ and the first-order closure compensation $J^+(m-g)$; the distribution is centred on zero (median -0.00).

resolved. A candidate account of the compensating drift is the closure's linear-response inverse, $\hat{\sigma} - \sigma \approx J^+(m - g(\sigma))$ (with $J = \partial g/\partial\sigma$), predictable a priori from J before any solubility training. On the matched set the per-molecule drift shows no signed alignment with $J^+(m-g)$ (median cosine -0.00; Fig. S7).[3] We read this as a signed negative only, and do not build on it, for two reasons intrinsic to the test. First, the matched set is sparse (median one pair per molecule), so each per-molecule Jacobian subspace is one-dimensional and the dimension-matched row-space controls are uninformative—a resolved direction common across molecules makes a molecule's "own" and "other" subspaces coincide. Second, and more fundamentally, the single-pair $J^+(m-g)$ is a poor *estimator* of the first-order compensation: the activity Fisher is near rank-one (PR $\approx$ 1.4 of 51, §3.3), so the closure's first-order inverse is ill-posed along all but one direction. A null here is therefore uninformative about whether the drift is first-order, not evidence that it is not. The load-bearing claim is the under-determination of σ by activity (§3.3), not the geometry of the drift.

The instrument behind §3.3—the rank of the activity Fisher $J^\top J$ ($J = \partial \ln \gamma_2^\infty/\partial\sigma$), read as a participation ratio PR $= (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$—is computationally inexpensive and oracle-free, which is its appeal for probing any physics-grounded latent. It is also easy to misread in three ways that inflate a null or a rank in the favourable direction without any warning sign; we record them because the corrections are not self-evident.

(i) Finite differences inflate the participation ratio. PR is a *tail* statistic, governed by the small eigenvalues. A closure evaluated by an unrolled fixed point carries a small, step-*independent* gradient noise (from the convergence tolerance, not truncation), so central finite differences lift the exactly- and near-zero eigenvalues onto a floor and PR rises. On the differentiable COSMO-SAC-2002 layer, autograd and finite differences on *identical* profiles give PR = 1.00 and 1.25—a 25% inflation from noise alone, though the per-gradient cosine is 0.995 (the direction is right, the tail is not). It is worse on a typed (3×51) profile basis. A closure that ignores the typing (2002) has 102 exactly-redundant coordinates that finite differences decorrelate into a spurious floor, inflating its PR from 1.0 toward 2.0, while a closure that uses the typing (2010) has no such redundancy. The same step therefore inflates the two *asymmetrically* and can manufacture a rank ordering with no physical content. *Autograd is the appropriate instrument where the closure is differentiable; with only finite differences,* PR *is interpretable against a known-rank or permuted control, never absolutely.*

(ii) "Own vs. other" controls are powerless when the resolved direction is common. A dimension- and smoothness-matched control aligns a molecule's drift against its own predicted compensation and against other molecules'. It has power only if the predictions differ. When the aggregate Fisher is rank-one (§3.3), however, the resolved direction is *common*, so "own" and "other" point almost the same way and the diagonal must sit inside the null—$p \approx 0.5$ regardless of the truth. On the

[3] The residual COSMO-SAC closure runs a segment-activity fixed point, so its autograd Jacobian is numerically unstable ($\|\partial g/\partial\sigma_{\mathrm{solvent}}\| \sim 10^{15}$ at infinite dilution); J is computed by finite differences on the converged closure.

matched set sparsity compounds this: a median of one pair per molecule makes each per-molecule Jacobian subspace one-dimensional, so "own" and "other" subspaces coincide exactly. *A control's null is interpretable only once its power is established: if the compared predictions have median cosine near one, the test cannot reject.*

(iii) Energy-in-subspace baselines need the effective rank and a cluster null. A random zero-mean vector in D dimensions places $k_{\text{eff}}/(D-1)$ of its energy in a subspace of *effective* rank k_{eff} (the participation ratio of its singular values), which for near-collinear Jacobian rows is far below the nominal row count. On the matched set the per-molecule subspace is effectively one-dimensional, so the baseline is $\approx$2% and the observed 5.5% is a two-to-threefold *enrichment*, not the "no concentration" a raw own-versus-other equality suggests: own equals other because both one-dimensional subspaces are the same common direction, not because the drift avoids the resolved one. With that direction common, N enriched molecules are not N independent draws—the enrichment needs a cluster null (§3.5), without which its significance is over-stated by $\sqrt{N/N_{\text{eff}}}$. We therefore report the 5.5% as evidence in neither direction. *The baseline must be set by effective rank, and the null clustered by the shared direction.*

Each failure converts a spectral concentration statistic into a structural claim it does not support. The surviving statements of §3.3 avoid that conversion by carrying their null model and a power check inline.

S7 A second chemical closure: pKa through a Hammett LFER

The measured result of this section is stated in §3.6; what follows is the construction, the estimator and the sweep behind it. The instrument is not specific to COSMO-SAC. In a second real domain—aqueous pK_a of substituted aromatics—the fixed closure is a Hammett linear-free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ over $z^\star = (s, \sigma^\star_{\mathrm{H}})$: the reaction series s, which fixes $(\mathrm{p}K_{a,0}, \rho)$, together with the Hammett substituent constant σ_{H}, written with its subscript throughout so that it cannot be confused with the σ-profile of §2.1, with Hansch–Leo tabulated constants as the external oracle. The two classic LFER failure modes fall cleanly onto the two channels of Lemma 1: *resonance saturation* (a curvature the linear closure cannot represent) lands in B_{closure}, while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. On the OPERA experimental compilation[S18]—the canonical full set under a uniform, closure-independent quality filter that drops pre-ionized microstates (§S8)—this yields a *measured* fidelity sweep. The clean meta/para series (n=158) is well specified (RMSE 0.55, B_{closure} below the insufficiency floor). The degraded pole (n=846)—an ortho substituent, or one outside the tabulated list, at some ring position (§3.6)—degrades to RMSE 2.26, of which a within-series estimator conditioning on $(s, \sigma^\star_{\mathrm{H}})$ places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure. That estimator is an out-of-fold residual, so both figures are one-sided, and the a-priori channel assignment—what the zeroed positions carry and the tabulated constant lacks—is consistent with the measurement rather than established by it (§S8); the sweep is the near-exact-to-degraded transition. The decisive test is the same oracle swap as on solubility—the learned $\hat{\sigma}_{\mathrm{H}}$ replaced by the tabulated $\sigma^\star_{\mathrm{H}}$ in the same closure, the sign of the change recorded per stratum—measured *in-distribution* across K=20 within-scaffold splits. It *flips*. On the clean meta/para pole feeding the reference constant *helps* (learned MAE 0.48 vs oracle 0.315; oracle better in 19/20 splits and in three of the four reaction series): the ceiling there is the noisy learned estimate, not the closure—a real-data pole on which grounding helps, and an internal positive control. On the degraded pole the same swap *hurts* (learned 0.83 vs oracle 1.62; oracle worse in **20/20** splits and in all four series, gap

90% CI $[+0.57, +0.95]$): the freely learned $\hat{\sigma}_{\mathrm{H}}$ absorbs what the zeroed positions carry and the tabulated constant lacks. What reverses here is the sign of the grounding penalty; it reverses through the insufficiency channel—consistent with the one-sided bound above, though not isolated by it—rather than through the closure-error cancellation of §3.3: a second instance of the sign rule, not of that mechanism. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the accompanying bootstrap resolves nothing finer than series-level sign agreement—read as a sign test on four units, p=1/16 on the degraded pole and $p\approx 0.31$ on the clean one—and is quoted as such (§S8).

Both signs are one comparison: the sign of the grounding gap is by construction that of the learned arm's error against the fidelity-set oracle error, and what a competence sweep adds is that the oracle side of it does not move with the training budget while the learned side does (§S8). The well-specified pole's threshold ($\approx$0.315) is never beaten by the learned arm, so grounding helps at all four budgets swept. The degraded pole's threshold (1.62) is high, so the learned arm beats it, and grounding hurts, across the whole interpolation range (down to 12% of the data), reverting to helps only under scaffold *extrapolation*, where $\hat{\sigma}_{\mathrm{H}}$ breaks (2.6±2.0) back above the threshold—in the mean over six splits, three of which still have grounding hurting. The flip is thus a within-distribution, fidelity-driven crossover whose one scope boundary is that same comparison read where the learned arm breaks. It is the sharp, two-sided form of the penalty sign rule (Prop. S1); its "a fitted intermediate can beat the true one" half is known—plug-in efficiency under correct specification,[S8] pseudo-true fitting under misspecification[S19]—sharpened here into a fidelity-set threshold. The trained physics-vs-DirectGNN comparison is weaker and is not load-bearing (§S8). On an independent chemical closure the instrument separates a pole where grounding helps from one where it hurts, each with a sign consistent across splits and series; the full construction, the estimator validation and the measured sweep are in §S8 (Fig. S8).

S8 The pKa closure: full construction, measured sweep, and trained comparison

This section gives the full construction behind §S7: a fixed Hammett linear-free-energy relationship (LFER) $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ over $z^\star = (s, \sigma^\star_{\mathrm{H}})$: the reaction series s, which fixes the per-scaffold Hansch–Leo–Taft constants $(\mathrm{p}K_{a,0}, \rho)$, together with the Hammett substituent constant σ_{H}, whose tabulated Hansch–Leo values are the external reference for the coordinate the network predicts. The analogy is close: as with the σ-profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 1: *resonance saturation*—a curvature in σ_{H} that the linear closure structurally cannot represent—is a mis-map of a function of $z^\star$ and lands in B_{closure}; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. The chemical scope is a fidelity sweep, over the four reaction series alike (benzoic acid, phenol, anilinium, pyridinium): records whose every substituted ring position is meta or para with a tabulated substituent are the near-exact ($B_{\mathrm{closure}} \approx 0$) regime, and records carrying an ortho substituent or one outside the tabulated list—both of which enter $\sigma^\star_{\mathrm{H}}$ as zero—the low-fidelity one. The assignment rule is a function of structure alone and never reads the residual, and it is stated in full next.

The assignment rule in full. The reference constant is assigned mechanically, and that rule is also what defines the two poles. The four kinds are tried in the fixed order benzoic acid, phenol, anilinium, pyridinium, and a record enters under the first of them whose ionizing group matches exactly once in the molecule, at a ring atom lying in exactly one ring and that ring six-membered and aromatic. An ionizing site the entering kind does not consume does not exclude the record: on the same ring it is typed as an ordinary substituent, tabulated or not. It enters only if its ring system is also unambiguous: no aromatic nitrogen anywhere in the molecule for the three carbon-ring series, exactly one and no N-oxide for pyridinium. The kind a record enters under names the reaction series and with it $(\mathrm{p}K_{a,0}, \rho)$. Ring positions are typed by shortest ring-bond distance from that ring atom—the one bearing the ionizing group, which for pyridinium is the ring nitrogen itself—one bond *ortho*, two *meta*, three *para*; and $\sigma^\star_{\mathrm{H}}$ is the sum over the substituted positions of the Hansch–Leo constant of that substituent *at that position*, σ_m meta and σ_p para, taken from the fixed list of nineteen substituents printed in §S8; an unsubstituted parent has $\sigma^\star_{\mathrm{H}} = 0$. An ortho position carries no tabulated constant and enters the sum as zero, and so does a substituent outside the list. The first pole—the *clean* one below—is the 158 records whose every substituted position is meta or para with a tabulated substituent; the *degraded* pole is the 846 where at least one position is neither—571 of them carrying an ortho substituent, the other 275 an untabulated one—and there $\sigma^\star_{\mathrm{H}}$ is the partial sum over the positions it can see. The split is not benzenoid against heteroaromatic: all four series occur on both sides, 32 of the clean pole's records being pyridinium.

The estimator, its identity, and its fold design. That estimator is not the eight-bin coarsening cell of §2.5, and its conditioning variable is not the constant alone; nor is the functional the same, the two figures just given being floors $\mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$ on B_{closure}, which when positive establish $B_{\mathrm{closure}} > 0$, and not the separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ of Table 1, which when positive would establish the stronger $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$. The closure reads the reaction series too—$\mathrm{p}K_{a,0}$ and ρ are per-series—so the argument Lemma 1 requires is the pair $z^\star = (s, \sigma^\star_{\mathrm{H}})$, series label and constant, and the estimand is $\mathbb{E}[\mathrm{Var}(\mathrm{p}K_a \mid s, \sigma^\star_{\mathrm{H}})]$. Neither coordinate needs binning: s takes four values and $\sigma^\star_{\mathrm{H}}$ is a scalar, so the conditional variance is estimated directly—a random forest of $\mathrm{p}K_a$ on $\sigma^\star_{\mathrm{H}}$ alone, fitted *within* each series and pooled over series by n, five-fold, its *out-of-fold* residual mean square. That regressor $\hat{r}$ reads $(s, \sigma^\star_{\mathrm{H}})$ and nothing else, and predicts each row from the folds that exclude it, which is what makes the split exact for it: $\mathbb{E}[(m - \hat{r})^2] = \mathbb{E}[\mathrm{Var}(m \mid s, \sigma^\star_{\mathrm{H}})] + \mathbb{E}[(\mathbb{E}[m \mid s, \sigma^\star_{\mathrm{H}}] - \hat{r})^2]$, so the estimate exceeds B_{insuff} by the regressor's own approximation error—an upper figure, leaving a lower one. Conditioning on s is not a free choice and it is not the conservative one: pool the four series and the same estimator returns 8.3 against the degraded pole's MSE of 5.1, a conditioning under which g is not measurable, so Lemma 1 does not license it and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on s belonging in $z^\star$, which is what the lemma demands of a closure that reads it. §3.5.1 declines fitted estimators on solubility because random folds let a held-out row borrow strength from a near-twin through the representation; here the regressor's only input is $\sigma^\star_{\mathrm{H}}$ within a series, so no fold can hand it what the closure's argument does not already carry. What the identity needs of the fold design is only that a row be held out of the fit that predicts it.

The nineteen substituents that rule can see, and the constants it sums, are Table S23. They are standard Hammett σ_m/σ_p values[S20] to two decimals; they are a constant of the builder, not fitted or refit on these data, which is the sense in which g is fixed for Lemma 1. The order matters: a ring position takes the constant of the first pattern in the list that matches at it, and the list is ordered specific before generic. A substituent outside this list makes the position untabulated, and the record degraded.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, p-nitrophenol reads $\mathrm{p}K_a \approx 8.3$ against an experimental ≈ 7.1 (and p-nitroaniline overshoots similarly): the linear closure is off by ~ 1 unit exactly where through-conjugation (σ^-_{H}) breaks it—B_{closure} observed in practice.

A controlled semi-synthetic construction on real tabulated σ_{H} (known ground-truth channels) validates the estimator and reproduces the

Table S23: The fixed substituent list behind $\sigma^\star_{\mathrm{H}}$: nineteen groups with the meta and para Hammett constants summed over the substituted positions (§3.6). Patterns are tried in the order shown—down the first block, then the second, then the third—and the first match at a ring position wins. An ortho position, and any group not in this list, enters the sum as zero.

group	σ_m	σ_p	group	σ_m	σ_p	group	σ_m	σ_p
NO_2	0.71	0.78	OCH_3	0.12	−0.27	Br	0.39	0.23
CF_3	0.43	0.54	OH	0.12	−0.37	I	0.35	0.18
CN	0.56	0.66	$N(CH_3)_2$	−0.16	−0.83	$C(CH_3)_3$	−0.10	−0.20
COOH	0.37	0.45	$NHC(O)CH_3$	0.21	0.00	C_2H_5	−0.07	−0.15
CO_2CH_3	0.37	0.45	NH_2	−0.16	−0.66	CH_3	−0.07	−0.17
$COCH_3$	0.38	0.50	F	0.34	0.06			
CHO	0.35	0.42	Cl	0.37	0.23			

paper's central distinction in this second closure (Fig. S8). Sweeping closure fidelity with a sufficient intermediate, the grounding gap $\mathcal{G}$ tracks B_{closure} (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem S1's proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives $B_{\mathrm{closure}} = 0$ at every knob while $\mathcal{G}$ rises with the insufficiency fraction, from -0.035 at zero insufficiency through zero near a fraction of 0.25 to $+0.34$ at 0.7: at the three fractions where it is positive it is positive with B_{closure} identically zero, which is the pKa realization of the conflation that makes $\mathcal{G}$ an indicator rather than a certificate. The two negative points are the same statement in the other direction—a well-specified closure with a nearly sufficient intermediate leaves the oracle marginally ahead. The oracle-σ_{H} pipeline (scaffold-resolved substituent$\rightarrow \sigma_{\mathrm{H}}$ assignment) and this validation extend directly to real data. On the OPERA experimental pK_a compilation[S18] (7,904 records under the same net-neutral filter), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep. On the clean meta/para series (n=158) the closure predicts experimental pK_a to RMSE 0.55, with the closure at the input-insufficiency floor: the leakage-immune lower bound MSE $-$ $B^{\mathrm{up}}_{\mathrm{insuff}}$ is -0.2 (95% CI $[-0.4, -0.0]$), i.e. below zero, as a lower bound on the non-negative B_{closure} must sit when the LFER predicts as well as a flexible within-scaffold regressor. There B_{closure} is not separately resolvable and is consistent with a well-specified closure. The per-scaffold systematic offsets are $\leq$ 0.19 in pK_a, the largest 0.184 on phenol; Lemma 1(b) bounds B_{closure} below by their *square*, so the Jensen bound here is $\leq$ 0.034 against a total of 0.303, or 11%—read unsquared it would be 63% and would contradict the sentence it supports. The degraded pole (n=846) degrades to RMSE 2.26, dominated by input insufficiency. Here $z^\star$ is the pair $(s, \sigma^\star_{\mathrm{H}})$—the reaction series, which fixes $(\mathrm{p}K_{a,0}, \rho)$, and the constant—and neither coordinate needs binning, four labels and a scalar, so—unlike the solubility regime—$B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(\mathrm{p}K_a \mid s, \sigma^\star_{\mathrm{H}})]$ is estimated *directly* rather than through a coarsening of the closure's argument (§2.1.1): a within-series out-of-fold regression estimator gives $B_{\mathrm{insuff}} = 4.5$ (95% CI $[4.0, 5.1]$; $\approx$ 89% of the error—$\sigma^\star_{\mathrm{H}}$ carries nothing from the zeroed positions, exactly the a-priori channel assignment), leaving a closure misspecification $B_{\mathrm{closure}} = 0.6$ ($[0.3, 0.8]$, a stratified within-scaffold molecule bootstrap; both intervals are percentile intervals of 4000 resamples of molecules within each series, at their sizes). Directly estimated is not point-identified: the estimator is an out-of-fold residual mean square, which is an upper estimate of $\mathbb{E}[\mathrm{Var}(\mathrm{p}K_a \mid s, \sigma^\star_{\mathrm{H}})]$, so 4.5 is an upper figure and the 0.6 it leaves is a lower one—one-sided in the same direction as everywhere else in this paper, and narrower only because a four-valued label and a scalar need no binning (§2.1.1, §S7). Conditioning on s is what Lemma 1 requires of a closure whose $(\mathrm{p}K_{a,0}, \rho)$ are per-series, and it is also what keeps the bound from being vacuous: pooled over the four series the same estimator returns 8.3, above the pole's own MSE of 5.1, and licenses nothing. This is the near-exact-to-degraded transition, measured rather than asserted. This places the pKa closure near the phase-diagram boundary in its clean regime.

Data quality control (uniform, closure-independent). The trained runs use the canonical full OPERA set (7904 records). A small number of entries are supplied as pre-ionized microstates—an $[NH^+]$-protonated aminopyridine at $pK_a = -7.78$, a pyrylium $[O^+]$, among others—inconsistent with a neutral-scaffold Hammett assignment, and they degrade the clean pole (pyridinium RMSE $0.63 \to 3.83$). We drop them with one rule, *net formal charge* $\neq 0$, applied uniformly to both strata. It is closure-independent (it never references the oracle residual, so it cannot be circular) and keeps net-neutral substituents such as nitro $[N^+][O^-]{=}O$. This removes 5 of 163 clean-pole rows (restoring that pole to RMSE $\approx$0.55) and 352 of 1198 degraded-pole rows (leaving 846; that pole's RMSE is essentially unchanged, so the filter does not manufacture the sign), for 1004 Hammett-amenable rows. The clean pole's floor does turn on it: MSE $- B^{\rm up}_{\rm insuff}$ is +0.55 on the 163 unfiltered rows against the −0.2 reported on the 158 filtered, so that pole reads as well specified only after the filter.

Evidence for the flip. We train the physics arm (encoder $\to$ $\hat{\sigma}_{\rm H}$ $\to$ fixed LFER) on K=20 stratified 80/20-within-scaffold splits—a Bemis–Murcko split collapses the meta/para pole to two ring systems and is unstable—and record the paired per-molecule error against the deterministic oracle, per stratum. Clean pole: learned MAE 0.48 ± 0.09 vs oracle 0.315 ± 0.07, gap −0.165, oracle better in 19/20 splits, 90% CI $[-0.28, -0.04]$. Degraded pole: learned 0.83 ± 0.07 vs oracle 1.62 ± 0.08, gap +0.79, oracle worse in 20/20 splits, 90% CI $[+0.57, +0.95]$ (both intervals the 5th and 95th percentiles of 4000 draws of the scaffold-cluster bootstrap, one cluster per reaction series).

What the two bootstraps do and do not resolve. We report the sign statements above rather than the bootstrap P values, because neither bootstrap here carries the resolution a two-decimal P implies. The cluster unit is the Hammett reaction series, and only G=4 exist (benzoic acid, phenol, anilinium, pyridinium): the resampling law then has $\binom{7}{4}$=35 distinct outcomes and a finest atom $(1/4)^4 \approx 0.004$, so the cluster-bootstrap P(oracle worse)=1.00 on the degraded pole is exactly the statement that all four series means favour the learned arm, and P=0.004 on the clean pole that exactly one of them does, the other three favouring the oracle. Read as a sign test on four units those are p=1/16 and $p\approx$0.31; the interval endpoints quoted above are percentiles of the same 35-outcome law and are no finer. This is the same small-G caution applied on the solubility side (§S3), where the P values are likewise read as a coarse band rather than as two-decimal quantities. The split bootstrap adds nothing independent: the 20 splits resample train/test assignments over the same molecules (each molecule lands in $\approx$4 test sets), so it prices split-to-split noise rather than molecule sampling, and with all 20 degraded-pole gaps positive it returns 1.000 by construction—a restatement of the 20/20, not a second confirmation. What the flip rests on is therefore the sign consistency across splits and series and the size of the degraded pole's gap (+0.79 against an oracle MAE of 1.62, $\approx$49%), not on a P value. A leave-one-series-out refit, the estimate that would settle the series-level question at G=4, was not run.

The crossover and its scope boundary. The sign of the grounding gap is by construction the sign of the learned arm's error against the oracle's. What a competence sweep (train fraction $\in \{1.0, 0.5, 0.25, 0.12\}$ and, as the low-competence extreme, scaffold extrapolation; six seeds each) adds is that the oracle side of that comparison is fixed by the fidelity set and not by the budget: its threshold is the same at each of the four budgets, while the learned side moves. On the clean pole the learned arm rises from 0.46 to 0.98 as data shrinks but never reaches the 0.31 oracle threshold, so grounding helps throughout. On the degraded pole the learned arm stays below the 1.62 threshold across the whole interpolation range (0.85 at full data, 1.31 at 12%), so grounding hurts throughout; only scaffold extrapolation, where $\hat{\sigma}_{\mathrm{H}}$ breaks to 2.6 ± 2.0, pushes it back above the threshold and reverts the sign. That reversion is a property of the mean over the six splits and not of every split: three of the six still have grounding hurting, and the negative mean gap (-1.04) is carried by two splits at -3.2 and -3.8. The flip is therefore a within-distribution, fidelity-driven crossover, and the extrapolation reversion follows the same comparison. A trained physics-vs-DirectGNN comparison agrees in direction overall but has weak absolute errors and an untuned control; the load-bearing second-domain results are the decomposition above and the sign-consistent flip.

S9 Data provenance and reproducibility

The broad IDAC set (n=477), and the UD profile database. The UD σ-profile database is the 2261-compound COSMO set reported by Fingerhut et al.[S10] and redistributed with the benchmark implementation of Ref.:[S1] 1941 compounds from the University of Delaware collection assembled in Sandler's group, which extends the freely available VT set with revised molecular conformations for about a third of its compounds, plus 320 generated by the authors of Ref.[S10] It ships both the Mullins-averaged single profile the 2002 kernel prescribes and the typed Hsieh three-profile set the 2010 kernel prescribes, with the cavity volume and, for most compounds, a tabulated dispersion parameter. The broad IDAC set is the intersection of that database with our expanded ThermoML infinite-dilution pull, at every temperature rather than only 298 K.

How the 14,900 *raw rows are extracted.* The pull walks the NIST ThermoML archive JSON records for four journals and keeps a row only where: the dataset has exactly two components; the property is an activity coefficient at infinite dilution, either declared so by a dataset constraint or carried at a solute mole fraction of zero; exactly one component is the non-solute; a temperature is resolvable for the value; and γ^{∞} is finite and positive. Components are mapped to SMILES through the record's own InChI. Duplicate (solute, solvent, temperature) records are then aggregated, with the record and DOI counts retained per row, which takes 14,910 extracted values to 14,900 rows. No solubility-side, method-side or quality-side filter is applied at this stage.

Construction ladder. 14,900 raw IDAC rows; 14,900 with a finite $\ln \gamma_2^{\infty}$ and temperature; 490 whose solute *and* solvent both resolve into the UD compound list by InChIKey (first-block fallback); 490 for which both the Mullins and the typed profile files exist and parse at the expected 51 and 153 bins; and 477 for which both components additionally carry a tabulated dispersion ϵ, which the *2010, dispersion on* arm needs and which we impose on every arm so that the three kernels are scored on identical rows. That leaves 78 solutes, 36 solvents and 35 distinct temperatures from 273.2 to 403.1 K, with $\mathrm{Var}(m) = 4.85$ and $\overline{m} = 3.42$—a harder and far less ideal regime than the VT-2005-matched set's $\overline{\ln \gamma_2^{\infty}} = 0.75$. It is not disjoint from the VT-2005-matched set: 57 of the 60 VT-2005-matched pairs appear in it at 298 K, evaluated there on UD rather than VT-2005 profiles. The two are therefore the same chemistry measured against two profile databases, not independent samples, and we never combine them. The 3 VT-2005-matched pairs it excludes are exactly the 3 that carry no tabulated dispersion ϵ: bromobenzene and iodobenzene in 1, 2-ethanediol, and 2-bromopropane in pyridine. The "dispersion-defined 57" of §3.5.3 and the "57 shared with the broad IDAC set" are therefore one set and not two.

Why neither set can observe B_{insuff} directly. $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^{\star}, T)]$ would be point-identified only by two or more independent determinations at a common closure argument, and neither design supplies one. VT-2005 assigns exactly one profile per compound; the 44 matched compounds have distinct profiles (closest pair L_2=3.4 against a median profile

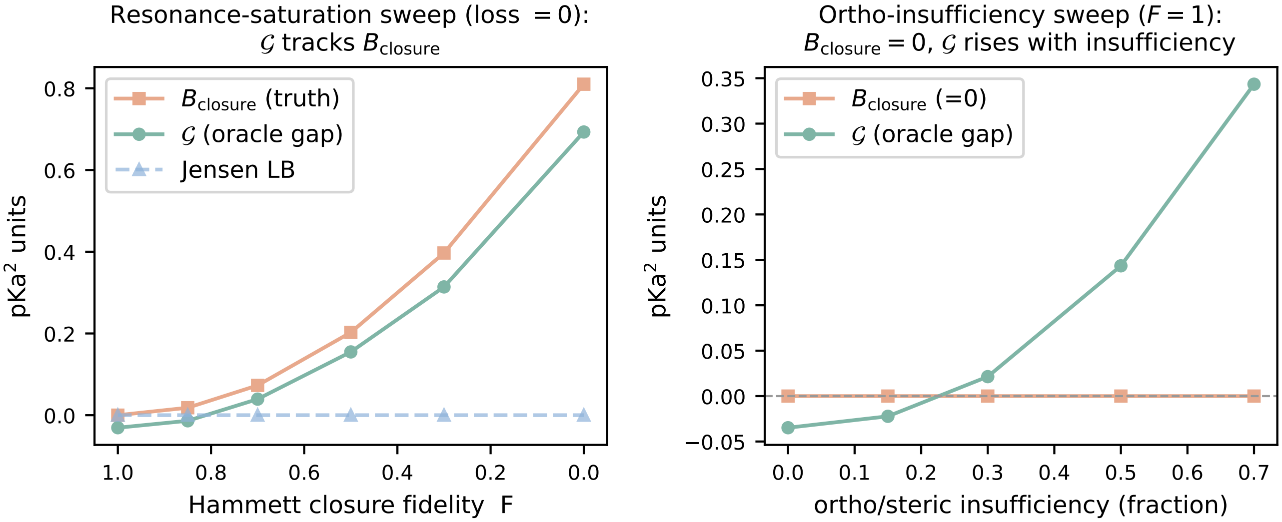

Figure S8: pKa through a Hammett LFER on a construction with known ground-truth channels. *Left:* closure fidelity varied with a sufficient intermediate; grounding gap $\mathcal{G}$, closure bias B_{closure}, and the assumption-light Jensen bound. *Right:* an ortho/steric insufficiency channel varied with a *well-specified* closure; $B_{\text{closure}} = 0$ at each knob while $\mathcal{G}$ rises with the insufficiency fraction, crossing zero near 0.25. Scope: §S8.

norm of 44), and no pair carries a replicate, so on the VT-2005-matched set no two rows sit at a common $z^\star$. The broad IDAC set comes closer without reaching it: 359 of its 477 rows share a molecule pair with another row, but at a different temperature and hence at a different $(z^\star, T)$. On these two sets B_{insuff} is accordingly reached only as an upper bound—by coarsening in the law-of-total-variance cells, and out of fold in the fifteen fitted rows of Table S9—and B_{closure} correspondingly only as a lower bound; where the latent is a scalar the headline estimate is the out-of-fold residual, one-sided in the same direction (§2.1.1, §S8).

The matching rule, and its asymmetry with the VT-2005-matched set. The VT-2005-matched set is matched by canonical SMILES and this set by InChIKey with a 14-character first-block fallback, which is looser and, since a $z^\star$ imported across a tautomer, protonation or stereochemical variant inflates MSE and hence B_{closure}, looser in the anti-conservative direction. We therefore count it. Of the 90 distinct molecules, 86 resolve by full InChIKey; four resolve only through the first block (`C=CC1CC=CCC1`, `CCC(C)N`, `CC(Cl)CCl`, `CCC(C)C(C)=O`), touching 12 of the 477 rows. In all four the discrepancy is the stereochemical layer and not a tautomer or a protonation state: each query key ends `-UHFFFAOYSA-N`, an unspecified stereocentre, against a stereo-labelled UD entry with the same connectivity. On the 465 exactly matched rows (178 distinct pairs) the decomposition is unchanged, read at the headline cell of the grid (eight equal-count bins, unbiased within-bin variance, edge values to the upper bin), which is what recomputes these two lines from the deposit: deployed convention MSE 1.930, $B_{\text{insuff}} \lesssim 0.699$, margin $+0.53$ at $P \approx 0.9$; full convention 1.926, 0.495, $+0.94$ at $P > 0.95$.

Composition, and where the squared error sits. The set's chemistry is small in both channels: n-alkanes, aromatic hydrocarbons and ketones supply 243 of the 477 solute rows, and water and the four ethylene-glycol oligomers 293 of the solvent rows. Its overlap with this paper's solubility corpus is correspondingly asymmetric—only 0.46% of corpus rows carry a solute from this set, while 34 of its 36 solvents cover 59% of corpus rows—which is what the main text means by the set covering the solvent side of deployment and not the solute side. The error concentrates in the same places: under the deployed convention 45% of the squared error sits in the four ethylene-glycol solvents and 32% in the 37 rows carrying water as solute, and under the full convention 49% sits in those same four solvents. The fidelity null of §3.5.3 is therefore a small-molecule and glycol-heavy null, and the same concentration is what makes the aggregate margin a composition statistic (Table S11). Broken out by the classification of that table, three solvent classes holding 47% of the rows carry 74% of the squared error while the acceptor-only aprotics carry 0.6% of it on 12% of the rows, and by publication the three largest contributors—32%, 24% and 20%—carry 76% between them.

Design and stratification. The 477 rows are 185 distinct solute–solvent pairs: 118 pairs carry one row, 38 carry five or more, and the largest carries 15. Counted per row the design ratio is $\dim(z^\star, T)/n = 103/477 \approx 0.22$; counted per distinct pair, 0.56, against 1.7 on the VT-2005-matched set. Molecule-pair identity coarsens $(z^\star, T)$, so $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$ is a valid B_{insuff} upper bound over the 359 rows that support it; it is 0.046, under 1% of $\mathrm{Var}(m)$ and thirteen times tighter than the binning bound on those same rows (0.61 deployed, 0.39 full). We report it and do not promote it, because all 67 multi-row pairs come from a *single* publication each: those rows are one laboratory's temperature series, so the bound omits the between-determination component of the superpopulation (§2.1.1) while the binning bound, whose eight bins mix all 15 sources, does not. It bounds a sub-component, so the reading that holds is that the binning bound is slack rather than that B_{insuff} is near zero. The same fact —that the temperature axis carries under 1% of $\mathrm{Var}(m)$ at a fixed profile pair—makes the choice between conditioning on $z^\star$ and on $(z^\star, T)$ numerically immaterial on this set, though only the latter is licensed by Lemma 1 and it is the one we use. Collapsing to one row per pair (each the row nearest 298 K) gives MSE $= 1.679$, $B_{\text{insuff}} \lesssim 0.495$ and a margin of $+0.69$ at $P > 0.95$ under the full convention (1.540, 0.635, $+0.27$, $P \approx 0.8$ deployed). By method, 334 rows are chromatographic, 37 chemical or heterogeneous equilibration and 106 carry no method field; by source, 15 publications in four journals (*J. Chem. Eng. Data* 203, *J. Chem. Thermodyn.* 114, *Fluid Phase Equilib.* 108, *Thermochim. Acta* 52), the largest single publication contributing 105 rows. On the 334 chromatographic rows alone, re-binned within the stratum, the activity error is MSE $= 1.783$ against $B_{\text{insuff}} \lesssim 0.36$ under the full convention—a margin of $+1.06$ ($P > 0.95$), wider than the full set's $+0.95$—and 1.474 against $B_{\text{insuff}} \lesssim 0.58$ under the deployed one, a margin of $+0.31$ ($P \approx 0.9$), narrower than the full set's $+0.51$. The cut therefore does not move the sign under either convention, and it does not uniformly widen the margin either. A DOI-clustered bootstrap of the full-set margin gives $P > 0.95$ with 90% $[+0.50, +1.57]$ (full) and $P \approx 0.9$ with $[-0.07, +1.74]$ (deployed), and a pair-clustered one $P > 0.95$ with $[+0.71, +1.27]$ and $[+0.23, +1.16]$ respectively. This is the stratification the 60-pair VT-2005-matched set cannot support, and it is why the broad IDAC set carries the map: the quality cut that *inverts* the enlarged 298 K set (§3.7, limitation (i)) leaves the sign alone here under either convention. What it does not buy is an aggregate verdict, for the reason Table S11 and §3.5.1 give: the chromatographic cut and the pair collapse each leave the sign standing, and the chemistry cut compounded with the pair collapse does not.

Scoring the deployed closure against the published evaluation. The deployed our differentiable COSMO-SAC layer—its own constants and damped fixed point—was run over the same three record files De Souza et al.[S21] score, with components resolved through the repository's CAS-to-InChIKey map and matched by exact InChIKey to this paper's VT-2005 artifact: 7889 of 8138 usable records (96.9%), 2279 distinct pairs, each at its own temperature. Those are the rows of Table 2's middle block, and the same 7889 rows are the out-of-sample set of §3.5.1.

Sources. Solubility: BigSolDB 2.0,[S22] partitioned by Bemis–Murcko scaffold—molecules grouped by ring system plus linker, side chains stripped. Crystal $T_m/\Delta H_{\text{fus}}$: an open melting-point pool plus a group-contribution prior. True σ-profiles: the VT-2005 database, the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for ∼1,400 compounds[S23] (ingested to 1432 compounds, with the COSMO cavity volume V_{cosmo} for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; mismatch to the experimental measurement conditions is noise in $z^\star$ that inflates B_{insuff}, but under a Berkson-type displacement it also loads onto B_{closure} through the curvature of g, so we treat it as an unsigned limitation and *not* as conservative for the closure-dominance verdict (§3.5). Concretely, if the true latent is $z^\star$ displaced by a mean-zero (Berkson-type) ε, then even a perfectly specified g has $B_{\text{closure}} = \mathbb{E}\big[(\mathbb{E}_\varepsilon[g(z^\star{+}\varepsilon)] - g(z^\star))^2\big]$, a pure curvature effect that vanishes only for affine g—and under classical rather than Berkson error, not even then. We do not bound that curvature share. B_{closure} is accordingly named for the closure and measured on the composite of closure and profile calculation, which is how §2.1.1 defines it and how every attribution in this work reads it; §3.5.3 sizes the exposure against the two-database bracket of the published evaluation and leaves the share unbounded. Infinite-dilution activity coefficients: ThermoML, the NIST/IUPAC archive of experimental thermophysical data.[S24] Enlarging the reference σ-profile set would not materially grow the matched activity set: of the 298 K ThermoML IDAC pairs, only two solutes lack a VT-2005 profile while their solvent has one—the binding constraint is joint IDAC×profile coverage, not profile availability. What does enlarge the VT-2005-matched set is a broader extraction on the IDAC side: the expanded ThermoML pull released with the code matches 115 pairs at 298 K. We do not adopt it as a replacement set because its verdict is quality-cut sensitive: it widens the margin from 0.24 to 0.35 but *inverts* it to −0.13 on the 100 pairs labelled gas chromatography, and that sensitivity is unresolved at 298 K. It is not a verdict on the extraction, which matters because the broad IDAC set is built from the same pull: there the same cut leaves the sign alone under either convention (+0.51 → +0.31 deployed, +0.95 → +1.06 full). What differs is resolution, not provenance—at 298 K a 15-row cut inverts a margin already inside the design's resolution, whereas on 477 rows a 143-row cut moves one without inverting it—so we use the enlarged pull where it is stratifiable and keep the strict $n{=}60$ set where the paradox's own oracle is defined. The deployed VT-2005-matched set is also narrowly sourced: the curated IDAC compilation yields 74 VT-2005-matchable pairs under canonical SMILES (60 at 298 K plus 14 off-temperature), all by gas chromatography and all from three publications, 33 pairs from one, so neither a method-stratified nor a source-clustered re-estimate is available on it. The VT-2005-matched set is matched to VT-2005 by RDKit canonical SMILES on solute and solvent, the rule the deployed σ-oracle itself uses; matching by InChIKey instead admits 85 pairs over 46 solutes and 19 solvents at 298 K in place of 60/41/17 (both counts are recorded in the deposited decomposition artifact). We keep the stricter rule so that the $z^\star$ charged against each label is the profile the deployed pipeline would supply, rather than one imported across a tautomer or protonation variant—the same mismatch that makes the reference imperfect above. The decomposition was not re-estimated on the InChIKey-matched set; whether the verdict moves on those 25 extra pairs is untested. The broad IDAC set uses the looser InChIKey rule, so the two sets are matched under different standards; that asymmetry is quantified above (four molecules, 12 rows, verdict unchanged on the exactly matched subset) rather than left implicit. Second-domain pKa: the OPERA experimental compilation[S18] (7904 QSAR-ready records, public-domain US EPA; cited, not redistributed); the uniform net-formal-charge filter reduces this to the 1004 Hammett-amenable rows (158 meta/para, 846 ortho/heteroaromatic) analysed in §S8. The Bemis–Murcko scaffold split follows the main text. Because the seeded split is not stable across pipeline versions (§2.2), the numbers in this paper are computed on a *single, pinned* split, whose per-file SHA-256 each run's manifest records and whose files are deposited with the checkpoints rather than versioned—with the one run whose split provenance was not logged, recorded in the footnote.[4] Melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

Grounding-stream provenance, and one build that violates the guard. The σ-stream builder drops any pool molecule whose Bemis–Murcko scaffold appears in the validation or test split and aborts rather than proceed when a split file is missing; a correct build retains 1319 of the 1427 VT-2005 molecules that parse into the pool. The stream file was subsequently rebuilt with the exclusion pointed at the splits of a small debug dataset: it carries 1425 rows, of which 106 (99 train, 7 validation) have a scaffold in the real test $\cup$ validation union of 2702 scaffolds. The overlap is not confined to scaffold relatives: 91 of those 106 rows are themselves held-out solutes by canonical SMILES — 65 rows carrying 64 distinct test-split molecules and 26 carrying 26 validation ones — and the correctly guarded build leaves none of them.

Those are counts over the split *files*, which carry the auxiliary melting-point rows (auxiliary-only, no solubility label) alongside the labelled ones and so span 2634 and 2467 distinct solutes. Every test metric in this paper is computed on the 5608 labelled rows, which span 147 solutes, and that is the set a leak can act on. Of those 147, 7 have their own reference profile in the defective stream and 8 have a scaffold relative in it, on 297 and 351 of the 5608 rows, 5.3% and 6.3%. For a scaffold split solute coverage is the load-bearing statistic rather than the fraction of stream rows, so these are the figures the disclosure of §2.2 carries; the correctly guarded build leaves zero of each. The remaining 57 leaked molecules appear in the test file only as melting-point rows and enter no reported metric, and the split holds nothing out on the solvent side, so the stream's coverage of test solvents violates no guarantee this split makes. The crystal stream's build on disk records the correct 2702-scaffold exclusion, so among the streams feeding the runs reported here the defect is confined to the σ one. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm.

What the file times establish is narrower than their ordering first suggests. The local stream path was created on 2026-06-21 and last written on 2026-07-06, and the content that write left is the defective one. Every prediction file of §3.1 postdates the creation, so the ordering separates the *last write* from the runs rather than the defective build from them: the prediction files behind Fig. 2 and Table S3 were created before 2026-07-06 and those of the two single-seed oracle-application controls (§S3) and the output-residual arms (§3.3) after it. Creation, not modification: the seed-44 σ-oracle copy carries a later write, from the recovery described below. A guarded build overwritten on 6 July and a build defective from 21 June are equally consistent with that. These are in any case modification times on local copies: the runs read an uploaded data bundle, the one bundle retained on disk was assembled on 2026-07-13, after all of these runs, and a local write does not date a read. The ordering therefore licenses no build provenance for any arm, and in particular does not establish that the two single-seed controls differ in build from their seed-42 comparator. The +0.18 and +0.27 of §3.1 are accordingly single-seed penalties of uncertified stream provenance, and "confound-free" names the test-time input swap they remove and nothing about the stream.

Three facts bound what the leak can do, and each bounds one contrast. It is common-mode across the *paradox* contrast, the learned-σ and σ-oracle rows being one checkpoint eval-

[4] The three-seed surrogate isolation (§3.3) was produced on a separate run whose per-run split provenance was not logged; its matched set is the same n=44 molecules, derivable from the committed split independently of the model, so it is consistent with the reported split. We record the split hash in the runner so future runs are auditable directly.

uated two ways, so it cannot generate the gap between them; DirectGNN uses no grounding stream, so in the *physics-versus-control* comparison a leak can only favour the physics arm that loses to it; and on the n=44 matched set of §3.3 a correct guard would have removed one molecule (piperidine). A fourth contrast none of the three reaches is the *grounded-versus-ungrounded* supervision contrast. The ungrounded arm sets its warm-up and stream budget to zero and so carries no σ stream either, which leaves the grounded arm as the only arm a leak can reach, in the direction the guard exists to prevent. The 0.20 supervision gain is therefore the one reported contrast that a leak could inflate rather than one it is bounded on, and it stays uncertified against the scaffold split until the leak-free re-run. What the defect does compromise is auditability, and with it the reach of the split guarantee: an arm trained on that build would have had reference σ-profiles — though no solubility labels — for the 7 evaluated test solutes above, and the guarantee stated in §2 is therefore a property of the released builder rather than of every stream file we trained on. Closing this requires hashing each stream build beside the split hash, which was not done for these runs.

What the leak-free re-run reports, and what it decides. The re-run is specified here rather than described afterwards, so that its protocol is fixed before its numbers exist. It trains the ungrounded and the grounded arm at five seeds (42 through 46) under the COSMO-SAC configuration of the published arms, unchanged. The σ stream is rebuilt with the scaffold guard pointed at the validation and test files of *this* split, and is certified inside each training run rather than at the builder: the run re-reads the stream it is about to consume and the split files it will be scored against, and asserts that they share no molecule by canonical SMILES and none by Bemis–Murcko scaffold. The stream's SHA-256 and row count are written into that run's manifest beside the SHA-256 of each pinned split file, so the artifact states which build fed which arm; the ungrounded arm's manifest records that it carried no stream at all, since here the absence is the thing asserted. Accuracy is read on the same 5608 labelled test rows as every other number in this paper, under the same cross-arm intersection rule (§2.4), and the five per-seed MAE values are reported for each arm rather than a mean and a standard deviation alone. Two further readings are taken on those same checkpoints: the evaluation-only substitution of §3.1, which is within-checkpoint and therefore a consistency check rather than a test, and the n=44 profile comparison of §3.3 with the one molecule the leak touches removed. Quoting the magnitude of the co-adaptation control of §3.1 as something other than one seed would need a separate pass at the same five seeds, which this protocol does not commit to running.

The decision the re-run settles is stated in advance and in one direction only, since it is the supervision contrast alone that hangs on it. If the five per-seed gains are of one sign, the two senses of grounding keep their opposite signs, §3.1 and Table S2 print the re-run's per-seed values in place of the three uncertified ones, and the abstract's scope clause moves from three seeds to five, with the refusal to order the two operations by magnitude unchanged. If the per-seed gains straddle zero, supervision has no established sign at five seeds: the paper then reports one operation with a sign—substituting the reference at inference hurts—and records the other as unresolved, and the abstract, the heading of §3.1, the opening sentence of the Conclusions (§4) and the supervision row of Table S2 are rewritten to say so. If the gains are of one sign and it is the opposite one, that is what is reported. In either of the last two cases the graphic for the table of contents, which prints the two senses of grounding and their opposite signs, is redrawn to what is then reported, and the freeze contrast of §3.3 is restated as well, because the warm-up state whose quality it turns on is the supervised one. The substitution result is unaffected in all three cases: it is one checkpoint evaluated two ways, and no σ-stream build can reach it.

Which displays the five-seed values replace is fixed here as well, and in all three branches alike, because a five-seed table printed beside a three-seed figure would be a contradiction, and which displays move is not a choice to be taken once the numbers are known. Three arms carry those values: the ungrounded arm, the grounded arm, and the σ-oracle arm, which is the grounded checkpoint re-evaluated with $z^\star$ substituted and therefore moves with it. Each display named next carries at least one of the three and is regenerated from the five-seed arms by the script that produced it—the abstract's seed count and its leak-free clause, which carry no value of their own; §3.1 entire, together with the two sections that carry the rest of its prose—§3.2, which carries the regression slopes read off Fig. 3, the ranking contrast and the solute-blind comparison, and §3.4, which carries its channel paragraph—the per-seed gains and the seed-42 values included wherever they now print; the five rows of Table S2 under *the two senses of grounding* and the +0.14 of its accuracy row; the seed list of the σ-grounding row of Table 3; the ungrounded, grounded and oracle bars of Fig. 2 with their R^2; panels (b), (c) and (d) of Fig. 3; the block-1 grounded and σ-oracle rows of Table 4 together with the clip sweep in its caption, the fuller sweep of §S4 and the readings §3.7 takes from both; Table S3 and the unrounded means, per-seed values and arm orderings §S3 reads off it; Table S18 with the ranking and per-group recalibration readings §S6.1 sets around it, whose two columns are the grounded checkpoint and its own σ-oracle re-evaluation and nothing else; §S6.4 with Tables S20 and S21, which resolve that same +0.14 along chemical axes and restate the two arm means it is taken from; the row-class split of the substitution in §S1, whose −0.000, +0.431 and +0.006 put the +0.41 on the solvent channel; and the graphic for the table of contents, whose caveat that training on the reference is not certified leak-free is removed in all three branches alike, the re-run certifying the stream inside each training run whichever way its per-seed gains fall. Four of those floats—Figs. 2 and 3, Tables 4 and S3—then carry re-run arms beside arms that were not re-run, so each states in its caption which of its arms came from the re-run—Table 4's block header already carries the form of words for it, and Fig. 3 remains one seed for all four of its panels, the new seed 42 in three of them and the published one in the control. Table S18 needs no such statement, both its arms being re-run; Tables S20 and S21 print no arm at all, every cell being a difference against the control, so each says in its caption that only the physics side is five-seed. Three numbers instead keep their published value with the five-seed one printed beside them and the caption saying so, because the re-run supplies one side of each comparison and not the other: the Staverman–Guggenheim contrast 1.8457 → 1.8788, whose combinatorial arm is not retrained; the channel swap's +0.27, a seed-42 checkpoint that is not retrained either; and the co-adaptation control's +0.18 together with the 0.59 share of §3.1, which move only if the separate five-seed pass above is run. Three passages carry no arm value but state that the 0.20 is uncertified—the disclosure above, limitation (xii) of §3.7 and the provenance paragraph of §S9—and the in-run certification answers all three, so each is rewritten to what the re-run certifies and limitation (xii)'s 0.20 moves with §3.1's. Nothing else moves, and the boundary is worth stating because two neighbouring numbers sit just outside it: neither the DirectGNN control nor the NRTL arm is retrained, so their rows in Tables 4 and S3, and the endpoint comparison of §S6.5—which is the NRTL arm against that control—stand as printed; and §3.5 uses no trained model at all.

The two decomposition sets, deposited row by row. Because the closure/insufficiency verdict rests on a newly constructed set, both sets are deposited with the Supporting Information as row-level tables that need no code to inspect. The results that otherwise require re-running the code are, specifically, the broad IDAC set with its design and provenance audit, the 2010 kernel's own-latent floors, the fitted kernel residual and the estimator grid. **Data Set S1** carries one row per measurement: pair identifier, solute and solvent SMILES, the UD compound key matched to each and which rule matched it (full InChIKey or first-block fallback), temperature, the experimental $\ln \gamma_2^\infty$, the *four* closure outputs—2002 and 2010-dispersion-off, each in the full and the residual-only combinatorial convention, so that the deployed-convention kernel contrast of §3.5.3 is recomputable from the file—the ThermoML method field, the source DOI and journal, and a flag for the 57 rows shared with the **Data Set S2** (printed in full as Table S24) carries the same fields for the 60 VT-2005-matched pairs, with three differences that follow from how the VT-2005-matched set is defined. Its closure outputs are the VT-2005 ones the deployed σ-oracle uses, in both combinatorial conventions, alongside the four UD-profile outputs that make it commensurable with Data Set S1. Its matching rule is canonical SMILES throughout, so it carries neither the per-row match-rule columns nor the UD compound keys. It also names each compound in words as well as in SMILES. It additionally carries a per-row record count, 1 throughout. All 60 rows sit at 298 K, all are chromatographic, and their three source DOIs divide the set 33/20/7. The counts, margins, stratified re-estimates and clustered bootstraps reported for either set are recomputable from these two files with the estimator definitions of §2.5; what they do *not* restore is the provenance of the closure evaluations themselves, which still requires the code.

Compute ledger. The headline σ-grounding comparison (Fig. 2, Table S3) is a full-corpus GPU run (three seeds); the closure/insufficiency decomposition (§3.5), the Fisher audit (§S6.2), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §S5 are CPU-reproducible from deposited artifacts. The prediction CSVs, the model checkpoints and the larger analysis artifacts will be deposited separately, in a Zenodo archive **[whose DOI is registered at submission]**, rather than in the repository. The broad IDAC set's design and provenance audit—distinct-pair counts, within-pair variance across temperature, pair-blocked out-of-fold estimators, method and source-DOI composition, the method-stratified and cluster-resampled margins, the kernel contrast split into the shared 57, the complementary 420 and the 80 complementary rows that are also at 298 K, the VT-2005-matched floor recomputed on UD profiles, and both sets' floors under the 2010 kernel's own typed latent—regenerates in one CPU run of the deposited design-and-provenance audit. A second CPU run, the convention audit, produces the convention-rule material: both sets' margins and clustered bootstraps under each combinatorial convention, the per-cell broad-set grid of Table S5, the InChIKey match-rule audit and its exactly matched re-estimate, the pair-conditional bound and its single-source diagnosis, the typed latent's dimensions, and the two deposited row-level tables. Those files are among the artifacts limitation (x) records as uncheckable until the repository is released; the two deposited tables are the part of that material which can be checked without it.

Seeds and determinism. The controlled comparisons of §3.1 use seeds $\{42, 43, 44\}$ and are reported on the labelled test rows (5608 of 8103; §S3.2); the surrogate isolation (§3.3) uses seeds $\{0, 1, 2\}$; the training-time and channel-swap oracle controls, the output-residual arms (§3.3), the data-efficiency curve (§S6.5) and the loss-simplification ablation (§S1) are seed 42 only. The decomposition is seed-independent (its closure has no learnable parameters).

One deposited file was partial, and has been repaired. The released per-row prediction file for the seed-44 oracle solubility arm carried 295 of the run's 8103 rows. The cause was not the run: the file the run wrote is intact on the compute volume, and the transfer that produced the deposited copy stopped mid-record—the deposited file is an exact byte-prefix of the complete one, 198,190 bytes of 5,213,388, ending part-way through the 295th line. The complete file has been recovered from that volume and now stands in the deposit, and it is the file the run scored: every field of that run's own summary—$n_{\text{rows}} = 8103$, n labelled $= 5608$, MAE 2.24199, RMSE 2.96058, $R^2 = 0.01768$, bias 1.00945, and the matched-subset block's $n = 5571$ with MAE 2.24373—recomputes from

Table S24: **Data Set S2** in full: the VT-2005-matched set (n=60), row by row, deposited so the set can be inspected without running code. Read in two blocks of 30, left then right.

#	solute[a]	solvent	m	g_{full}	g_{res}	broad	#	solute	solvent	m	g_{full}	g_{res}	broad
1	`Brc1ccccc1`	`OCCO`	3.81	2.33	2.40	n	31	`CCc1ccccc1`	`c1ccncc1`	0.36	0.35	0.39	y
2	`C#CCCCCC`	`c1ccncc1`	0.44	-0.42	-0.34	y	32	`CO`	`C1CCNCC1`	-0.84	-2.38	-2.21	y
3	`C1=CCCCC1`	`c1ccncc1`	0.91	0.75	0.75	y	33	`CO`	`CC(C)(C)N`	-1.24	-3.15	-2.99	y
4	`C1CCOC1`	`c1ccncc1`	-0.07	0.03	0.03	y	34	`CO`	`CCC(C)N`	-0.97	-3.20	-3.02	y
5	`C=CCCCCC`	`c1ccncc1`	1.27	0.93	1.03	y	35	`CO`	`CCCCCCN`	-0.97	-3.23	-2.87	y
6	`CC#N`	`c1ccncc1`	0.38	0.25	0.30	y	36	`CO`	`CCCCN`	-1.05	-3.00	-2.79	y
7	`CC(C)(C)N`	`CO`	-1.39	-2.07	-1.85	y	37	`CO`	`CCCNCCC`	-0.37	-2.33	-1.97	y
8	`CC(C)(C)N`	`c1ccccc1`	0.28	0.39	0.38	y	38	`CO`	`CCNCC`	-0.67	-2.35	-2.15	y
9	`CC(C)=O`	`c1ccncc1`	0.17	-0.13	-0.12	y	39	`CO`	`Cc1cccc(C)n1`	-0.49	-0.78	-0.49	y
10	`CC(C)Br`	`c1ccncc1`	0.30	0.22	0.21	n	40	`CO`	`Cc1ccccn1`	-0.24	-1.07	-0.86	y
11	`CC(Cl)CCl`	`c1ccncc1`	-0.07	-0.05	-0.05	y	41	`CO`	`Cc1cccnc1`	-0.06	-1.06	-0.85	y
12	`CCC(C)N`	`CO`	-0.29	-1.99	-1.71	y	42	`CO`	`Cc1ccncc1`	-0.39	-1.11	-0.90	y
13	`CCC(C)N`	`c1ccccc1`	0.14	0.42	0.42	y	43	`CO`	`NC1CCCCC1`	-1.11	-3.21	-2.98	y
14	`CCCCCCC`	`c1ccncc1`	1.91	1.60	1.71	y	44	`Cc1ccc(C)cc1`	`c1ccncc1`	0.44	0.39	0.45	y
15	`CCCCCCCC`	`c1ccncc1`	2.07	1.74	1.91	y	45	`Cc1cccc(C)c1`	`c1ccncc1`	0.43	0.37	0.43	y
16	`CCCCCCCCC`	`c1ccncc1`	2.17	1.86	2.12	y	46	`Cc1cccc(C)n1`	`CO`	-0.06	0.14	0.64	y
17	`CCCCCCCCCC`	`c1ccncc1`	2.26	1.97	2.32	y	47	`Cc1ccccc1`	`OCCO`	4.53	2.61	2.67	y
18	`CCCCCCCCCCC`	`c1ccncc1`	2.32	2.08	2.52	y	48	`Cc1ccccc1C`	`c1ccncc1`	0.34	0.33	0.37	y
19	`CCCCCCN`	`CO`	2.62	-2.35	-1.66	y	49	`Cc1ccccn1`	`CO`	-0.29	-0.21	0.12	y
20	`CCCCCCN`	`c1ccccc1`	2.79	0.51	0.59	y	50	`Cc1cccnc1`	`CO`	-0.20	-0.36	-0.03	y
21	`CCCCCOC(C)=O`	`c1ccncc1`	0.11	0.09	0.26	y	51	`Cc1cccnc1`	`c1ccccc1`	-0.04	-0.01	-0.01	y
22	`CCCCCl`	`c1ccncc1`	0.43	0.37	0.38	y	52	`Cc1ccncc1`	`CO`	0.00	-0.41	-0.08	y
23	`CCCCN`	`CO`	-0.78	-2.18	-1.85	y	53	`Clc1ccccc1`	`OCCO`	3.69	2.29	2.35	y
24	`CCCCOC(C)=O`	`c1ccncc1`	0.14	0.04	0.14	y	54	`Fc1ccccc1`	`OCCO`	3.47	2.03	2.06	y
25	`CCCN(CCC)CCC`	`CCCCCCCCCCCCCCCC`	0.01	-0.03	0.07	y	55	`Ic1ccccc1`	`OCCO`	3.74	2.45	2.54	n
26	`CCCN(CCC)CCC`	`CO`	2.17	1.96	3.15	y	56	`NC1CCCCC1`	`CO`	1.57	-2.17	-1.81	y
27	`CCCN(CCC)CCC`	`c1ccccc1`	1.03	0.54	0.79	y	57	`NC1CCCCC1`	`c1ccccc1`	1.86	0.45	0.46	y
28	`CCCNCCC`	`CO`	-0.09	-0.31	0.40	y	58	`c1ccc2ccccc2c1`	`OCCO`	5.47	2.71	2.85	y
29	`CCCNCCC`	`c1ccccc1`	0.39	0.28	0.35	y	59	`c1ccccc1`	`OCCO`	3.47	2.11	2.12	y
30	`CCNCC`	`CO`	-1.11	-0.64	-0.30	y	60	`c1ccncc1`	`CO`	0.18	-0.28	-0.09	y

[a] Columns: m, the experimental $\ln\gamma_2^\infty$ at 298 K; g_{full} and g_{res}, the VT-2005-input COSMO-SAC-2002 output under the full and residual-only conventions; *broad*, the 57 pairs that also appear in the broad IDAC set, the three that do not being the bromine- and iodine-containing pairs of §3.5.3. The 477-row broad IDAC set is the companion machine-readable Data Set S1.

it to the last printed digit, at zero absolute difference in float64. Its 8103 rows carry no non-finite $\ln x_2$ prediction, its labelled subset is the same 5608 rows in the same order as all five other seed-44 arms, and on the 37 labelled rows where the oracle matched neither molecule its $\ln\gamma_2$ agrees with the grounded arm's to 5×10^{-6}, the two being the same checkpoint forwarded twice. Those checks are not a claim to be taken on trust: a deposited verification pass runs all of them over the deposit in one CPU pass—row count, labelled count, whether the last record is complete, non-finite predictions, cross-arm row order, and each file against its own summary—and exits non-zero if any file fails. All twenty-two pass.

Nothing computed at run time moves: Fig. 2, Table S3 and the +0.41 were aggregated from the in-memory predictions before any file was written, and they are unchanged. What the repair changes is what could be recomputed afterwards. The σ-oracle row of Table 4 had been the mean over the two intact deposits and is now a three-seed mean like the rest of its block, moving from 2.26 ± 0.02 to 2.25 ± 0.02 in MAE, from 1.57 ± 0.17 to 1.60 ± 0.14 in $\log_{10} S$ RMSE, and from -0.05 ± 0.02 to -0.03 ± 0.04 in R^2—the last of which is the same -0.03 ± 0.04 Table S3 reads, the truncation having been the whole of the disagreement between the two tables. The clip sweep of §S4 likewise carries three seeds for that arm.

The learned-σ and DirectGNN solubility files and the activity-level (n=60, n=44) artifacts were complete for every seed throughout.

S10 Black-box probe programme: full tables

All three studies use the same current-split DirectGNN checkpoints (seeds 42/43/44, test MAE 1.749/1.674/1.684) whose manifests pin the train/val/test SHA-256 to the split on disk. The probed object throughout is the control's 788-dimensional pair representation. Every number here is CPU-computed; none required a GPU.

Crystal-term decodability (Table S25). Probe fitted on 152 solutes / 14,348 rows, scored on 13 held-out solutes / 527 rows, solute overlap zero. Φ^* is 92.2% between-solute variance, so the effective n is 13 clusters.
Controls: positive—model's own prediction $R^2 = 0.980$, temperature $R^2 = 0.9997$; negative—random target -0.0006, molecule-blind constant profile -0.015, raw RDKit descriptors -0.514 (set A) and -0.928 (set B). Null $p_{95} = +0.105/+0.117/+0.096$ and max $= +0.316/+0.380/+0.346$ over 1000 draws.

Two features of Table S25 bear on how such probes should be read. First, the between-seed spread (0.616 range, sd 0.334) exceeds every individual value: a single-seed run would have been unreadable, and seed 43 alone would have read as a near-miss. Second, a decision rule of the common form $R^2(\text{model}) - R^2(\text{raw}) \geq 0.10$ would have declared success here on all three seeds under descriptor set B—including seed 42, whose probe is worse than a molecule-blind constant—and on two of three under set A. The subtrahend is an unregistered analyst choice worth more than the effect; we report $R^2(\text{raw})$ as a level beside $R^2(\text{model})$ and never as a difference.

Temperature structure (Table S26). Slopes are fitted at fixed (solute, solvent); a slope pooled across solvents would confound activity differences with temperature. Ceiling, measured on the data before any model: $r = +0.239$ on train (1353 pairs / 151 solutes) and -0.125 held out (66 pairs / 13 solutes, permutation $p = 0.581$).

Organisation of the representation (Table S27). Excess over a permutation null matched to the factor's level, on the full test set.
As a share of the solute-identity ceiling: scaffold 100.3/99.9/99.8%, $\log P$ 13.6/8.2/4.7%, TPSA 9.0/10.3/7.1%, MolWt 5.3/2.3/3.1%. That 99.8–100.3% scaffold range is the organisation measure §3.7 points here for. The scaffold share is pinned near one by the split rather than by the representation: of the 124 scaffolds spanned by the 147 test solutes only 11 hold more than one solute, so only 34 solutes share a Bemis–Murcko scaffold with another, carrying 492 of the 5608 rows (8.8%), and the two partitions coincide on 91% of the data. The raw variance shares run the other way (seed 42: scaffold 0.636 against solute 0.639), so the *raw* ratio falls short of one by 0.4/0.7/0.9% across the three seeds, and it reaches one only through the differing size-matched permutation nulls (0.022 against 0.026); no interval is attached to that 0.4–0.9% gap. The row therefore bounds what the representation adds beyond scaffold and does not measure within-scaffold discriminability (§3.7). A direct measure on the 34 solutes that do share a scaffold sizes what it bounds: two same-scaffold solutes in a common solvent sit at 0.21–0.40 of the between-scaffold representation distance across the three seeds (112 pairs over 11 scaffolds; 90% scaffold-cluster intervals inside $[0.14, 0.57]$). Molecules sharing a scaffold are compressed rather than unresolved.

Temperature is decodable from this vector at $R^2 = 0.9997$ yet occupies only 3–4% of its variance (rows above): decodability of a quantity is not evidence that a representation is organised around it, which is why the crystal-term probe is run as a held-out transfer test rather than an in-sample decode.

On the 331 rows carrying a measured Φ^*, Φ^* accounts for 18.3/42.5/18.0% of that subset's own solute-identity ceiling. This does not contradict the held-out null above. Φ^* is a solute-level quantity and the representation is organised by solute identity, so any solute-level variable shares variance with it in sample; the decodability probe above asks the different and sharper question of whether the mapping trans-

Table S25: Crystal-term decodability per seed, with the permutation null, the drop-one-solute jackknife and the leverage carried by the worst solutes.

seed	R^2(model)	perm. p	jackknife range	sign flips	top-3 leverage
42	−0.506	0.894	[−1.151, −0.228]	0/13	56.2%
43	+0.111	0.055	[−0.286, +0.333]	3/13	59.8%
44	+0.027	0.177	[−0.253, +0.237]	6/13	57.2%

Table S26: The model's temperature response against the empirical slope and against $-\Delta H_{\text{fus}}/R$, with the curvature of both in $1/T$.

	seed 42	seed 43	seed 44
corr(model, empirical slope)	+0.294	+0.406	+0.195
corr(model, $-\Delta H/R$)	+0.066	−0.283	−0.028
median |model − empirical| slope	1329	1088	1007
curvature in $1/T$ (data: 6.8×10^5)	2.0×10^6	2.1×10^6	2.3×10^6

Table S27: Share of representation variance per organising factor, as an excess over a permutation null matched to the level at which that factor varies.

factor	seed 42	seed 43	seed 44	null level
solute identity (147 levels)	+0.613	+0.587	+0.597	row
solute scaffold (124 levels)	+0.614	+0.586	+0.595	row
scaffold beyond arbitrary solute grouping	+0.088	+0.082	+0.082	solute
solvent identity (70 levels)	+0.278	+0.301	+0.271	row
temperature	+0.040	+0.029	+0.037	row
$\log P$ (solute)	+0.083	+0.048	+0.028	solute
TPSA (solute)	+0.055	+0.061	+0.042	solute
MolWt (solute)	+0.032	+0.013	+0.018	solute

fers to solutes the probe never saw. In sample the organisation is partly Φ^*-aligned; out of sample that alignment does not carry.

S11 Closure fidelity sets the reference-input penalty: a controlled synthetic family

To isolate the mechanism we construct a controlled synthetic experiment in which only the closure's fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent $z^\star$ and a target through a known map T; a fixed candidate closure g_F reconstructs it under a tunable misspecification of scalar fidelity F ($F = 1$ recovers T exactly, smaller F pushes g_F away). The reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ is a measure of B_{closure} in this construction—the free head recovers $\mathbb{E}[m \mid z^\star]$ while g_F is the misspecified map. Because ΔR^2 *equals* B_{closure} by design here, the sweep verifies that the instrument responds to fidelity as constructed; it is a check on the construction, not an independent test of the hypothesis. As F falls the penalty grows monotonically: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term) F=1.00 gives $\Delta R^2 \approx 0$, F=0.38 gives ≈ -0.33, and an over-rotation stress test (F=−0.50) gives ≈ -2.0 (Fig. S9); all 24 family × form sweeps are individually monotone, and the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), whose curves are visually coincident. That coincidence says the teacher factor carries no information *in this*

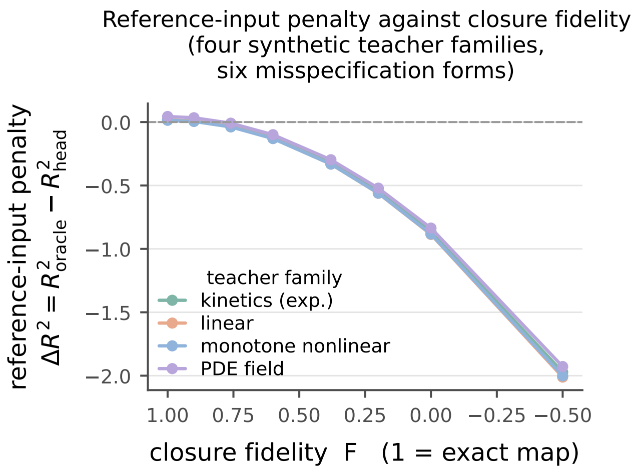

Figure S9: Closure-fidelity sweep on synthetic, non-chemical teacher functions. As the closure is made more misspecified (fidelity F falling from 1; axis reversed), the reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ falls monotonically below zero. Each curve is one of four teacher families, averaged over six misspecification forms, which the panel does not resolve. No point is marked for the real closure. *Caveat:* here ΔR^2 *equals* B_{closure} by design, so the sweep is a check on the construction and not an independent test of the hypothesis; and the four curves are visually coincident because the teacher factor carries no information in this construction, not because the effect is domain-general. Scope: §S11.

construction; it is not evidence that the effect is domain-general, and the panel does not resolve the six misspecification forms, which do spread (Fig. S9). Placing the real system on the sweep only illustratively (the real-closure-to-F map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted σ-profiles falls at $F \lesssim 0.4$: its solubility $\Delta R^2 \approx -0.36$ is what a closure of this fidelity predicts, consistent with B_{closure} dominating B_{insuff}. The two ΔR^2 values are measured on different targets—synthetic m on the sweep, $\ln x_2$ through the SLE solver and the crystal head in the real system—which is the other reason the placement carries no calibration.

The same construction shows why the grounding gap is an indicator, not a certificate (§2.1.2). Closure fidelity and input sufficiency are varied separately: when $z^\star$ is sufficient, $\mathcal{G}$ tracks B_{closure} as fidelity varies (Theorem S1's proxy is exact); when $z^\star$ is lossy but the closure is *well specified*, $B_{\text{closure}} = 0$ throughout yet $\mathcal{G} > 0$ grows with the discarded variance. A positive grounding gap can thus be pure input insufficiency, not closure error—so what closure dominance is claimed on rests on the law-of-total-variance bound of the decomposition, computed under the deployed convention (the bound is convention-specific even though the estimand it bounds is not; §2.1.1). That claim is made stratum by stratum and on one row set, the glycol-ether solvents, and through a one-sided bound; the aggregate does not stand (§3.5.1). $\mathcal{G}$ only corroborates it.

S12 Positioning against adjacent literatures

§1 places the instrument against the model-discrepancy problem of Kennedy and O'Hagan[S25] and its calibration critique,[S14] which is the frame the paper works in. Four further literatures bear on it without being that frame. Separating whether predictions are right on average from whether they are sharp is Murphy and DeGroot's reliability–resolution decomposition:[S26,S27] the closure/insufficiency split is an object of the same kind, an identity that partitions a squared error, but on a *fixed* operator with an external reference for the intermediate rather than on a forecast–outcome pair. A misspecified model still converges to a pseudo-true limit under White's quasi-maximum-likelihood account,[S19] which is what Δ_∞ of Proposition S1 names for a composed predictor. That a decomposition may be only weakly constrained by the data is the subject of parameter sloppiness[S28,S29] and of structural versus practical identifiability[S30,S31]—the setting of Lemma S1. That apparent skill may reflect pipeline choices rather than physics is the concern of underspecification,[S32] leakage[S33] and compositional risk,[S34] which is why every arm difference in this paper is reported as a difference between two separately tuned pipelines

(§3.7). Finally, that a constrained prior lowers variance at the cost of a bias confined to the directions the data leave undetermined is the null-space reading of regularisation[S35,S36] and of null-space priors in inverse problems,[S37] which is the frame in which Proposition S1 is stated.

References

(S1) Bell, I. H.; Mickoleit, E.; Hsieh, C.-M.; Lin, S.-T.; Vrabec, J.; Breitkopf, C.; Jäger, A. A Benchmark Open-Source Implementation of COSMO-SAC. *Journal of Chemical Theory and Computation* **2020**, *16*, 2635–2646.

(S2) Hsieh, C.-M.; Sandler, S. I.; Lin, S.-T. Improvements of COSMO-SAC for Vapor–Liquid and Liquid–Liquid Equilibrium Predictions. *Fluid Phase Equilibria* **2010**, *297*, 90–97.

(S3) Hsieh, C.-M.; Lin, S.-T.; Vrabec, J. Considering the Dispersive Interactions in the COSMO-SAC Model for More Accurate Predictions of Fluid Phase Behavior. *Fluid Phase Equilibria* **2014**, *367*, 109–116.

(S4) Renon, H.; Prausnitz, J. M. Local Compositions in Thermodynamic Excess Functions for Liquid Mixtures. *AIChE Journal* **1968**, *14*, 135–144.

(S5) Fredenslund, A.; Jones, R. L.; Prausnitz, J. M. Group-Contribution Estimation of Activity Coefficients in Nonideal Liquid Mixtures. *AIChE Journal* **1975**, *21*, 1086–1099.

(S6) Rampášek, L.; Galkin, M.; Dwivedi, V. P.; Luu, A. T.; Wolf, G.; Beaini, D. Recipe for a General, Powerful, Scalable Graph Transformer. Advances in Neural Information Processing Systems (NeurIPS). 2022.

(S7) van der Vaart, A. W. *Asymptotic Statistics*; Cambridge Series in Statistical and Probabilistic Mathematics; Cambridge University Press, 1998; Chapter 25 (semiparametric efficiency: efficient score and information).

(S8) Hirano, K.; Imbens, G. W.; Ridder, G. Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score. *Econometrica* **2003**, *71*, 1161–1189.

(S9) Robins, J. M.; Ritov, Y. Toward a Curse of Dimensionality Appropriate (CODA) Asymptotic Theory for Semi-Parametric Models. *Statistics in Medicine* **1997**, *16*, 285–319.

(S10) Fingerhut, R.; Chen, W.-L.; Schedemann, A.; Cordes, W.; Rarey, J.; Hsieh, C.-M.; Vrabec, J.; Lin, S.-T. Comprehensive Assessment of COSMO-SAC Models for Predictions of Fluid-Phase Equilibria. *Industrial & Engineering Chemistry Research* **2017**, *56*, 9868–9884.

(S11) Lin, S.-T.; Sandler, S. I. A Priori Phase Equilibrium Prediction from a Segment Contribution Solvation Model. *Industrial & Engineering Chemistry Research* **2002**, *41*, 899–913.

(S12) Attia, L.; Burns, J. W.; Doyle, P. S.; Green, W. H. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications* **2025**, *16*, 7497, FASTSOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.

(S13) Vermeire, F. H.; Chung, Y.; Green, W. H. Predicting Solubility Limits of Organic Solutes for a Wide Range of Solvents and Temperatures. *Journal of the American Chemical Society* **2022**, *144*, 10785–10797.

(S14) Brynjarsdóttir, J.; O'Hagan, A. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems* **2014**, *30*, 114007.

(S15) Takeishi, N.; Kalousis, A. Physics-Integrated Variational Autoencoders for Robust and Interpretable Generative Modeling. Advances in Neural Information Processing Systems (NeurIPS). 2021.

(S16) Internò, C.; Yamaguchi, J.; Amdahl-Culleton, L.; Olhofer, M.; Klindt, D.; Hammer, B. The Observer Effect in

World Models: Invasive Adaptation Corrupts Latent Physics. *arXiv preprint arXiv:2602.12218* **2026**,

(S17) Sanchez Medina, E. I.; Linke, S.; Stoll, M.; Sundmacher, K. Gibbs–Helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution. *Digital Discovery* **2023**, *2*, 781–798.

(S18) Mansouri, K.; Cariello, N. F.; Korotcov, A.; Tkachenko, V.; Grulke, C. M.; Sprankle, C. S.; Allen, D.; Casey, W. M.; Kleinstreuer, N. C.; Williams, A. J. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics* **2019**, *11*, 60.

(S19) White, H. Maximum Likelihood Estimation of Misspecified Models. *Econometrica* **1982**, *50*, 1–25.

(S20) Hansch, C.; Leo, A.; Taft, R. W. A Survey of Hammett Substituent Constants and Resonance and Field Parameters. *Chemical Reviews* **1991**, *91*, 165–195.

(S21) de Souza, E. T., Jr.; Alcantara, M. L.; Staudt, P. B.; Coutinho, J. a. A. P.; Soares, R. d. P. Development of a COSMO-SAC Parametrization with Advanced QM Method TZVPD-FINE. *Industrial & Engineering Chemistry Research* **2025**, *64*, 14700–14711.

(S22) Krasnov, L.; Malikov, D.; Kiseleva, M.; Tatarin, S.; Sosnin, S.; Bezzubov, S. BigSolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data* **2025**, *12*, 1236.

(S23) Mullins, E.; Oldland, R.; Liu, Y. A.; Wang, S.; Sandler, S. I.; Chen, C.-C.; Zwolak, M.; Seavey, K. C. Sigma-Profile Database for Using COSMO-Based Thermodynamic Methods. *Industrial & Engineering Chemistry Research* **2006**, *45*, 4389–4415.

(S24) Frenkel, M.; Chirico, R. D.; Diky, V. V.; Yan, X.; Dong, Q.; Muzny, C. ThermoML: An XML-Based Approach for Storage and Exchange of Experimental and Critically Evaluated Thermophysical and Thermochemical Property Data. *Journal of Chemical Information and Computer Sciences* **2003**, *43*, 1344–1359.

(S25) Kennedy, M. C.; O'Hagan, A. Bayesian Calibration of Computer Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **2001**, *63*, 425–464.

(S26) Murphy, A. H. A New Vector Partition of the Probability Score. *Journal of Applied Meteorology* **1973**, *12*, 595–600.

(S27) DeGroot, M. H.; Fienberg, S. E. The Comparison and Evaluation of Forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)* **1983**, *32*, 12–22.

(S28) Transtrum, M. K.; Machta, B. B.; Brown, K. S.; Daniels, B. C.; Myers, C. R.; Sethna, J. P. Perspective: Sloppiness and emergent theories in physics, biology, and beyond. *The Journal of Chemical Physics* **2015**, *143*, 010901.

(S29) Gutenkunst, R. N.; Waterfall, J. J.; Casey, F. P.; Brown, K. S.; Myers, C. R.; Sethna, J. P. Universally Sloppy Parameter Sensitivities in Systems Biology Models. *PLoS Computational Biology* **2007**, *3*, 1871–1878.

(S30) Raue, A.; Kreutz, C.; Maiwald, T.; Bachmann, J.; Schilling, M.; Klingmüller, U.; Timmer, J. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **2009**, *25*, 1923–1929.

(S31) Villaverde, A. F.; Barreiro, A.; Papachristodoulou, A. Structural Identifiability of Dynamic Systems Biology Models. *PLoS Computational Biology* **2016**, *12*, e1005153.

(S32) D'Amour, A.; Heller, K.; Moldovan, D.; Adlam, B.; Alipanahi, B.; Beutel, A.; Chen, C.; Deaton, J.; Eisenstein, J.; Hoffman, M. D.; Hormozdiari, F.; Houlsby, N.; Hou, S.; Jerfel, G.; Karthikesalingam, A.; Lucic, M.; Ma, Y.; McLean, C.; Mincu, D.; Mitani, A.; Montanari, A.; Nado, Z.; Natarajan, V.; Nielson, C.; Osborne, T. F.; Raman, R.; Ramasamy, K.; Sayres, R.; Schrouff, J.; Seneviratne, M.; Sequeira, S.; Suresh, H.; Veitch, V.; Vladymyrov, M.; Wang, X.; Webster, K.; Yadlowsky, S.; Yun, T.; Zhai, X.; Sculley, D. Underspecification Presents Challenges for Credibility in Modern Machine Learning. *Journal of Machine Learning Research* **2022**, *23*, 1–61.

(S33) Kapoor, S.; Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **2023**, *4*, 100804.

(S34) Mahajan, D.; Pezeshki, M.; Arnal, C.; Mitliagkas, I.; Ahuja, K.; Vincent, P. Compositional Risk Minimization. Proceedings of the 42nd International Conference on Machine Learning. 2025; pp 42503–42553.

(S35) Engl, H. W.; Hanke, M.; Neubauer, A. *Regularization of Inverse Problems*; Mathematics and Its Applications; Kluwer Academic Publishers, 1996; Vol. 375.

(S36) Hansen, P. C. *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion*; SIAM Monographs on Mathematical Modeling and Computation; SIAM, 1998.

(S37) Schwab, J.; Antholzer, S.; Haltmeier, M. Deep Null Space Learning for Inverse Problems: Convergence Analysis and Rates. *Inverse Problems* **2019**, *35*, 025008.